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1 Registers

This chapter lists the registers in the chip.

Unspecified bits in the registers listed in this section must be written as zeroes and can be ignored on reads.

The following table lists all register targets and associated base addresses. The next level lists registers groups and offsets within targets, and the deepest level lists registers within the register groups.

Both register groups and registers may be replicated (repeated) a number of times. The repeat-count and the distance between two repetitions is listed in the "Instances and Address Spacing" column of the tables. If there is only one instance, the spacing is omitted. The "Offset within Target"/"Offset within Register Group" columns hold the offset of the first instance of the register group/register.

In order to calculate the absolute address of a given register, multiply the register group's replication number by the register group's address spacing and add it to the register group's offset within the target. Then multiply the register's replication number with the register's address spacing and add it to the register's offset within the register group. Finally add these two numbers to the absolute address of the target in question.

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Table 1. List of Targets and Base Addresses

MMD Name	MMD Number	Description	Details
CL22_BASIC	0x00 (=MMD)	cl22_basic_regs registers.	Page 1
PMA_COMMON_100T1	0x01 (=MMD)	chiptop_regs registers.	Page 4
1000T1_PMA	0x01 (=MMD)	pma_regs registers.	Page 9
PCS_COMMON_100T1	0x03 (=MMD)	pcs_common_regs registers.	Page 11
1000T1_PCS	0x03 (=MMD)	pcs_regs registers.	Page 14
100T1_PCS_VENDOR	0x03 (=MMD)	pcs_100_regs registers.	Page 30
ANEG	0x07 (=MMD)	aneg_regs registers.	Page 51
MIS	0x1E (=MMD)	mis_regs registers.	Page 61
CHIPTOP_COMMON	0x1E (=MMD)	dev30_common registers.	Page 84
MX_CHIP_TOP	0x1E (=MMD)	mx_chip_top_regs registers.	Page 90
MX_UVOV	0x1E (=MMD)	mx_uvov_regs registers.	Page 129
MX_PVT	0x1E (=MMD)	mx_pvt_regs registers.	Page 142

1.1 cl22_basic_regs

Table 2. Register Groups in MMD cl22_basic_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
cl22_basic_regs	0x00000000	1		Page 1

1.1.1 cl22_basic_regs:cl22_basic_regs

Parent: cl22_basic_regs

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Table 3. Registers in cl22_basic_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
CL22_BASIC_CONTROL	0x00000000	1		Page 2
CL22_BASIC_STATUS	0x00000001	1		Page 2
PHY_ID_1	0x00000002	1		Page 3
PHY_ID_2	0x00000003	1		Page 3
CL45_INDIRECT_ACCESS_REG13	0x0000000D	1		Page 3
CL45_INDIRECT_ACCESS_REG14	0x0000000E	1		Page 3

1.1.1.1 cl22_basic_regs:cl22_basic_regs:CL22_BASIC_CONTROL

Parent: cl22_basic_regs:cl22_basic_regs

Instances: 1

Table 4. Fields in CL22_BASIC_CONTROL

Field Name	Bit	Access	Description	Default
RESET	15	R/W	(self-clearing)	0x0
PMA_LOCAL_LB	14	R/W		0x0
SPEED_SEL_LSB	13	R/W		0x1
CL22_BASIC_AUTO_EN	12	R/O		0x0
LOW_POWER	11	R/W		0x0
ISOLATE	10	R/W		0x0
CL22_RESTART_ANEG	9	R/W		0x0
DUPLEX_MODE	8	R/O		0x1
COLLISION_TEST	7	R/O		0x0
SPEED_SEL_MSB	6	R/W		0x0
RESERVED1_CL22_BASIC_CONTROL_1_10	5:0	R/O		0x00

1.1.1.2 cl22_basic_regs:cl22_basic_regs:CL22_BASIC_STATUS

Parent: cl22_basic_regs:cl22_basic_regs

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Table 5. Fields in CL22_BASIC_STATUS

Field Name	Bit	Access	Description	Default
RESERVED1_CL22_BASIC_STATUS_1_11	15:6	R/O		0x000
MR_AUTONEG_COMPLETE	5	R/O		0x0
RESERVED2_CL22_BASIC_STATUS_1_13	4	R/O		0x0
CFG_ANEG_IMPL	3	R/O		0x0
RCV_LINK_STATUS	2	R/O		0x0
JABBER_DET	1	R/O	Note: Reading this field will clear it.	0x1
RESERVED3_CL22_BASIC_STATUS_1_17	0	R/O		0x0

1.1.1.3 cl22_basic_regs:cl22_basic_regs:PHY_ID_1

Parent: cl22_basic_regs:cl22_basic_regs

Instances: 1

Table 6. Fields in PHY_ID_1

Field Name	Bit	Access	Description	Default
OUI_3_18	15:0	R/W		0x0007

1.1.1.4 cl22_basic_regs:cl22_basic_regs:PHY_ID_2

Parent: cl22_basic_regs:cl22_basic_regs

Instances: 1

Table 7. Fields in PHY_ID_2

Field Name	Bit	Access	Description	Default
OUI_19_24	15:10	R/W		0x30
MODEL_NUMBER	9:4	R/W		0x1F
REVISION_NUMBER	3:0	R/W		0x1

1.1.1.5 cl22_basic_regs:cl22_basic_regs:CL45_INDIRECT_ACCESS_REG13

Parent: cl22_basic_regs:cl22_basic_regs

Instances: 1

Table 8. Fields in CL45_INDIRECT_ACCESS_REG13

Field Name	Bit	Access	Description	Default
REG_13_CL45_ACC	15:0	R/W		0x0000

1.1.1.6 cl22_basic_regs:cl22_basic_regs:CL45_INDIRECT_ACCESS_REG14

Parent: cl22_basic_regs:cl22_basic_regs

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Instances: 1

Table 9. Fields in CL45_INDIRECT_ACCESS_REG14

Field Name	Bit	Access	Description	Default
REG_14_CL45_ACC	15:0	R/W		0x0000

1.2 chiptop_regs

Table 10. Register Groups in MMD chiptop_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
chiptop_regs	0x00000000	1		Page 4

1.2.1 chiptop_regs:chiptop_regs

Parent: chiptop_regs

Instances: 1

Table 11. Registers in chiptop_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PMA_CONTROL_1	0x00000000	1		Page 4
PMA_STATUS_1	0x00000001	1		Page 5
PMA_DEV_ID_1	0x00000002	1		Page 5
PMA_DEV_ID_2	0x00000003	1		Page 5
PMA_SPEED_ABILITY	0x00000004	1		Page 6
PMA_DEV_PKG1	0x00000005	1		Page 6
PMA_DEV_PKG2	0x00000006	1		Page 6
PMA_CONTROL_2	0x00000007	1		Page 7
PMA_STATUS_2	0x00000008	1		Page 7
PMA_TX_DISABLE	0x00000009	1		Page 7
PMA_EXT_ABILITY	0x0000000B	1		Page 8
PMA_EXT_ABILITY_2	0x00000012	1		Page 8
PMA_CONTROL_T1	0x00000834	1		Page 8
PMA_TEST_CONTROL_T1	0x00000836	1		Page 8

1.2.1.1 chiptop_regs:chiptop_regs:PMA_CONTROL_1

Parent: chiptop_regs:chiptop_regs

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Instances: 1

Table 12. Fields in PMA_CONTROL_1

Field Name	Bit	Access	Description	Default
RESET	15	R/W	(self-clearing)	0x0
RESERVED1_PMA_CONTROL_1_1	14	R/O		0x0
SPEED_SEL_LSB	13	R/O	N/A for T1 PHYs	0x0
RESERVED2_PMA_CONTROL_1_3	12	R/O		0x0
LOW_POWER	11	R/W	Always 0	0x0
RESERVED3_PMA_CONTROL_1_5	10:7	R/O		0x0
SPEED_SEL_MSB	6	R/O	N/A for T1 PHYs	0x0
SPEED_SELECTION	5:2	R/O	N/A for T1 PHYs	0x0
PMA_REMOTE_LB	1	R/W	PMA remote loopback	0x0
PMA_LOCAL_LB	0	R/W	PMA local loopback	0x0

1.2.1.2 chiptop_regs:chiptop_regs:PMA_STATUS_1

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 13. Fields in PMA_STATUS_1

Field Name	Bit	Access	Description	Default
RESERVED1_PMA_STATUS_1_10	15:8	R/O		0x00
FAULT	7	R/O		0x0
RESERVED2_PMA_STATUS_1_12	6:3	R/O		0x0
RCV_LINK_STATUS	2	R/O		0x0
LOW_POWER_ABILITY	1	R/O		0x1
RESERVED3_PMA_STATUS_1_15	0	R/O		0x0

1.2.1.3 chiptop_regs:chiptop_regs:PMA_DEV_ID_1

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 14. Fields in PMA_DEV_ID_1

Field Name	Bit	Access	Description	Default
OUI_3_18	15:0	R/W		0x0007

1.2.1.4 chiptop_regs:chiptop_regs:PMA_DEV_ID_2

Parent: chiptop_regs:chiptop_regs

Instances: 1

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Table 15. Fields in PMA_DEV_ID_2

Field Name	Bit	Access	Description	Default
OUI_19_24	15:10	R/W		0x30
MODEL_NUMBER	9:4	R/W		0x1F
REVISION_NUMBER	3:0	R/W		0x1

1.2.1.5 chiptop_regs:chiptop_regs:PMA_SPEED_ABILITY

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 16. Fields in PMA_SPEED_ABILITY

Field Name	Bit	Access	Description	Default
RESERVED1_PMA_SPEED_ABILITY_20	15:6	R/O		0x000
CAPABLE_100M	5	R/O		0x1
CAPABLE_1000M	4	R/O		0x1
RESERVED2_PMA_SPEED_ABILITY_23	3:0	R/O		0x0

1.2.1.6 chiptop_regs:chiptop_regs:PMA_DEV_PKG1

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 17. Fields in PMA_DEV_PKG1

Field Name	Bit	Access	Description	Default
RESERVED1_PMA_DEV_PKG1_24	15:4	R/O		0x000
PCS_PRESENT	3	R/O		0x1
RESERVED2_PMA_DEV_PKG1_26	2	R/O		0x0
PMA_PMD_PRESENT	1	R/O		0x1
CL22_REG_PRESENT	0	R/O		0x0

1.2.1.7 chiptop_regs:chiptop_regs:PMA_DEV_PKG2

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 18. Fields in PMA_DEV_PKG2

Field Name	Bit	Access	Description	Default
VENDOR_SPEC_2	15	R/O		0x0
VENDOR_SPEC_1	14	R/O		0x1

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Field Name	Bit	Access	Description	Default
RESERVED1_PMA_DEV_PKG2_31	13:0	R/O		0x0000

1.2.1.8 chiptop_regs:chiptop_regs:PMA_CONTROL_2

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 19. Fields in PMA_CONTROL_2

Field Name	Bit	Access	Description	Default
RESERVED1_PMA_CONTROL_2_32	15:10	R/O		0x00
PMA_INGR_AUI_STOP_ENAB	9	R/W		0x0
PMA_EGR_AUI_STOP_ENAB	8	R/W		0x0
RESERVED2_PMA_CONTROL_2_35	7	R/O		0x0
PMA_PMD_SEL_TYPE	6:0	R/W	BASE-T1 PMA/PMD	0x3D

1.2.1.9 chiptop_regs:chiptop_regs:PMA_STATUS_2

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 20. Fields in PMA_STATUS_2

Field Name	Bit	Access	Description	Default
DEV_PRESENT	15:14	R/O		0x2
TX_FAULT_ABILITY	13	R/O		0x0
RCV_FAULT_ABILITY	12	R/O		0x1
TX_FAULT	11	R/O		0x0
RCV_FAULT	10	R/O		0x1
EXT_ABILITIES	9	R/O	Has extended abilities listed in 1.11	0x1
PMD_TX_DISABLE_ABILITY	8	R/O		0x0
RESERVED1_PMA_STATUS_2_44	7:1	R/O		0x00
PMA_LOCAL_LPBK_ABILITY	0	R/O		0x1

1.2.1.10 chiptop_regs:chiptop_regs:PMA_TX_DISABLE

Parent: chiptop_regs:chiptop_regs

Instances: 1

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Table 21. Fields in PMA_TX_DISABLE

Field Name	Bit	Access	Description	Default
RESERVED_TX_DISABLE_46	15:1	R/O		0x0000
TRANSMIT_DISABLE	0	R/W		0x0

1.2.1.11 chiptop_regs:chiptop_regs:PMA_EXT_ABILITY

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 22. Fields in PMA_EXT_ABILITY

Field Name	Bit	Access	Description	Default
RESERVED1_PMA_EXT_ABILITY_48	15:12	R/O		0x0
BASET1_EXT_ABILITIES	11	R/O		0x1
RESERVED2_PMA_EXT_ABILITY_50	10:0	R/O		0x000

1.2.1.12 chiptop_regs:chiptop_regs:PMA_EXT_ABILITY_2

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 23. Fields in PMA_EXT_ABILITY_2

Field Name	Bit	Access	Description	Default
RESERVED_PMA_EXT_ABILITY_2_51	15:2	R/O		0x0000
ABILITY_1000T1	1	R/O		0x1
ABILITY_100T1	0	R/O		0x1

1.2.1.13 chiptop_regs:chiptop_regs:PMA_CONTROL_T1

Parent: chiptop_regs:chiptop_regs

Instances: 1

Table 24. Fields in PMA_CONTROL_T1

Field Name	Bit	Access	Description	Default
RESERVED_MAS_SLV_MAN_CFG_EN_54	15	R/O		0x1
MAS_SLV_CFG_VAL	14	R/W	depends on cfg_phy_mode	0x1
RESERVED_PMA_CONTROL_T1_56	13:4	R/O		0x000
TYPE_SEL	3:0	R/W		0x1

1.2.1.14 chiptop_regs:chiptop_regs:PMA_TEST_CONTROL_T1

Parent: chiptop_regs:chiptop_regs

Instances: 1

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Table 25. Fields in PMA_TEST_CONTROL_T1

Field Name	Bit	Access	Description	Default
TEST_MODE_CONTROL_T1	15:13	R/W		0x0
RESERVED_PMA_TEST_CONTROL_T1_59	12:0	R/O		0x0000

1.3 pma_regs

Table 26. Register Groups in MMD pma_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
pma_regs	0x00000000	1		Page 9

1.3.1 pma_regs:pma_regs

Parent: pma_regs

Instances: 1

Table 27. Registers in pma_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PMA_CONTROL	0x00000900	1	1000BASE-T1 PMA Control Reg.	Page 9
PMA_STATUS	0x00000901	1	1000BASE-T1 PMA Status Reg.	Page 10
PMA_TRAINING_CONFIG	0x00000902	1	1000BASE-T1 Training Reg.	Page 10
LP_TRAINING_STAT	0x00000903	1	1000BASE-T1 LP Training Status register	Page 10
TEST_MODE_CTRL	0x00000904	1	1000BASE-T1 Test Mode Control Reg.	Page 11

1.3.1.1 pma_regs:pma_regs:PMA_CONTROL

Parent: pma_regs:pma_regs

Instances: 1

Table 28. Fields in PMA_CONTROL

Field Name	Bit	Access	Description	Default
PMA_RST	15	R/W	1 - PMA/PMD reset; 0 - Normal Operation	0x0
TX_DISABLE	14	R/W	1 - Transmit disable ; 0 - Normal Operation	0x0
RESERVED1_2	13:12	R/O		0x0
LOW_PWR_MODE	11	R/W	1 - Low power mode; 0 - Normal operation	0x0
RESERVED2_4	10:0	R/O		0x000

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1.3.1.2 pma_regs:pma_regs:PMA_STATUS

Parent: pma_regs:pma_regs

Instances: 1

Table 29. Fields in PMA_STATUS

Field Name	Bit	Access	Description	Default
RESERVED1_5	15:12	R/O		0x0
GIG_T1_OAM_ABILITY	11	R/O	1 - PHY has 1000T1 OAM ability; 0 - PHY does not have 1000T1 OAM ability.	0x1
EEE_ABILITY	10	R/O	1 - PHY has EEE ability; 0 - PHY does not have EEE ability.	0x0
RX_FAULT_ABILITY	9	R/O	1 - PMA/PMD has the ability to detect a fault condition on the receive path; 0 - PMA/PMD does not have the ability to detect a fault condition on the receive path.	0x0
LOW_PWR_ABILITY	8	R/O	1 - PMA/PMD has the low power ability ; 0 - PMA/PMD does not have the low-power ability.	0x0
RESERVED2_10	7:3	R/O		0x00
RX_POLARITY	2	R/O	1 - Receive polarity is reversed; 0 - Receive polarity is not reversed.	0x0
RX_FAULT	1	R/O	1 - Fault condition detected; 0 - Fault condition not detected.	0x0
RX_LINK_STAT	0	R/O	1 - PMA/PMD receive link up; 0 - PMA/PMD receive link down. This is a Latch-low bit.	0x0

1.3.1.3 pma_regs:pma_regs:PMA_TRAINING_CONFIG

Parent: pma_regs:pma_regs

Instances: 1

Table 30. Fields in PMA_TRAINING_CONFIG

Field Name	Bit	Access	Description	Default
RESERVED1_14	15:11	R/O		0x00
TRN_USR_FIELD	10:4	R/W	7-bit user defined field to send to the link-partner	0x00
RESERVED2_16	3:2	R/O		0x0
GIG_T1_OAM_ADV	1	R/W	1 - 1000T1 OAM ability advertised to the link partner; 0 - 1000T1 OAM ability not advertised to the link partner	0x1
EEE_ADV	0	R/W	1 - 1000T1 EEE ability advertised to the link partner; 0 - 1000T1 EEE ability not advertised to the link partner	0x0

1.3.1.4 pma_regs:pma_regs:LP_TRAINING_STAT

Parent: pma_regs:pma_regs

Instances: 1

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Table 31. Fields in LP_TRAINING_STAT

Field Name	Bit	Access	Description	Default
RESERVED1_19	15:11	R/O		0x00
LP_TRN_USR_FIELD	10:4	R/O	7-bit user defined field received from the link partner	0x00
RESERVED2_21	3:2	R/O		0x0
LP_GIG_T1_OAM_ADV	1	R/O	1 - Link partner has the 1000T1 OAM ability; 0 - Link Partner does not have 1000T1 OAM ability.	0x0
LP_EEE_ADV	0	R/O	1 - Link partner has the 1000T1 EEE ability; 0 - Link Partner does not have 1000T1 EEE ability.	0x0

1.3.1.5 pma_regs:pma_regs:TEST_MODE_CTRL

Parent: pma_regs:pma_regs

Instances: 1

Table 32. Fields in TEST_MODE_CTRL

Field Name	Bit	Access	Description	Default
TEST_MODE	15:13	R/W	111 - Test Mode 7; 110 - Test Mode 6; 101 - Test Mode 5; 100 - Test Mode 4; 011 - Reserved; 010 - Test Mode 2;001 - Test Mode 1; 000 - Normal (Non-test) operation.	0x0
RESERVED_25	12:0	R/O		0x0000

1.4 pcs_common_regs

Table 33. Register Groups in MMD pcs_common_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
pcs_common_regs	0x00000000	1		Page 11

1.4.1 pcs_common_regs:pcs_common_regs

Parent: pcs_common_regs

Instances: 1

Table 34. Registers in pcs_common_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PCS_CTRL1	0x00000000	1		Page 12
PCS_STS1	0x00000001	1		Page 12
PCS_DEV_ID1	0x00000002	1		Page 12
PCS_DEV_ID2	0x00000003	1		Page 13

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PCS_SPD_ABIL	0x00000004	1		Page 13
PCS_DEV_PKG1	0x00000005	1		Page 13
PCS_DEV_PKG2	0x00000006	1		Page 13

1.4.1.1 pcs_common_regs:pcs_common_regs:PCS_CTRL1

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 35. Fields in PCS_CTRL1

Field Name	Bit	Access	Description	Default
PCS_RESET	15	R/W	1=PCS Reset; 0=Normal operation (self-clearing)	0x0
PCS_LPBK	14	R/W	PCS loopback is enabled when this bit is set to 1.	0x0
RESERVED_13_12_2	13:12	R/O	Reserved for future use.	0x0
PCS_LOW_PWR	11	R/W	Enter low-power mode when this bit is set to 1.	0x0
RESERVED_10_0_4	10:0	R/O	Reserved for future use.	0x000

1.4.1.2 pcs_common_regs:pcs_common_regs:PCS_STS1

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 36. Fields in PCS_STS1

Field Name	Bit	Access	Description	Default
RESERVED_15_3_5	15:3	R/O	Reserved for future use.	0x0000
PCS_LINK_STAT	2	R/O	PCS receive link status (latch-low) Note: Reading this field will set it.	0x0
PCS_LOW_PWR_ABIL	1	R/O	Low-power ability. 1 indicated low-power mode support by PCS.	0x1
RESERVED_0_8	0	R/O	Always 0.	0x0

1.4.1.3 pcs_common_regs:pcs_common_regs:PCS_DEV_ID1

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 37. Fields in PCS_DEV_ID1

Field Name	Bit	Access	Description	Default
OUI_3_18	15:0	R/W	Bits [3:18] of the OUI. WH uses the SMSC OUI of 00-80-0F (or 00:01:F0)	0x0007

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1.4.1.4 pcs_common_regs:pcs_common_regs:PCS_DEV_ID2

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 38. Fields in PCS_DEV_ID2

Field Name	Bit	Access	Description	Default
OUI_19_24	15:10	R/W	Bits [19:24] of the OUI.	0x30
MODEL_NUM	9:4	R/W	The device model number.	0x1F
REV_NUM	3:0	R/W	Revision number (0 for WH A0)	0x1

1.4.1.5 pcs_common_regs:pcs_common_regs:PCS_SPD_ABIL

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 39. Fields in PCS_SPD_ABIL

Field Name	Bit	Access	Description	Default
RESERVED_13	15:0	R/O	N/A for T1 PHYs.	0x0000

1.4.1.6 pcs_common_regs:pcs_common_regs:PCS_DEV_PKG1

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 40. Fields in PCS_DEV_PKG1

Field Name	Bit	Access	Description	Default
RESERVED_15_8_14	15:8	R/O	Reserved for future use.	0x00
ANEG_PRESENT	7	R/O	If 1, auto-negotiation is present in the package.	0x1
RESERVED_6_4_16	6:4	R/O	Reserved for future use.	0x0
PCS_PRESENT	3	R/O	If 1, a PCS is present in the package.	0x1
RESERVED_2_18	2	R/O	Reserved for future use.	0x0
PMA_PMD_PRESENT	1	R/O	If 1, a PMA/PMD is present in the package.	0x1
CL22_REGS_PRESENT	0	R/O	If 1, Clause 22 registers are present in the package.	0x0

1.4.1.7 pcs_common_regs:pcs_common_regs:PCS_DEV_PKG2

Parent: pcs_common_regs:pcs_common_regs

Instances: 1

Table 41. Fields in PCS_DEV_PKG2

Field Name	Bit	Access	Description	Default
VEND_SPEC_DEV2_PRESENT	15	R/O	If 1, vendor-specific device 2 is present in the package.	0x0
VEND_SPEC_DEV1_PRESENT	14	R/O	If 1, vendor-specific device 1 is present in the package.	0x1

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Field Name	Bit	Access	Description	Default
CL22_EXT_PRESENT	13	R/O	If 1, Clause 22 extension is present in the package.	0x0
RESERVED_24	12:0	R/O	Reserved for future use.	0x0000

1.5 pcs_regs

Table 42. Register Groups in MMD pcs_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
pcs_regs	0x00000000	1		Page 14

1.5.1 pcs_regs:pcs_regs

Parent: pcs_regs

Instances: 1

Table 43. Registers in pcs_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PCS_CONTROL	0x00000900	1		Page 16
PCS_STATUS_1	0x00000901	1		Page 16
PCS_STATUS_2	0x00000902	1		Page 17
OAM_TX	0x00000904	1		Page 17
OAM_MSG_0_1	0x00000905	1		Page 17
OAM_MSG_2_3	0x00000906	1		Page 18
OAM_MSG_4_5	0x00000907	1		Page 18
OAM_MSG_6_7	0x00000908	1		Page 18
OAM_RX	0x00000909	1		Page 18
OAM_LP_MSG_0_1	0x0000090A	1		Page 19
OAM_LP_MSG_2_3	0x0000090B	1		Page 19
OAM_LP_MSG_4_5	0x0000090C	1		Page 19
OAM_LP_MSG_6_7	0x0000090D	1		Page 19
DATA_MODE_SCRAM_SEED	0x00008000	1		Page 19
TR_SCRAM_SEED_0	0x00008001	1		Page 20
TR_SCRAM_SEED_1	0x00008002	1		Page 20

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
SCR_CONFIG	0x00008003	1		Page 20
RSFEC_ERR0_CNT	0x00008015	1		Page 20
RSFEC_ERR1_CNT	0x00008016	1		Page 21
RSFEC_ERR2_CNT	0x00008017	1		Page 21
RSFEC_ERR3_CNT	0x00008018	1		Page 21
RSFEC_ERR4_CNT	0x00008019	1		Page 21
RSFEC_ERR5_CNT	0x0000801A	1		Page 22
RSFEC_ERR6_CNT	0x0000801B	1		Page 22
RSFEC_ERR7_CNT	0x0000801C	1		Page 22
RSFEC_ERR8_CNT	0x0000801D	1		Page 22
RSFEC_ERR9_CNT	0x0000801E	1		Page 22
RSFEC_ERR10_CNT	0x0000801F	1		Page 23
RSFEC_ERR11_CNT	0x00008020	1		Page 23
RSFEC_ERR12_CNT	0x00008021	1		Page 23
RSFEC_ERR13_CNT	0x00008022	1		Page 23
RSFEC_ERR14_CNT	0x00008023	1		Page 24
RSFEC_ERR15_CNT	0x00008024	1		Page 24
RSFEC_ERR16_CNT	0x00008025	1		Page 24
RSFEC_ERR17_CNT	0x00008026	1		Page 24
RSFEC_ERR18_CNT	0x00008027	1		Page 24
RSFEC_ERR19_CNT	0x00008028	1		Page 25
RSFEC_ERR20_CNT	0x00008029	1		Page 25
RSFEC_ERR21_CNT	0x0000802A	1		Page 25
RSFEC_ERR22_CNT	0x0000802B	1		Page 25
RSFEC_ERR23_CNT	0x0000802C	1		Page 26
DEBUG_BUS_CONFIG	0x0000802F	1		Page 26
OAM_SNR_CONFIG1	0x00008030	1		Page 26
OAM_SNR_CONFIG2	0x00008031	1		Page 26

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
FEC_CONFIG	0x00008032	1		Page 26
RSFEC_ERR_CNT_FLB	0x00008033	1		Page 27
RX_DATA_SCRAM_SEED	0x00008034	1		Page 27
PCS_RX_DEC_CTRL	0x00008035	1		Page 27
LFL_COMM_COUNT	0x00008036	1		Page 27
LFL_COUNT	0x00008037	1		Page 28
PCS_INTR_1	0x00008038	1		Page 28
PCS_INTR_2	0x00008039	1		Page 28
PCS_INTR_3	0x0000803A	1		Page 29
PCS_INTR_4	0x0000803B	1		Page 29
OAM_PCS_DBG_STS	0x0000803C	1		Page 29
LQ_COM_TC12	0x0000803D	1		Page 30

1.5.1.1 pcs_regs:pcs_regs:PCS_CONTROL

Parent: pcs_regs:pcs_regs

Instances: 1

Table 44. Fields in PCS_CONTROL

Field Name	Bit	Access	Description	Default
PCS_RST	15	R/W	PCS Reset	0x0
PCS_LPBK	14	R/W	PCS Loopback	0x0
RESERVED_2	13:0	R/O	Always 0	0x0000

1.5.1.2 pcs_regs:pcs_regs:PCS_STATUS_1

Parent: pcs_regs:pcs_regs

Instances: 1

Table 45. Fields in PCS_STATUS_1

Field Name	Bit	Access	Description	Default
RESERVED1_3	15:12	R/O	Always 0	0x0
TX_LPI_RCVD	11	R/O	TX LPI received. Latch-High (Set until read) Note: Reading this field will clear it.	0x0
RX_LPI_RCVD	10	R/O	RX LPI received. Latch-High Note: Reading this field will clear it.	0x0
TX_LPI_RCVNG	9	R/O	TX LPI indication	0x0

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Field Name	Bit	Access	Description	Default
RX_LPI_RCVNG	8	R/O	RX LPI indication	0x0
FAULT	7	R/O	RX Fault detected	0x0
RESERVED2_9	6:3	R/O	Always 0	0x0
PCS_LINK_STAT_1	2	R/O	PCS RX link status. Latch Low	0x0
RESERVED3_11	1:0	R/O	Always 0	0x0

1.5.1.3 pcs_regs:pcs_regs:PCS_STATUS_2

Parent: pcs_regs:pcs_regs

Instances: 1

Table 46. Fields in PCS_STATUS_2

Field Name	Bit	Access	Description	Default
RESERVED_12	15:11	R/O	Always 0	0x00
PCS_LINK_STAT_2	10	R/O	PCS RX link status	0x0
PCS_HI_BER	9	R/O	PCS high BER	0x0
PCS_BLK_LOCK	8	R/O	PCS block lock	0x0
PCS_HI_BER_LH	7	R/O	Latched high BER	0x0
PCS_BLK_LOCK_LL	6	R/O	Latched low block lock	0x0
BER_CNT	5:0	R/O	BER counter (Clear on read counter)	0x00

1.5.1.4 pcs_regs:pcs_regs:OAM_TX

Parent: pcs_regs:pcs_regs

Instances: 1

Table 47. Fields in OAM_TX

Field Name	Bit	Access	Description	Default
OAM_MSG_VLD	15	R/W	1000BASE-T1 OAM message valid. Hardware clears this register when the message is loaded into the OAM State machine.	0x0
OAM_TOG_VAL	14	R/O	Toggle value	0x0
OAM_MSG_RCVD	13	R/O	1000BASE-T1 OAM message received.Latch high .This bit clears on Read.	0x0
OAM_RX_TOG_VAL	12	R/O	Received message Toggle value	0x0
OAM_MSG_NUM	11:8	R/W	User-defined message number to send	0x0
RESERVED_24	7:4	R/O	Always 0	0x0
OAM_PING_RX	3	R/O	OAM Ping received	0x0
OAM_PING_TX	2	R/W	OAM Ping transmit	0x0
OAM_LOC_SNR	1:0	R/O	Local SNR	0x0

1.5.1.5 pcs_regs:pcs_regs:OAM_MSG_0_1

Parent: pcs_regs:pcs_regs

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Instances: 1

Table 48. Fields in OAM_MSG_0_1

Field Name	Bit	Access	Description	Default
OAM_MSG_1	15:8	R/W	1000BASE-T1 OAM message 1	0x00
OAM_MSG_0	7:0	R/W	1000BASE-T1 OAM message 0	0x00

1.5.1.6 pcs_regs:pcs_regs:OAM_MSG_2_3

Parent: pcs_regs:pcs_regs

Instances: 1

Table 49. Fields in OAM_MSG_2_3

Field Name	Bit	Access	Description	Default
OAM_MSG_3	15:8	R/W	1000BASE-T1 OAM message 3	0x00
OAM_MSG_2	7:0	R/W	1000BASE-T1 OAM message 2	0x00

1.5.1.7 pcs_regs:pcs_regs:OAM_MSG_4_5

Parent: pcs_regs:pcs_regs

Instances: 1

Table 50. Fields in OAM_MSG_4_5

Field Name	Bit	Access	Description	Default
OAM_MSG_5	15:8	R/W	1000BASE-T1 OAM message 5	0x00
OAM_MSG_4	7:0	R/W	1000BASE-T1 OAM message 4	0x00

1.5.1.8 pcs_regs:pcs_regs:OAM_MSG_6_7

Parent: pcs_regs:pcs_regs

Instances: 1

Table 51. Fields in OAM_MSG_6_7

Field Name	Bit	Access	Description	Default
OAM_MSG_7	15:8	R/W	1000BASE-T1 OAM message 7	0x00
OAM_MSG_6	7:0	R/W	1000BASE-T1 OAM message 6	0x00

1.5.1.9 pcs_regs:pcs_regs:OAM_RX

Parent: pcs_regs:pcs_regs

Instances: 1

Table 52. Fields in OAM_RX

Field Name	Bit	Access	Description	Default
OAM_LP_MSG_VLD	15	R/O	1000BASE-T1 OAM LP message valid. Reading this bit clears this field.	0x0
OAM_LP_TOG_VAL	14	R/O	LP Toggle value	0x0
RESERVED1_38	13:12	R/O	Always 0	0x0

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Field Name	Bit	Access	Description	Default
OAM_LP_MSG_NUM	11:8	R/O	Message number from link partner	0x0
RESERVED2_40	7:2	R/O	Always 0	0x00
OAM_LP_SNR	1:0	R/O	Link Partner SNR	0x0

1.5.1.10 pcs_regs:pcs_regs:OAM_LP_MSG_0_1

Parent: pcs_regs:pcs_regs

Instances: 1

Table 53. Fields in OAM_LP_MSG_0_1

Field Name	Bit	Access	Description	Default
OAM_LP_MSG_1	15:8	R/O	1000BASE-T1 OAM LP message 1	0x00
OAM_LP_MSG_0	7:0	R/O	1000BASE-T1 OAM LP message 0	0x00

1.5.1.11 pcs_regs:pcs_regs:OAM_LP_MSG_2_3

Parent: pcs_regs:pcs_regs

Instances: 1

Table 54. Fields in OAM_LP_MSG_2_3

Field Name	Bit	Access	Description	Default
OAM_LP_MSG_3	15:8	R/O	1000BASE-T1 OAM LP message 3	0x00
OAM_LP_MSG_2	7:0	R/O	1000BASE-T1 OAM LP message 2	0x00

1.5.1.12 pcs_regs:pcs_regs:OAM_LP_MSG_4_5

Parent: pcs_regs:pcs_regs

Instances: 1

Table 55. Fields in OAM_LP_MSG_4_5

Field Name	Bit	Access	Description	Default
OAM_LP_MSG_5	15:8	R/O	1000BASE-T1 OAM LP message 5	0x00
OAM_LP_MSG_4	7:0	R/O	1000BASE-T1 OAM LP message 4	0x00

1.5.1.13 pcs_regs:pcs_regs:OAM_LP_MSG_6_7

Parent: pcs_regs:pcs_regs

Instances: 1

Table 56. Fields in OAM_LP_MSG_6_7

Field Name	Bit	Access	Description	Default
OAM_LP_MSG_7	15:8	R/O	1000BASE-T1 OAM LP message 7	0x00
OAM_LP_MSG_6	7:0	R/O	1000BASE-T1 OAM LP message 6	0x00

1.5.1.14 pcs_regs:pcs_regs:DATA_MODE_SCRAM_SEED

Parent: pcs_regs:pcs_regs

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Table 57. Fields in DATA_MODE_SCRAM_SEED

Field Name	Bit	Access	Description	Default
USE_RAND_DP_SCR_SEED	15	R/W	1 - Use an internally generated random seed for Datapath scrambler (default). 0 - Use the value configured in the register dm_scram_seed as seed for datapath scrambler.	0x1
DM_SCRAM_SEED	14:0	R/W	Data Mode Scrambler Seed. The default value would be random if use_rand_dp_scr_seed is set.	0x7EEF

1.5.1.15 pcs_regs:pcs_regs:TR_SCRAM_SEED_0

Parent: pcs_regs:pcs_regs

Instances: 1

Table 58. Fields in TR_SCRAM_SEED_0

Field Name	Bit	Access	Description	Default
TR_SCR_SEED_15_0	15:0	R/W	Training Mode Scrambler Seed bits 15:0	0xBEEF

1.5.1.16 pcs_regs:pcs_regs:TR_SCRAM_SEED_1

Parent: pcs_regs:pcs_regs

Instances: 1

Table 59. Fields in TR_SCRAM_SEED_1

Field Name	Bit	Access	Description	Default
TR_SCR_SEED_31_16	15:0	R/W	Training Mode Scrambler Seed bits 31:16	0xDEAD

1.5.1.17 pcs_regs:pcs_regs:SCR_CONFIG

Parent: pcs_regs:pcs_regs

Instances: 1

Table 60. Fields in SCR_CONFIG

Field Name	Bit	Access	Description	Default
PCS_REG_RST	15	R/W	This bit resets the register contents in 1000T1-PCS block. 1 - Reset the registers - Self-clears after the operation is complete.	0x0
RESERVED	14	R/W	Must be set to its default.	0x1
RESERVED	13	R/W	Must be set to its default.	0x1
RESERVED	8	R/W	Must be set to its default.	0x1
RESERVED	7:4	R/W	Must be set to its default.	0x5
RESERVED	3:0	R/W	Must be set to its default.	0x7

1.5.1.18 pcs_regs:pcs_regs:RSFEC_ERR0_CNT

Parent: pcs_regs:pcs_regs

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Table 61. Fields in RSFEC_ERR0_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR0_FRM_CNT	15:0	R/O	RS FEC frames with 0 symbol error (clear on read counter/Saturate based on fec_err_cntr_saturate field)	0x0000

1.5.1.19 pcs_regs:pcs_regs:RSFEC_ERR1_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 62. Fields in RSFEC_ERR1_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR1_FRM_CNT	15:0	R/O	RS FEC frames with 1 symbol error (clear on read counter/Saturate based on fec_err_cntr_saturate field)	0x0000

1.5.1.20 pcs_regs:pcs_regs:RSFEC_ERR2_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 63. Fields in RSFEC_ERR2_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR2_FRM_CNT	15:0	R/O	RS FEC frames with 2 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.21 pcs_regs:pcs_regs:RSFEC_ERR3_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 64. Fields in RSFEC_ERR3_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR3_FRM_CNT	15:0	R/O	RS FEC frames with 3 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.22 pcs_regs:pcs_regs:RSFEC_ERR4_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 65. Fields in RSFEC_ERR4_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR4_FRM_CNT	15:0	R/O	RS FEC frames with 4 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

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1.5.1.23 pcs_regs:pcs_regs:RSFEC_ERR5_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 66. Fields in RSFEC_ERR5_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR5_FRM_CNT	15:0	R/O	RS FEC frames with 5 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.24 pcs_regs:pcs_regs:RSFEC_ERR6_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 67. Fields in RSFEC_ERR6_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR6_FRM_CNT	15:0	R/O	RS FEC frames with 6 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.25 pcs_regs:pcs_regs:RSFEC_ERR7_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 68. Fields in RSFEC_ERR7_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR7_FRM_CNT	15:0	R/O	RS FEC frames with 7 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.26 pcs_regs:pcs_regs:RSFEC_ERR8_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 69. Fields in RSFEC_ERR8_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR8_FRM_CNT	15:0	R/O	RS FEC frames with 8 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.27 pcs_regs:pcs_regs:RSFEC_ERR9_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

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Table 70. Fields in RSFEC_ERR9_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR9_FRM_CNT	15:0	R/O	RS FEC frames with 9 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.28 pcs_regs:pcs_regs:RSFEC_ERR10_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 71. Fields in RSFEC_ERR10_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR10_FRM_CNT	15:0	R/O	RS FEC frames with 10 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.29 pcs_regs:pcs_regs:RSFEC_ERR11_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 72. Fields in RSFEC_ERR11_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR11_FRM_CNT	15:0	R/O	RS FEC frames with 11 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.30 pcs_regs:pcs_regs:RSFEC_ERR12_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 73. Fields in RSFEC_ERR12_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR12_FRM_CNT	15:0	R/O	RS FEC frames with 12 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.31 pcs_regs:pcs_regs:RSFEC_ERR13_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 74. Fields in RSFEC_ERR13_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR13_FRM_CNT	15:0	R/O	RS FEC frames with 13 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

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1.5.1.32 pcs_regs:pcs_regs:RSFEC_ERR14_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 75. Fields in RSFEC_ERR14_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR14_FRM_CNT	15:0	R/O	RS FEC frames with 14 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.33 pcs_regs:pcs_regs:RSFEC_ERR15_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 76. Fields in RSFEC_ERR15_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR15_FRM_CNT	15:0	R/O	RS FEC frames with 15 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.34 pcs_regs:pcs_regs:RSFEC_ERR16_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 77. Fields in RSFEC_ERR16_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR16_FRM_CNT	15:0	R/O	RS FEC frames with 16 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.35 pcs_regs:pcs_regs:RSFEC_ERR17_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 78. Fields in RSFEC_ERR17_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR17_FRM_CNT	15:0	R/O	RS FEC frames with 17 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.36 pcs_regs:pcs_regs:RSFEC_ERR18_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

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Table 79. Fields in RSFEC_ERR18_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR18_FRM_CNT	15:0	R/O	RS FEC frames with 18 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.37 pcs_regs:pcs_regs:RSFEC_ERR19_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 80. Fields in RSFEC_ERR19_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR19_FRM_CNT	15:0	R/O	RS FEC frames with 19 symbol errors(clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.38 pcs_regs:pcs_regs:RSFEC_ERR20_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 81. Fields in RSFEC_ERR20_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR20_FRM_CNT	15:0	R/O	RS FEC frames with 20 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.39 pcs_regs:pcs_regs:RSFEC_ERR21_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 82. Fields in RSFEC_ERR21_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR21_FRM_CNT	15:0	R/O	RS FEC frames with 21 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.40 pcs_regs:pcs_regs:RSFEC_ERR22_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 83. Fields in RSFEC_ERR22_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR22_FRM_CNT	15:0	R/O	RS FEC frames with 22 symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

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1.5.1.41 pcs_regs:pcs_regs:RSFEC_ERR23_CNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 84. Fields in RSFEC_ERR23_CNT

Field Name	Bit	Access	Description	Default
RSFEC_ERR23_FRM_CNT	15:0	R/O	RS FEC frames with 23 or more symbol errors (clear on read counter/Saturate based on fec_err_cntr_saturate field).	0x0000

1.5.1.42 pcs_regs:pcs_regs:DEBUG_BUS_CONFIG

Parent: pcs_regs:pcs_regs

Instances: 1

Table 85. Fields in DEBUG_BUS_CONFIG

Field Name	Bit	Access	Description	Default
RESERVED_140	15:4	R/O	Reserved	0x000

1.5.1.43 pcs_regs:pcs_regs:OAM_SNR_CONFIG1

Parent: pcs_regs:pcs_regs

Instances: 1

Table 86. Fields in OAM_SNR_CONFIG1

Field Name	Bit	Access	Description	Default
RESERVED1_142	15:10	R/O	Reserved	0x00
SNR_RFER_THRESH2	9:5	R/W	RFER Threshold2 for SNR Calculation from DSP logic	0x0A
SNR_RFER_THRESH1	4:0	R/W	RFER Threshold1 for SNR Calculation from DSP logic	0x06

1.5.1.44 pcs_regs:pcs_regs:OAM_SNR_CONFIG2

Parent: pcs_regs:pcs_regs

Instances: 1

Table 87. Fields in OAM_SNR_CONFIG2

Field Name	Bit	Access	Description	Default
RESERVED1_145	15:6	R/O	Reserved	0x000
SNR_GOOD_THRESH	5:3	R/W	SQI Threshold for SNR Calculation from DSP logic	0x5
SNR_MARG_THRESH	2:0	R/W	SQI Threshold for SNR Calculation from DSP logic	0x2

1.5.1.45 pcs_regs:pcs_regs:FEC_CONFIG

Parent: pcs_regs:pcs_regs

Instances: 1

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Table 88. Fields in FEC_CONFIG

Field Name	Bit	Access	Description	Default
RESERVED1_148	15:2	R/O	Reserved	0x0000
RESERVED	1	R/W	Must be set to its default.	0x1
FEC_BYPASS_ENA	0	R/W	Enable RS FEC in bypass mode - With this decoder will stream the data to its output without performing parity checks.	0x0

1.5.1.46 pcs_regs:pcs_regs:RSFEC_ERR_CNT_FLB

Parent: pcs_regs:pcs_regs

Instances: 1

Table 89. Fields in RSFEC_ERR_CNT_FLB

Field Name	Bit	Access	Description	Default
RESERVED1_151	14:10	R/O	Reserved	0x00
FEC_CNTR_LAST_BLK	9:0	R/O	FEC counter as needed for Cable diagnostics. Reading this register will clear this counter. Writing to this register does not have any effect.	0x000

1.5.1.47 pcs_regs:pcs_regs:RX_DATA_SCRAM_SEED

Parent: pcs_regs:pcs_regs

Instances: 1

Table 90. Fields in RX_DATA_SCRAM_SEED

Field Name	Bit	Access	Description	Default
RESERVED_153	15	R/O	Reserved	0x0
RX_DATA_SCRAM_SEED	14:0	R/O	Received Data Scrambler Seed from the link-partner	0x0000

1.5.1.48 pcs_regs:pcs_regs:PCS_RX_DEC_CTRL

Parent: pcs_regs:pcs_regs

Instances: 1

Table 91. Fields in PCS_RX_DEC_CTRL

Field Name	Bit	Access	Description	Default
RESERVED_155	15:2	R/O	Reserved	0x0000
RESERVED	1	R/W	Must be set to its default.	0x1
RESERVED	0	R/W	Must be set to its default.	0x1

1.5.1.49 pcs_regs:pcs_regs:LFL_COMM_COUNT

Parent: pcs_regs:pcs_regs

Instances: 1

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Table 92. Fields in LFL_COMM_COUNT

Field Name	Bit	Access	Description	Default
LFL_COMM_LINK_LOSS_CNT	15:10	R/O	Number of link-losses that occurred since last power cycle (0 to 63). This counter caps at 63 and does not rollover until a reset event. This register uses the link_ready field of DSP Link-Monitor as a source for the counter.	0x00
LFL_COMM_LINK_FAIL_CNT	9:0	R/O	Number of Link Failures causing NOT a link loss since last power cycle (0 to 1023).	0x000

1.5.1.50 pcs_regs:pcs_regs:LFL_COUNT

Parent: pcs_regs:pcs_regs

Instances: 1

Table 93. Fields in LFL_COUNT

Field Name	Bit	Access	Description	Default
LFL_LINK_LOSS_CNT	15:10	R/O	Number of link-losses that occurred since last power cycle (0 to 63). This counter caps at 63 and does not rollover until a reset event. This register uses the link_status as the source for the counter.	0x00
LFL_LINK_FAIL_CNT	9:0	R/O	Number of Link Failures causing NOT a link loss since last power cycle (0 to 1023).	0x000

1.5.1.51 pcs_regs:pcs_regs:PCS_INTR_1

Parent: pcs_regs:pcs_regs

Instances: 1

Table 94. Fields in PCS_INTR_1

Field Name	Bit	Access	Description	Default
PCS_INTR_MASK_1	15:8	R/W	PCS Interrupt Mask register 1 NOTE: The meaning of each interrupt mask bit is: 1: enable the interrupt 0: disable the interrupt	0x00
PCS_INTR_STAT_1	7:0	Sticky	PCS Interrupt Status register 1. This is a Write-1 to clear register. Bit-7: locally maintained PFC count mismatch; bit-6 - invalid infofield preamble received; bit-5-invalid info-field crc error rcvd; bit-4-remote receiver status went high; bit-3-local receiver status went high; bit-2-received uncorrectable frame; bit-1-high_rfer found; bit-0 - link is up.	0x00

1.5.1.52 pcs_regs:pcs_regs:PCS_INTR_2

Parent: pcs_regs:pcs_regs

Instances: 1

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Table 95. Fields in PCS_INTR_2

Field Name	Bit	Access	Description	Default
PCS_INTR_MASK_2	15:8	R/W	PCS Interrupt Mask register 2 NOTE: The meaning of each interrupt mask bit is: 1: enable the interrupt 0: disable the interrupt	0x00
PCS_INTR_STAT_2	7:0	Sticky	PCS Interrupt Status register 2. This is a Write-1 to clear register. Bit-7: RX OAM LP message valid; bit-6: rx oam LPS received; bit-5: RX OAM WUR received; bit-4: RX OAM frame CRC not OK received; bit-3: Remote PHY ready went high; bit-2: rx block lock went high; bit-1: rx training descrambler is locked; bit-0: restarting the rx training and datapath.	0x00

1.5.1.53 pcs_regs:pcs_regs:PCS_INTR_3

Parent: pcs_regs:pcs_regs

Instances: 1

Table 96. Fields in PCS_INTR_3

Field Name	Bit	Access	Description	Default
PCS_INTR_MASK_3	15:8	R/W	PCS Interrupt Mask register 3 NOTE: The meaning of each interrupt mask bit is: 1: enable the interrupt 0: disable the interrupt	0x00
PCS_INTR_STAT_3	7:0	Sticky	PCS Interrupt Status register 3. This is a Write-1 to clear register. Bit-0 : RX OAM ping received.	0x00

1.5.1.54 pcs_regs:pcs_regs:PCS_INTR_4

Parent: pcs_regs:pcs_regs

Instances: 1

Table 97. Fields in PCS_INTR_4

Field Name	Bit	Access	Description	Default
PCS_INTR_MASK_4	15:8	R/W	PCS Interrupt Mask register 4 NOTE: The meaning of each interrupt mask bit is: 1: enable the interrupt 0: disable the interrupt	0x00
PCS_INTR_STAT_4	7:0	Sticky	PCS Interrupt Status register 4. This is a Write-1 to clear register. Bit-2: link went down; Bit-1: block lock went down; Bit-0 : Remote PHY ready went low.	0x00

1.5.1.55 pcs_regs:pcs_regs:OAM_PCS_DBG_STS

Parent: pcs_regs:pcs_regs

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Instances: 1

Table 98. Fields in OAM_PCS_DBG_STS

Field Name	Bit	Access	Description	Default
RESERVED_170	15:9	R/O	Reserved	0x00
OAM_RX_CRC_NOT_OK	8	R/O	CRC Not-OK for the received OAM frame. Latch-High and Clear on Read.	0x0
WUR_RECV	7	R/O	WUR received in the OAM message. Latch-High and Clear on Read	0x0
LPS_RECV	6	R/O	LPS received in the OAM message. Latch-High and Clear on Read	0x0
OAM_RX_CUR_STATE	5:3	R/O	OAM RX SM Current state	0x0
OAM_TX_CUR_STATE	2:0	R/O	OAM TX SM Current state	0x0

1.5.1.56 pcs_regs:pcs_regs:LQ_COM_TC12

Parent: pcs_regs:pcs_regs

Instances: 1

Table 99. Fields in LQ_COM_TC12

Field Name	Bit	Access	Description	Default
RESERVED_176	15:1	R/O	Reserved	0x0000
LQ_COM_READY	0	R/O	information on communication ready.	0x0

1.6 pcs_100_regs

Table 100. Register Groups in MMD pcs_100_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
pcs_100_regs	0x00000000	1		Page 30

1.6.1 pcs_100_regs:pcs_100_regs

Parent: pcs_100_regs

Instances: 1

Table 101. Registers in pcs_100_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
DESCM_CNTRL0	0x00008200	1		Page 32
DESCM_CNTRL1	0x00008201	1		Page 34
EQ_TIMER_CNTRL	0x00008202	1		Page 34
PCS_RX_CNTRL	0x00008203	1		Page 35

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
IDLE_ERR_TIMER_WIN	0x00008204	1		Page 35
PHY_MIN_TIMER_CNTRL	0x00008205	1		Page 35
PHY_MAX_TIMER_CNTRL0	0x00008206	1		Page 36
PHY_MAX_TIMER_CNTRL1	0x00008207	1		Page 36
PCS_CNTRL3	0x00008208	1		Page 36
LINK_CNTRL	0x00008209	1		Page 36
PCS_TX_CNTRL	0x0000820A	1		Page 36
ED_HYST_WIN	0x0000820B	1		Page 37
ED_HYST_CNT	0x0000820C	1		Page 37
MII_FC_ERR_CNT_STS	0x0000820D	1		Page 37
PCS_RX_ERR_CNT_STS	0x0000820E	1		Page 37
PCS_TX_SSD_ESD_CNTRL0	0x0000820F	1		Page 38
PCS_TX_SSD_ESD_CNTRL1	0x00008210	1		Page 38
PCS_RX_SSD_ESD_CNTRL0	0x00008211	1		Page 38
PCS_RX_SSD_ESD_CNTRL1	0x00008212	1		Page 39
IDLE_ERR_CNT_THRESH	0x00008213	1		Page 39
HW_INIT_WAIT_TIME	0x00008214	1		Page 39
CHK_IDLE_TIMER_WIN	0x00008215	1		Page 40
DEBUG_BUS_STS	0x00008216	1		Page 40
ZERO_SYMB_TIMER_WIN	0x00008217	1		Page 40
PCS_DEBUG_CNTRL	0x00008218	1		Page 41
LINKUP_TIME_STS	0x00008219	1		Page 41
LINKUP_TIMEOUT_TIMER	0x0000821A	1		Page 42
SLEEP_WAKE_CNTRL	0x0000821B	1		Page 42
AVOID_SLEEP_TIMER	0x0000821C	1		Page 42
GOTO_SLEEP_TIMER	0x0000821D	1		Page 43
SEND_IDLE_TIMER	0x0000821E	1		Page 43
LINKUP_FAIL_COUNT	0x0000821F	1		Page 43

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
SLEEP_WAKE_DET	0x00008220	1		Page 43
SLEEP_WAKE_SEND	0x00008221	1		Page 44
ED_HYST_MAST_MODE_WIN	0x00008222	1		Page 44
CHK_REM_SENDN_WIN	0x00008223	1		Page 44
SLAVE_TO_SS_TIME_ON_MAST_RST	0x00008224	1		Page 44
SLAVE_TO_SS_TIME_ON_TRNG_RESTART	0x00008225	1		Page 45
ED_DET_STABILITY_WIN	0x00008226	1		Page 45
SLAVE_HD_DESCR_LOCK_TIME	0x00008227	1		Page 45
PCS_CNTRL_STS	0x00008228	1		Page 46
DSP_EQ_LOCK_ASSERT_CNT	0x00008229	1		Page 46
DESCR_LOCK_ASSERT_CNT	0x0000822A	1		Page 46
LOC_RCVR_STS_ASSERT_CNT	0x0000822B	1		Page 47
REM_RCVR_STS_ASSERT_CNT	0x0000822C	1		Page 47
PCS_FSM_STS0	0x0000822D	1		Page 47
PCS_FSM_STS1	0x0000822E	1		Page 47
PCS_FSM_STS2	0x0000822F	1		Page 48
MAX_WAIT_TIMER_EVNT_CNT	0x00008230	1		Page 48
TXPD_CMN_MODE_SETTLE_TIMER	0x00008231	1		Page 48
POLFLIP_FIX_TIMER	0x00008232	1		Page 49
DESCR_LOCK_WIN_EXTEND	0x00008233	1		Page 49
SLAVE_TRNG_IDLE_ERR_CHECK	0x00008234	1		Page 49
SLAVE_TRNG_IDLE_ERR_CHECK_WIN	0x00008235	1		Page 50
SLAVE_TRNG_IDLE_ERR_CNT_THRESH	0x00008236	1		Page 50
WUP_LEN	0x00008240	1		Page 50
SLEEP_ABRT	0x00008241	1		Page 50

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1.6.1.1 pcs_100_regs:pcs_100_regs:DESCM_CNTRL0

Parent: pcs_100_regs:pcs_100_regs

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Instances: 1

Table 102. Fields in DESCM_CNTRL0

Field Name	Bit	Access	Description	Default
SEND_IDLE_LOC_RCV_STS_CNTRL	15	R/W	0 = No effect 1 = Debug purpose When PHY_CNTRL is in SEND_IDLE state, by enabling this bit it makes transition to SEND_Z state only when dsp_eq_sts = 0 by allowing the PCSTX to send the IDLE codes with loc_rcv_sts = 0 when descrambler lock = 0.So allowing descrambler to lock within SEND_IDLE state and allow the PHY_CNTRL to move to SEND_IDLE_OR_DATA state	0x0
DESCR_LOCK_ON_IDLE_ERR_THRES_DET	14	R/W	0 = No effect (Debug only) 1 = when rcv_fsm_state is in BAD_SSD state or IDLE state It has detected programmed number of SEND_N-IDLE errors within the programmable window and asserted the idle_thresh_det. The descrambler LFSR loads fresh and check for consecutive received symbol match to qualify the descr_lock status. If it is not matched deasserts the descr_lock and repeat the steps of descrambler lock (load_lfsr followed by check the number of consecutive decoded ternary symbols match within the programmable window)	0x1
DESCR_RELOCK_ON_RCV_JAB_STATE	13	R/W	0 = no effect (Debug only) 1 = load the LFSR fresh and check for consecutive receive symbol match to qualify the descry_lock status. If it is not matched deasserts the descry_lock and repeat the steps of descrambler lock (load_lfsr followed by check the number of consecutive decoded ternary symbols match within the programmable window)	0x1
DESCR_RELOCK_ON_EQ_UNLOCK	12	R/W	0 = no effect (Debug only) 1 = Relock descrambler on DSP eqauliser unlock	0x1
DESCR_LOCK_SYMB_CNT	11:6	R/W	This programmable symbol count defines the number of consecutive decoded symbols A & B are matched within programmed descr_lock_chk_win pcsclock cycles to assert the descr_lock indication. (Ref for symbol expected against LFSR Sdn[2:0] is Table 36 of BRSPEC v3.2) If it is not matched within this window loop starts again with the option swap_symbol or polarity_flip, or delayed_symbols w.r.t 66.66mhz (To get the symbol boundary aligned w.r.t 33.33Mhz clock edges)	0x3F
DESCR_LOCK_CHK_WIN	5:0	R/W	Value programmed are in pcs_clk cycles Once receive LFSR loaded with decoded ternary symbols of configured number of clocks from feed_dscr_lfsr_0_win, this configured window is started to check the validity of received	0x3F

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Field Name	Bit	Access	Description	Default
			ternary symbols A & B against Sdn[2:0] derived from descrambler LFSR.	

1.6.1.2 pcs_100_regs:pcs_100_regs:DESCM_CNTRL1

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 103. Fields in DESCM_CNTRL1

Field Name	Bit	Access	Description	Default
RESERVED_0X201_2_6	15:11	R/O	Reserved	0x00
DESCR_PIPE_DELAY	10:8	R/W	Pipeline delay for the locked descrambler LFSR for the decoded ternary symbols descrambling. (In hardware actual dealy of locked lfsr bits used for descrambling ternary symbols is descry_pipe_delay + 2)	0x3
RESERVED_0X201_1_8	7:6	R/O	Reserved	0x0
FEED_DESCR_LFSR_0_WIN	5:0	R/W	Value programmed are in pcs_clk cycles Programmable register window to load the LFSR with expected decoded ternary symbol-A. (Refer Table 3-6 of BR-SPEC v3.2 for expected SDN[0]). Programmed to default value of 40 pcs clock cycles window. In this time the decoded ternary symbols are loaded in to LFSR to lock with remote LFSR.	0x28

1.6.1.3 pcs_100_regs:pcs_100_regs:EQ_TIMER_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 104. Fields in EQ_TIMER_CNTRL

Field Name	Bit	Access	Description	Default
EQ_STATUS_TIMER_CONTROL	15:0	R/W	Value programmed in microseconds This is the window for stabilization time for equalizer status. In steps of 1 s time. Used when PHY configured as SLAVE When PHY configured as SLAVE, the DSP equalizer and descrambler locked with received symbols from remote PHY. Once the descrambler is locked, the local PHY starts with training. In this time DSP equalizer unlocks and locks it again with fresh equalizer filter coefficients. During this programmable time the loc_rcvr_sts is not removed even though the equalizer is unlocked. Once this programmable window timer expires, loc_rcvr_sts is de-asserted if DSP-equalizer is not achieved lock.	0x07FF

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1.6.1.4 pcs_100_regs:pcs_100_regs:PCS_RX_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 105. Fields in PCS_RX_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X203_1_11	15:12	R/O	Reserved	0x0
REM_RCVR_STS_IDLES_CNT	11:6	R/W	This programmable IDLE count defines the number of consecutive decoded symbols A & B are matched within programmed rem_rcvr_sts_chk_win pcs-clock cycles against {~Sdn[2], Sdn[1:0]} derived from descrambler LFSR to assert the rem_rcvr_sts indication. (Ref for symbol expected against LFSR Sdn[2:0] is Table 3-6 of BR-SPEC v3.2)	0x0F
REM_RCVR_STS_CHK_WIN	5:0	R/W	Value programmed is in pcs_clk cycles.Once the loc_rcv_sts is asserted and the receive FSM is in IDLE state, this configured window is started to check the validity of received ternary symbols A & B against {~Sdn[2], Sdn[1:0]} derived from descrambler LFSR	0x17

1.6.1.5 pcs_100_regs:pcs_100_regs:IDLE_ERR_TIMER_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 106. Fields in IDLE_ERR_TIMER_WIN

Field Name	Bit	Access	Description	Default
IDLE_ERR_TIMER_EN	15	R/W	0 = Idle_err_timer_win is not enabled 1 = Idle_err_timer_win is enabled	0x1
IDLE_ERR_TIMER_WIN	14:0	R/W	Value programmed is in microseconds. Programmable timer value to detect SEND_IDLE-errors when recive FSM state stays in BAD_SSD or IDLE state. When this timer expires it checks for the programmable threshold count to enable the idle_err_det_sts. Note: If en_phycntrl_minwt_test = 1 hardware itself override this register value to 1 and when software reads it always gets value	0x0001

1.6.1.6 pcs_100_regs:pcs_100_regs:PHY_MIN_TIMER_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 107. Fields in PHY_MIN_TIMER_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X205_1_16	15:6	R/O	Reserved	0x000
PHY_CTRL_MIN_WAIT_TIMER	5:0	R/W	Value programmed is in counts of 25 MHz clock period	0x2D

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1.6.1.7 pcs_100_regs:pcs_100_regs:PHY_MAX_TIMER_CNTRL0

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 108. Fields in PHY_MAX_TIMER_CNTRL0

Field Name	Bit	Access	Description	Default
RESERVED_0X206_1_18	15:11	R/O	Reserved	0x00
PHY_CNTRL_MAX_WAIT_TIMER_MST	10:0	R/W	Value programmed is in counts of 1 ms	0x0C8

1.6.1.8 pcs_100_regs:pcs_100_regs:PHY_MAX_TIMER_CNTRL1

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 109. Fields in PHY_MAX_TIMER_CNTRL1

Field Name	Bit	Access	Description	Default
RESERVED_0X207_1_20	15:11	R/O	Reserved	0x00
PHY_CNTRL_MAX_WAIT_TIMER_SLV	10:0	R/W	Value programmed is in counts of 1 ms	0x0C8

1.6.1.9 pcs_100_regs:pcs_100_regs:PCS_CNTRL3

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 110. Fields in PCS_CNTRL3

Field Name	Bit	Access	Description	Default
RESERVED_0X208_1_22	15:11	R/O	Reserved	0x00
RCVR_JAB_MAX_TIMER	10:0	R/W	Receiver jabber max timer. Runs at 1 s event	0x438

1.6.1.10 pcs_100_regs:pcs_100_regs:LINK_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 111. Fields in LINK_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X209_1_24	15:6	R/O	Reserved	0x000
LINK_MONITOR_TIMER	5:0	R/W	Value programmed is in counts of 25 MHz clock.	0x2E

1.6.1.11 pcs_100_regs:pcs_100_regs:PCS_TX_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

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Table 112. Fields in PCS_TX_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X20A_1_26	15:6	R/O	Reserved	0x000
TST_MODE1_SYMB_CNT	5:0	R/W	Defines the number of consecutive symbols to be sent in TST_MODE1	0x28

1.6.1.12 pcs_100_regs:pcs_100_regs:ED_HYST_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 113. Fields in ED_HYST_WIN

Field Name	Bit	Access	Description	Default
ED_HYST_WIN_CNT	15:0	R/W	Window to be programmed to detect the number consecutive high pulses Note: When en_energy_det_master_mode = 1 and while changing this window make sure that ratio of ed_hyst_win_cnt and ed_hyst_mast_mode_win_cnt is 1:1.5.	0x03FF

1.6.1.13 pcs_100_regs:pcs_100_regs:ED_HYST_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 114. Fields in ED_HYST_CNT

Field Name	Bit	Access	Description	Default
ED_HYST_LOW_CNT	15:8	R/W	Low hysteresis count to detect consecutive low pulses	0xFF
ED_HYST_HIGH_CNT	7:0	R/W	High hysteresis count to detect high edge pulses within the programmable window ed_hyst_win_cnt	0x40

1.6.1.14 pcs_100_regs:pcs_100_regs:MII_FC_ERR_CNT_STS

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 115. Fields in MII_FC_ERR_CNT_STS

Field Name	Bit	Access	Description	Default
MII_FC_ERR_CNT	15:0	R/O	False carrier error count when pcs rx FSM transition to BAD_SSD state. Clear on read.	0x0000

1.6.1.15 pcs_100_regs:pcs_100_regs:PCS_RX_ERR_CNT_STS

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

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Table 116. Fields in PCS_RX_ERR_CNT_STS

Field Name	Bit	Access	Description	Default
PCS_RX_ERR_CNT	15:0	R/O	Number of rx_er assertion count when PCS-RX-FSM transition to BAD SSD,LINK FAILED, RX ERROR, BAD END, BAD ESD2 states. Clear on read.	0x0000

1.6.1.16 pcs_100_regs:pcs_100_regs:PCS_TX_SSD_ESD_CNTRL0

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 117. Fields in PCS_TX_SSD_ESD_CNTRL0

Field Name	Bit	Access	Description	Default
PCS_TX_SSD_ESD_CNTRL_EN	15	R/W	0 = SSD, ESD & ERR_ESD values are taken as per the spec for PSC_TX_FSM 1 = SSD, ESD & ERR_ESD values are controlled from register address 0x0F & 0x10 for PCS_TX_FSM	0x0
RESERVED_0X20F_1_34	14:12	R/O	Reserved	0x0
TX_SSD3	11:8	R/W	SSD3 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
TX_SSD2	7:4	R/W	SSD2 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
TX_SSD1	3:0	R/W	SSD1 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0

1.6.1.17 pcs_100_regs:pcs_100_regs:PCS_TX_SSD_ESD_CNTRL1

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 118. Fields in PCS_TX_SSD_ESD_CNTRL1

Field Name	Bit	Access	Description	Default
TX_ERR_ESD3	15:12	R/W	ERR_ESD3 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0xF
TX_ESD3	11:8	R/W	ESD3 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x5
TX_ESD2	7:4	R/W	ESD2 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
TX_ESD1	3:0	R/W	ESD1 value for TX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0

1.6.1.18 pcs_100_regs:pcs_100_regs:PCS_RX_SSD_ESD_CNTRL0

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

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Table 119. Fields in PCS_RX_SSD_ESD_CNTRL0

Field Name	Bit	Access	Description	Default
PCS_RX_SSD_ESD_CNTRL_EN	15	R/W	0 = SSD, ESD & ERR_ESD values are taken as per the spec for PSC_RX_FSM 1 = SSD, ESD & ERR_ESD values are controlled from register address 0x11 & 0x12 for PCS_RX_FSM	0x0
RESERVED_0X211_1_43	14:12	R/O	Reserved	0x0
RX_SSD3	11:8	R/W	SSD3 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
RX_SSD2	7:4	R/W	SSD2 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
RX_SSD1	3:0	R/W	SSD1 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0

1.6.1.19 pcs_100_regs:pcs_100_regs:PCS_RX_SSD_ESD_CNTRL1

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 120. Fields in PCS_RX_SSD_ESD_CNTRL1

Field Name	Bit	Access	Description	Default
RX_ERR_ESD3	15:12	R/W	ERR_ESD3 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0xF
RX_ESD3	11:8	R/W	ESD3 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x5
RX_ESD2	7:4	R/W	ESD2 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0
RX_ESD1	3:0	R/W	ESD1 value for RX_FSM (valid only when pcs_tx_ssd_esd_cntrl_en = 1)	0x0

1.6.1.20 pcs_100_regs:pcs_100_regs:IDLE_ERR_CNT_THRESH

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 121. Fields in IDLE_ERR_CNT_THRESH

Field Name	Bit	Access	Description	Default
IDLE_ERR_CNT_THRESH	15:0	R/W	Valid only when idle_err_timer_en is enabled This calculated SEND_N-IDLE error count is checked against this programmable err_cnt_threshold within the programmable Idle_err_timer_win to assert the idle_err_cnt_sts det. Note: If en_phycntrl_minwt_test = 1 hardware itself override this register value to 8 and when software reads it always gets value 8	0x0008

1.6.1.21 pcs_100_regs:pcs_100_regs:HW_INIT_WAIT_TIME

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

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Table 122. Fields in HW_INIT_WAIT_TIME

Field Name	Bit	Access	Description	Default
RESERVED_0X214_1_52	15:6	R/O	Reserved	0x000
HW_INIT_WAIT_TIME	5:0	R/W	Hardware initialization wait time for the PLL lock (Stable clocks from PLL).	0x32

1.6.1.22 pcs_100_regs:pcs_100_regs:CHK_IDLE_TIMER_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 123. Fields in CHK_IDLE_TIMER_WIN

Field Name	Bit	Access	Description	Default
CHK_IDLE_TIMER_WIN_EN	15	R/W	1 = chk_idle_timer wndow enable 0 = chk_idle_timer window disable Note: When this control register is 0 it selects the hardware remote PHY_CNTRL- SEND_N detection logic status to check only the SEND_N-IFG idle codes (When local PHY is in SEND_I state: it checks three types of IDLE codes (IDLE codes with or without loc_rcv_sts set and IFG-IDLE codes). If none of the IDLE code matches it indicates as an error IDLE code. When local PHY is in SEND_N state: it checks 2-types of IDLE codes (IDLE code with loc_rcv_sts set and IFG-IDLE codes). If none of the IDLE code matches it indicates as an error IDLE code	0x0
CHK_IDLE_TIMER_WIN	14:0	R/W	Value programmed in microseconds: Depending on free running 1 s event, actual timer value may be between 7 s to 8 s. This timer is ran when PHY_CNTRL enters into the SEND_N state Within the window time: checked for IDLE codes with loc_rcvr_sts is set and SEND_N IFG Idle codes After this window time: checked only the SEND_N-IFG Idle codes	0x04B0

1.6.1.23 pcs_100_regs:pcs_100_regs:DEBUG_BUS_STS

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 124. Fields in DEBUG_BUS_STS

Field Name	Bit	Access	Description	Default
RESERVED_0X216_1_56	15:8	R/O	Reserved	0x00
DEBUG_BUS_STS	7:0	R/O	Debug bus output on this register on the basis of debug selection from register debug_bus_sel.	0x00

1.6.1.24 pcs_100_regs:pcs_100_regs:ZERO_SYMB_TIMER_WIN

Parent: pcs_100_regs:pcs_100_regs

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Table 125. Fields in ZERO_SYMB_TIMER_WIN

Field Name	Bit	Access	Description	Default
ZERO_SYMB_TIMER_EN	15	R/W	0 = zero_symb_timer_win is not enabled 1 = zero_symb_timer_win is enabled	0x1
ZERO_SYMB_TIMER_WIN	14:0	R/W	Value programmed is in microseconds: valid only zero_symb_timer_en = 1 Programmable timer value to restart the training sequence if PCS-RX receives consecutive zero ternary symbols from DSP up to the programmable timer (Remote-PHY in SEND_Z state)	0x0001

1.6.1.25 pcs_100_regs:pcs_100_regs:PCS_DEBUG_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 126. Fields in PCS_DEBUG_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X218_3_60	15:5	R/O	Reserved	0x000
DIS_PCNTRL_RST_ON_MWTIMR_EVNT	4	R/W	1=Disables PHY control reset on min_wait_timer event	0x0
EN_OVERRIDE_EQ_STATUS_TIMER_CONTROL	3	R/W	1=Enables the "eq_status_timer_control" default register value override to 0x0001 by hardware itself.	0x0
EN_PHYCNTRL_MINWT_TEST	2	R/W	0 = no effect 1 = enables the min_wait_timer related tests in PHY Control Test Suite	0x0
RESERVED_0X218_1_64	1	R/O	Reserved	0x0
EN_OVERRIDE_RESTART_DSP_EQ	0	R/W	1= Enables Override Restart for DSP Equalizer	0x0

1.6.1.26 pcs_100_regs:pcs_100_regs:LINKUP_TIME_STS

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 127. Fields in LINKUP_TIME_STS

Field Name	Bit	Access	Description	Default
LINK_UP_TIME	15:8	R/O	Link up timer status (msec). If link up time is more than 8'hFF msecs, this timer latches to max value of 8'hFF. This timer value is updated on each linkup process. The timer value captured from hardware initialization done to rem_rcvr_lock assertion in both master and slave mode.	0x00
LINK_STS_TIME	7:0	R/O	Link status assertion time (msec). If link up time is more than 8'hFF msecs, this timer latches to max value of 8'hFF. This timer value is updated on each linkup process. The timer value	0x00

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Field Name	Bit	Access	Description	Default
			captured from hardware initialization done to loc_rcvr_lock assertion in both master and slave mode.	

1.6.1.27 pcs_100_regs:pcs_100_regs:LINKUP_TIMEOUT_TIMER

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 128. Fields in LINKUP_TIMEOUT_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X21A_1_68	15:8	R/O	Reserved	0x00
LINK_UP_TIMEOUT_THRESH	7:0	R/W	Link up time out timer threshold (ms). 1. In master mode: After the timeout event in Master mode, check no signal using AFE energy detect and toggle the DSP start_equalizer signal. This feature in master mode is enabled with en_energy_det_master_mode = 1 The timeout counter should be counting after the HW_INIT/the energy detect has detected no signal presence (start_dsp_eqaualiser = 1).Timeout counter should stop counting upon achieving loc_rcvr_sts set. 2. In Slave mode: Remove the start_equaliser, check signal presence using AFE energy detect and enable the start equalizer. Timeout counter should start counting after HW_INIT done/the energy detect has detected signal presence ((start_dsp_eqaualiser = 1). Timeout counter should stop counting upon achieving loc_rcvr_sts set.	0xC8

1.6.1.28 pcs_100_regs:pcs_100_regs:SLEEP_WAKE_CNTRL

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 129. Fields in SLEEP_WAKE_CNTRL

Field Name	Bit	Access	Description	Default
RESERVED_0X21B_3_70	15:4	R/O	Reserved	0x000
SLV_SEND_IDLE_EN	3	R/W	1 = Slaving sending IDLE codes is enabled. 0 = Slaving sending IDLE codes is not enabled.	0x0
RESERVED_0X21B_2_72	2	R/O	Reserved	0x0
SLEEP_IND_EN	1	R/W	1 = Sleep indication signal output is enabled. 0 = Sleep indication signal output is not enabled	0x0
RESERVED_0X21B_1_74	0	R/O	Reserved	0x0

1.6.1.29 pcs_100_regs:pcs_100_regs:AVOID_SLEEP_TIMER

Parent: pcs_100_regs:pcs_100_regs

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Table 130. Fields in AVOID_SLEEP_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X21C_1_75	15:9	R/O	Reserved	0x00
AVOID_SLEEP_TIMER	8:0	R/W	Avoid Sleep Timer value in milliseconds.	0x064

1.6.1.30 pcs_100_regs:pcs_100_regs:GOTO_SLEEP_TIMER

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 131. Fields in GOTO_SLEEP_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X21D_1_77	15:12	R/O	Reserved	0x0
GOTO_SLEEP_TIMER	11:0	R/W	Go to sleep timer value in milliseconds.	0x5DC

1.6.1.31 pcs_100_regs:pcs_100_regs:SEND_IDLE_TIMER

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 132. Fields in SEND_IDLE_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X21E_1_79	15:11	R/O	Reserved	0x00
SEND_IDLE_TIMER	10:0	R/W	Send idle timer value in microseconds	0x3E8

1.6.1.32 pcs_100_regs:pcs_100_regs:LINKUP_FAIL_COUNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 133. Fields in LINKUP_FAIL_COUNT

Field Name	Bit	Access	Description	Default
RESERVED_0X21F_1_81	15:8	R/O	Reserved	0x00
LINKUP_FAIL_COUNT	7:0	R/O	The number of expiration of Avoid Sleep Timer. The counter stops at max value 255. (RC)	0x00

1.6.1.33 pcs_100_regs:pcs_100_regs:SLEEP_WAKE_DET

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 134. Fields in SLEEP_WAKE_DET

Field Name	Bit	Access	Description	Default
WUR_DET_LENGTH	15:8	R/W	WUR code detect length. This value is used by detection logic that does not operate in the VBAT domain.	0x3F

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Field Name	Bit	Access	Description	Default
			This field has units of 30 ns.	
LPS_DET_LENGTH	7:0	R/W	LPS code detect length. This field has units of 30 ns.	0x3F

1.6.1.34 pcs_100_regs:pcs_100_regs:SLEEP_WAKE_SEND

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 135. Fields in SLEEP_WAKE_SEND

Field Name	Bit	Access	Description	Default
WUR_SEND_LENGTH	15:8	R/W	WUR code send length. Note: This field has units of 30 ns.	0x3F
LPS_SEND_LENGTH	7:0	R/W	LPS length. Note: This field has units of 30 ns	0x3F

1.6.1.35 pcs_100_regs:pcs_100_regs:ED_HYST_MAST_MODE_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 136. Fields in ED_HYST_MAST_MODE_WIN

Field Name	Bit	Access	Description	Default
ED_HYST_MAST_MODE_WIN_CNT	15:0	R/W	No energy detect window in master mode (running on reference clock of 40ns period). Window to be programmed to check no-energy on line in master mode during hardware initailisation, training re]start and timeout event to toggle the start equaliser to DSP. Note : This is valid only when en_energy_det_master_mode = 1. This window should always to be programmed greater than (1.5 times) ed_hyst_win_cnt.	0x0640

1.6.1.36 pcs_100_regs:pcs_100_regs:CHK_REM_SENDN_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 137. Fields in CHK_REM_SENDN_WIN

Field Name	Bit	Access	Description	Default
CHK_REM_SENDN_IDLES_CNT_THRESH	15:8	R/W	Threshold count for the detection of remote PHY SEND_N consecutive Idles codes match within the chk_rem_sendn_win.	0xE6
CHK_REM_SENDN_WIN	7:0	R/W	Window to be programmed to check the remote PHY SEND_N Idles codes to know that remote PHY enters in to SEND_N state or not.	0xFF

1.6.1.37 pcs_100_regs:pcs_100_regs:SLAVE_TO_SS_TIME_ON_MAST_RST

Parent: pcs_100_regs:pcs_100_regs

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Instances: 1

Table 138. Fields in SLAVE_TO_SS_TIME_ON_MAST_RST

Field Name	Bit	Access	Description	Default
SLAVE_TO_SS_TIME_ON_MAST_RST	15:0	R/O	Slave entering to Slave silent state time when reset is applied to Master and the Master energy detect feature is enabled. This timer is valid when en_energy_det_master_mode = 1) Timer saturates to value 16hFFFF and it is in usec and reset to 16h0000 on hard or soft reset.	0x0000

1.6.1.38 pcs_100_regs:pcs_100_regs:SLAVE_TO_SS_TIME_ON_TRNG_RESTART

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 139. Fields in SLAVE_TO_SS_TIME_ON_TRNG_RESTART

Field Name	Bit	Access	Description	Default
SLAVE_TO_SS_TIME_ON_TRNG_RESTART	15:0	R/O	Slave entering to Slave silent state time when Master is re-started the Training and Master energy detect feature is enabled. This timer is valid when en_energy_det_master_mode = 1) Timer saturate to value 16hFFFF and it is in usec and reset to 16h0000 on each Training re-start.	0x0000

1.6.1.39 pcs_100_regs:pcs_100_regs:ED_DET_STABILITY_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 140. Fields in ED_DET_STABILITY_WIN

Field Name	Bit	Access	Description	Default
ED_DET_STABILITY_WIN	15:0	R/W	Value programmed are in microseconds. Programmable timer value to wait for the digital hysteresis energy detect logic to start its operation after enabling the AFE-Squelch circuit to detect the energy on/energy loss on MDI line. The default value was change for C0.	0x1770

1.6.1.40 pcs_100_regs:pcs_100_regs:SLAVE_HD_DESCR_LOCK_TIME

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 141. Fields in SLAVE_HD_DESCR_LOCK_TIME

Field Name	Bit	Access	Description	Default
RESERVED_0X227_1_93	15:8	R/O	Reserved	0x00
SLAVE_HD_DESCR_LOCK_TIME	7:0	R/O	Slave half duplex descrambler lock time (msec). This is the time from hardware init done to	0x00

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Field Name	Bit	Access	Description	Default
			pcs-descrambler lock in half duplex	

1.6.1.41 pcs_100_regs:pcs_100_regs:PCS_CNTRL_STS

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 142. Fields in PCS_CNTRL_STS

Field Name	Bit	Access	Description	Default
RESERVED_0X228_1_95	15:12	R/O	Reserved	0x0
SLEEP_IND	11	R/O	1=PHY entering to sleep mode indication valid only when auto_sleep_wake is enabled	0x0
PHY_CNTRL_SEND_N	10	R/O	1= PHY control is in SEND_IDLE_OR_DATA_STATE	0x0
PHY_CNTRL_SEND_I	9	R/O	1= PHY control is in TRAINING or SEND_IDLE state	0x0
PHY_CNTRL_SEND_Z	8	R/O	1=PHY control is in slave silent state indication	0x0
REM_RCVR_STS	7	R/O	1=Remote receiver status asserted	0x0
LOC_RCVR_STS	6	R/O	1=Local receiver status asserted	0x0
DESCR_LOCK_STS	5	R/O	1= Descrambler lock status asserted	0x0
SLAVE_IN_FULL_DUPLEX	4	R/O	1=SLAVE in full duplex mode indication from DSP	0x0
DSP_EQ_STS	3	R/O	1=DSP equalizer lock assertion status	0x0
DSP_START_EQ_STS	2	R/O	1= DSP start equalixation is asserted	0x0
HW_INIT_DONE	1	R/O	1= Hardware installation done indication	0x0
ED_DET	0	R/O	1 = Energy detect valid when AFE-squelch is enabled	0x0

1.6.1.42 pcs_100_regs:pcs_100_regs:DSP_EQ_LOCK_ASSERT_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 143. Fields in DSP_EQ_LOCK_ASSERT_CNT

Field Name	Bit	Access	Description	Default
DSP_EQ_LOCK_ASSERT_CNT	15:0	R/O	Count of DSP equalizer lock assertion count not linked to soft reset or hard reset. Reset only on writing to register bit "pcs_sts_counter_rst"	0x0000

1.6.1.43 pcs_100_regs:pcs_100_regs:DESCR_LOCK_ASSERT_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

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Table 144. Fields in DESCR_LOCK_ASSERT_CNT

Field Name	Bit	Access	Description	Default
DESCR_LOCK_ASSERT_CNT	15:0	R/O	Count of local descrambler lock assertion count not linked to soft reset or hard reset. Reset only on writing to register bit "pcs_sts_counter_rst"	0x0000

1.6.1.44 pcs_100_regs:pcs_100_regs:LOC_RCVR_STS_ASSERT_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 145. Fields in LOC_RCVR_STS_ASSERT_CNT

Field Name	Bit	Access	Description	Default
LOC_RCVR_STS_ASSERT_CNT	15:0	R/O	Count of local receiver status assertion count not linked to soft reset or hard reset. Reset only on writing to register bit "pcs_sts_counter_rst"	0x0000

1.6.1.45 pcs_100_regs:pcs_100_regs:REM_RCVR_STS_ASSERT_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 146. Fields in REM_RCVR_STS_ASSERT_CNT

Field Name	Bit	Access	Description	Default
REM_RCVR_STS_ASSERT_CNT	15:0	R/O	Count of remotel receiver status assertion count not linked to soft reset or hard reset. Reset only on writing to register bit "pcs_sts_counter_rst"	0x0000

1.6.1.46 pcs_100_regs:pcs_100_regs:PCS_FSM_STS0

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 147. Fields in PCS_FSM_STS0

Field Name	Bit	Access	Description	Default
PHYCNTRL_FSM_STATE	15:13	R/O	PHY control FSM state.	0x0
LINK_MON_FSM_STATE	12:11	R/O	Link monitor FSM state.	0x0
DSCR_FSM_STATE	10:8	R/O	escrambler FSM state.	0x0
RESTART_EQ_FSM_STATE	7:6	R/O	DSP equalization restart FSM state	0x0
HW_INIT_FSM_STATE	5:2	R/O	Hardware initialization FSM state	0x0
ED_HYST_FSM_STATE	1:0	R/O	Energy detect Hystersis FSM state	0x0

1.6.1.47 pcs_100_regs:pcs_100_regs:PCS_FSM_STS1

Parent: pcs_100_regs:pcs_100_regs

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Table 148. Fields in PCS_FSM_STS1

Field Name	Bit	Access	Description	Default
LINKUP_TIMEOUT_FSM_STATE	15:14	R/O	Linkup timeout FSM state.	0x0
PCS_RX_FSM_STATE	13:10	R/O	PCS-RX FSM state.	0x0
PCS_RX_FSM_JAB_STATE	9:8	R/O	PCS-RX Jab detection FSM state.	0x0
SLAVE_EQ_TIMER_FSM_STATE	7:6	R/O	Waiting for slave full duplex to assert the loc_rcvr_sts FSM state	0x0
REM_RCVR_FSM_STATE	5:4	R/O	Remote receiver detection FSM state.	0x0
IDLE_ERR_FSM_STATE	3:2	R/O	IDLE error detection FS, state.	0x0
REM_SENDN_FSM_STATE	1:0	R/O	Remote entering to SEND_N detection FSM state.	0x0

1.6.1.48 pcs_100_regs:pcs_100_regs:PCS_FSM_STS2

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 149. Fields in PCS_FSM_STS2

Field Name	Bit	Access	Description	Default
RESERVED_0X299_1_125	15:9	R/O	Reserved	0x00
PCS_TX_FSM_STATE	8:5	R/O	PCS-TX FSM state.	0x0
AUTO_SLEEP_WAKE_FSM_STATE	4:2	R/O	Auto Sleep Wake FSM state	0x0
EDPWRDWN_FSM_STATE	1:0	R/O	Energy detect power down FSM state	0x0

1.6.1.49 pcs_100_regs:pcs_100_regs:MAX_WAIT_TIMER_EVNT_CNT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 150. Fields in MAX_WAIT_TIMER_EVNT_CNT

Field Name	Bit	Access	Description	Default
MAX_WAIT_TIMER_EVNT_CNT	15:0	R/O	max wait timer event assertion count not linked to soft reset or hard reset. Reset only on writing to register bit "pcs_sts_counter_rst".	0x0000

1.6.1.50 pcs_100_regs:pcs_100_regs:TXPD_CMN_MODE_SETTLE_TIMER

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 151. Fields in TXPD_CMN_MODE_SETTLE_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X231_1_130	15:8	R/O	Reserved	0x00
TXPD_CMN_MODE_SETTLE_TIMER	7:0	R/W	Timer in usec for the AFE common mode to settle after the TXPD removal and before asserting the start_equalizer to DSP during Training restart.	0x64

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1.6.1.51 pcs_100_regs:pcs_100_regs:POLFLIP_FIX_TIMER

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 152. Fields in POLFLIP_FIX_TIMER

Field Name	Bit	Access	Description	Default
RESERVED_0X232_1_132	15:8	R/O	Reserved	0x00
POLFLIP_FIX_ENABLE	7	R/W	Enable the feature of polarity flip check even on link training restart condition after link-down.	0x0
POLFLIP_FIX_ENABLE_TIMER	6:0	R/W	Timer to enable the polarity flip check in slave mode upon reset removal or link training restart. This timer is in millisecond and valid only when "polflip_fix_enable" register control is set to logic 1.	0x06

1.6.1.52 pcs_100_regs:pcs_100_regs:DESCR_LOCK_WIN_EXTEND

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 153. Fields in DESCR_LOCK_WIN_EXTEND

Field Name	Bit	Access	Description	Default
DESCR_LOCK_SYMB_CNT_EXTEND	15:8	R/W	Descrambler lock idle symbol count matched threshold count. This register should be used along with "descr_lock_symb_cnt"	0x00
DESCR_LOCK_CHK_WIN_EXTEND	7:0	R/W	Descrambler lock checking window extend. This regiser should be used along with "descr_lock_chk_win"	0x00

1.6.1.53 pcs_100_regs:pcs_100_regs:SLAVE_TRNG_IDLE_ERR_CHECK

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 154. Fields in SLAVE_TRNG_IDLE_ERR_CHECK

Field Name	Bit	Access	Description	Default
RESERVED_0X234_1_137	15:8	R/O	Reserved	0x00
EN_SLAVE_TRNG_IDLE_ERR_CHK	7	R/W	Enable control bit to check the IDLE error in Slave mode when PHY control is in Training state and the "dsp_slv_full_duplex_eq_lock_time" timer expires.	0x0
DSP_SLV_FULL_DUPLEX_EQ_LOCK_TIME	6:0	R/W	Expected full duplex equalizer lock time. This timer is in mili seconds. In training state, within this timer if full duplex equalizer lock is not asserted, the logic checks the IDLE error if any to reset the PHY control FSM to move to SLAVE_SILENT state. This timer is valid only when	0x19

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Field Name	Bit	Access	Description	Default
			"en_slave_tmg_idle_err_chk = 1"	

1.6.1.54 pcs_100_regs:pcs_100_regs:SLAVE_TRNG_IDLE_ERR_CHECK_WIN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 155. Fields in SLAVE_TRNG_IDLE_ERR_CHECK_WIN

Field Name	Bit	Access	Description	Default
SLV_TRNG_IDLE_ERR_CHK_WINDOW	15:0	R/W	Programmable window to check the Idle errors in Slave mode Training state. This window counter runs on 33.33Mhz clock domain.	0x0080

1.6.1.55 pcs_100_regs:pcs_100_regs:SLAVE_TRNG_IDLE_ERR_CNT_THRESH

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 156. Fields in SLAVE_TRNG_IDLE_ERR_CNT_THRESH

Field Name	Bit	Access	Description	Default
SLAVE_TRNG_IDLE_ERR_CNT_THRESHOLD	15:0	R/W	Programmable threshold count to check the Idle erros in Slave training state	0x0078

1.6.1.56 pcs_100_regs:pcs_100_regs:WUP_LEN

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 157. Fields in WUP_LEN

Field Name	Bit	Access	Description	Default
RESERVED_0X240_1_142	15:8	R/O	Reserved	0x00
WUP_SEND_LEN	7:0	R/W	WUP code send length. Note: This field has units of 8 ns.	0x7F

1.6.1.57 pcs_100_regs:pcs_100_regs:SLEEP_ABRT

Parent: pcs_100_regs:pcs_100_regs

Instances: 1

Table 158. Fields in SLEEP_ABRT

Field Name	Bit	Access	Description	Default
RESERVED_0X241_2_144	15:8	R/O	Reserved	0x00
TC10_SEL_OSW_CLK	7:6	R/W	Configuration to select osw_clk	0x0
SLEEP_ABORT	5	R/W	When set a pending sleep request, received via partner LPS signaling, is aborted. Note: This bit was added for revision B0.	0x0
SLEEP_REJECT	4	R/W	When set all sleep requests shall be rejected. Note: This bit was added for revision B0.	0x0

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Field Name	Bit	Access	Description	Default
RESERVED_0X241_1_148	3:0	R/O	Reserved	0x0

1.7 aneg_regs

Table 159. Register Groups in MMD aneg_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
aneg_regs	0x00000000	1		Page 51

1.7.1 aneg_regs:aneg_regs

Parent: aneg_regs

Instances: 1

Table 160. Registers in aneg_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
ANEG_DEV_ID1	0x00000002	1		Page 52
ANEG_DEV_ID2	0x00000003	1		Page 52
ANEG_DEV_PKG1	0x00000005	1		Page 52
ANEG_DEV_PKG2	0x00000006	1		Page 53
ANEG_CNTRL_REG1	0x00000200	1		Page 53
ANEG_STAT_REG1	0x00000201	1		Page 53
ANEG_ADV_REG1	0x00000202	1		Page 54
ANEG_ADV_REG2	0x00000203	1		Page 54
ANEG_ADV_REG3	0x00000204	1		Page 54
ANEG_LP_AB_REG1	0x00000205	1		Page 55
ANEG_LP_AB_REG2	0x00000206	1		Page 55
ANEG_LP_AB_REG3	0x00000207	1		Page 55
ANEG_NP_ADV_REG1	0x00000208	1		Page 56
ANEG_NP_ADV_REG2	0x00000209	1		Page 56
ANEG_NP_ADV_REG3	0x0000020A	1		Page 56
ANEG_LP_NPAB_REG1	0x0000020B	1		Page 56
ANEG_LP_NPAB_REG2	0x0000020C	1		Page 57

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
ANEG_LP_NPAB_REG3	0x0000020D	1		Page 57
VEND_ARB_STAT	0x00008000	1		Page 57
VEND_AB_MATCH_WORD1	0x00008001	1		Page 58
VEND_AB_MATCH_WORD2	0x00008002	1		Page 58
VEND_AB_MATCH_WORD3	0x00008003	1		Page 58
VEND_INTR_MASK	0x00008004	1		Page 59
VEND_INTR_STAT	0x00008005	1		Page 59
VEND_DBG_CTRL_STAT_REG	0x00008013	1		Page 60

1.7.1.1 aneg_regs:aneg_regs:ANEG_DEV_ID1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 161. Fields in ANEG_DEV_ID1

Field Name	Bit	Access	Description	Default
OUI_3_18	15:0	R/O	Bits [3:18] of the OUI. WH uses the SMSC OUI of 00-80-0F (or 00:01:F0)	0x0007

1.7.1.2 aneg_regs:aneg_regs:ANEG_DEV_ID2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 162. Fields in ANEG_DEV_ID2

Field Name	Bit	Access	Description	Default
OUI_19_24	15:10	R/O	Bits [19:24] of the OUI.	0x30
MODEL_NUM	9:4	R/O	The device model number.	0x1F
REV_NUM	3:0	R/O	Revision number (0 for WH A0; 1 for Molineux A0)	0x1

1.7.1.3 aneg_regs:aneg_regs:ANEG_DEV_PKG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 163. Fields in ANEG_DEV_PKG1

Field Name	Bit	Access	Description	Default
RESERVED_15_8_4	15:8	R/O	Reserved for future use.	0x00
ANEG_PRESENT	7	R/O	If 1, auto-negotiation is present in the package.	0x1

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Field Name	Bit	Access	Description	Default
RESERVED_6_4_6	6:4	R/O	Reserved for future use.	0x0
PCS_PRESENT	3	R/O	If 1, a PCS is present in the package.	0x1
RESERVED_2_8	2	R/O	Reserved for future use.	0x0
PMA_PMD_PRESENT	1	R/O	If 1, a PMA/PMD is present in the package.	0x1
CL22_REGS_PRESENT	0	R/O	If 1, Clause 22 registers are present in the package.	0x0

1.7.1.4 aneg_regs:aneg_regs:ANEG_DEV_PKG2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 164. Fields in ANEG_DEV_PKG2

Field Name	Bit	Access	Description	Default
VEND_SPEC_DEV2_PRESENT	15	R/O	If 1, vendor-specific device 2 is present in the package.	0x0
VEND_SPEC_DEV1_PRESENT	14	R/O	If 1, vendor-specific device 1 is present in the package.	0x1
CL22_EXT_PRESENT	13	R/O	If 1, Clause 22 extension is present in the package.	0x0
RESERVED_14	12:0	R/O	Reserved for future use.	0x0000

1.7.1.5 aneg_regs:aneg_regs:ANEG_CNTRL_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 165. Fields in ANEG_CNTRL_REG1

Field Name	Bit	Access	Description	Default
AN_RESET	15	R/W	ANEG Reset (Self-clearing)	0x0
RESERVED_1_16	14:13	R/O	Always 0	0x0
AN_ENABLE	12	R/W	ANEG Enable	0x0
RESERVED_18	11:10	R/O	Always 0	0x0
AN_RESTART	9	R/W	ANEG Restart (Self-clearing)	0x0
RESERVED_0_20	8:0	R/O	Always 0	0x000

1.7.1.6 aneg_regs:aneg_regs:ANEG_STAT_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 166. Fields in ANEG_STAT_REG1

Field Name	Bit	Access	Description	Default
RESERVED_1_21	15:7	R/O	Always 0	0x000
AN_PAGE_RCVD	6	R/O	AN Page Received - Latch High	0x0

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Field Name	Bit	Access	Description	Default
			Note: Reading this field will clear it.	
AN_COMP	5	R/O	ANEG Complete	0x0
AN_REMOTE_FAULT_STATUS	4	R/O	AN Remote Fault - Latch High (no remote fault detection implemented at this time) Note: Reading this field will clear it.	0x0
AN_ABILITY	3	R/O	AN Ability	0x1
AN_LINK_STATUS	2	R/O	AN Link Status - Latch Low	0x0
RESERVED_0_27	1:0	R/O	Always 0	0x0

1.7.1.7 aneg_regs:aneg_regs:ANEG_ADV_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 167. Fields in ANEG_ADV_REG1

Field Name	Bit	Access	Description	Default
AN_NXT_PAGE	15	R/W	ANEG Next page	0x0
AN_ACK	14	R/O	ANEG Acknowledge (Set by arbiter; always 0 on read)	0x0
AN_REMOTE_FAULT	13	R/W	ANEG Remote Fault	0x0
AN_MS	12	R/W	ANEG Force Master/Slave	0x0
AN_C	11:10	R/W	ANEG Pause Ability	0x0
AN_E	9:5	R/O	ANEG Echoed Nonce Field -- captured from link partner's Transmitted Nonce Field	0x00
AN_SEL	4:0	R/W	ANEG Selector Field	0x01

1.7.1.8 aneg_regs:aneg_regs:ANEG_ADV_REG2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 168. Fields in ANEG_ADV_REG2

Field Name	Bit	Access	Description	Default
AN_ADV_10_0	15:5	R/W	ANEG Technology Ability[10:0] -- Advertise both 100BASE-T1 and 1000BASE-T1 by default (A[0] and A[2])	0x005
AN_T_4	4	R/W	ANEG Transmitted Nonce Field T[4] -- indicates M/S preference when an_ms is 0, Forces M/S when an_ms is 1	0x0
AN_T_3_0	3:0	R/O	ANEG Transmitted Nonce Field T[3:0] -- random value captured by arbiter	0x0

1.7.1.9 aneg_regs:aneg_regs:ANEG_ADV_REG3

Parent: aneg_regs:aneg_regs

Instances: 1

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Table 169. Fields in ANEG_ADV_REG3

Field Name	Bit	Access	Description	Default
AN_ADV_26_11	15:0	R/W	ANEG Technology Ability[26:11]	0x0000

1.7.1.10 aneg_regs:aneg_regs:ANEG_LP_AB_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 170. Fields in ANEG_LP_AB_REG1

Field Name	Bit	Access	Description	Default
AN_LP_NXT_PAGE	15	R/O	ANEG Link Partner Next Page (a read of reg 517 latches the contents of reg 518 and reg 519)	0x0
AN_LP_ACK	14	R/O	ANEG Link Partner Acknowledge	0x0
AN_LP_REMOTE_FAULT	13	R/O	ANEG Link Partner Remote Fault	0x0
AN_LP_MS	12	R/O	ANEG Link Partner Force Master/Slave	0x0
AN_LP_C	11:10	R/O	ANEG Link Partner Pause Ability	0x0
AN_LP_E	9:5	R/O	ANEG Link Partner Echoed Nonce Field -- captured from local device's Transmitted Nonce Field	0x00
AN_LP_SEL	4:0	R/O	ANEG Link Partner Selector Field	0x00

1.7.1.11 aneg_regs:aneg_regs:ANEG_LP_AB_REG2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 171. Fields in ANEG_LP_AB_REG2

Field Name	Bit	Access	Description	Default
AN_LP_AB_10_0	15:5	R/O	ANEG Link Partner Technology Ability[10:0] -- A[0] => 100BASE-T1 and A[2] => 1000BASE-T1	0x000
AN_LP_T_4	4	R/O	ANEG Link Partner Transmitted Nonce Field T[4] -- indicates M/S preference when an_ms is 0, Forces M/S when an_ms is 1	0x0
AN_LP_T_3_0	3:0	R/O	ANEG Link Partner Transmitted Nonce Field T[3:0] -- random value captured by arbiter	0x0

1.7.1.12 aneg_regs:aneg_regs:ANEG_LP_AB_REG3

Parent: aneg_regs:aneg_regs

Instances: 1

Table 172. Fields in ANEG_LP_AB_REG3

Field Name	Bit	Access	Description	Default
AN_LP_AB_26_11	15:0	R/O	ANEG Link Partner Ability Register Bits[26:11]	0x0000

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1.7.1.13 aneg_regs:aneg_regs:ANEG_NP_ADV_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 173. Fields in ANEG_NP_ADV_REG1

Field Name	Bit	Access	Description	Default
AN_NP_NXT_PAGE	15	R/W	ANEG Next Page (mr_next_page_loaded set when reg 520 is written and cleared by Arb FSM in the NEXT PAGE WAIT state)	0x0
AN_NP_ACK	14	R/O	ANEG Acknowledge	0x0
AN_MSG_PAGE	13	R/W	ANEG Message Page	0x1
AN_ACK2	12	R/O	ANEG Acknowledge 2	0x0
AN_TOGGLE	11	R/O	ANEG Toggle Field	0x0
AN_MSG_UNFRMT	10:0	R/W	ANEG Message/Unformatted Field [10:0] (defaults to Null Msg.)	0x001

1.7.1.14 aneg_regs:aneg_regs:ANEG_NP_ADV_REG2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 174. Fields in ANEG_NP_ADV_REG2

Field Name	Bit	Access	Description	Default
AN_NP_ADV_UNFRMT_MID	15:0	R/W	ANEG Unformatted field [26:11] or [15:0] depending on whether the message page bit is clear or set.	0x0000

1.7.1.15 aneg_regs:aneg_regs:ANEG_NP_ADV_REG3

Parent: aneg_regs:aneg_regs

Instances: 1

Table 175. Fields in ANEG_NP_ADV_REG3

Field Name	Bit	Access	Description	Default
AN_NP_ADV_UNFRMT_HI	15:0	R/W	ANEG Unformatted field [42:27] or [31:0]	0x0000

1.7.1.16 aneg_regs:aneg_regs:ANEG_LP_NPAB_REG1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 176. Fields in ANEG_LP_NPAB_REG1

Field Name	Bit	Access	Description	Default
AN_LP_NP_NXT_PAGE	15	R/O	ANEG Link Partner Next Page (a read of reg 523 latches the contents of reg 524 and reg 525)	0x0
AN_LP_NP_ACK	14	R/O	ANEG Link Partner Acknowledge	0x0
AN_LP_MSG_PAGE	13	R/O	ANEG Link Partner Message Page	0x0

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Field Name	Bit	Access	Description	Default
AN_LP_ACK2	12	R/O	ANEG Link Partner Acknowledge 2	0x0
AN_LP_TOGGLE	11	R/O	ANEG Link Partner Toggle Field	0x0
AN_LP_MSG_UNFRMT	10:0	R/O	ANEG Link Partner Message/Unformatted Field [10:0]	0x000

1.7.1.17 aneg_regs:aneg_regs:ANEG_LP_NPAB_REG2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 177. Fields in ANEG_LP_NPAB_REG2

Field Name	Bit	Access	Description	Default
AN_LP_NP_ADV_UNFRMT_MID	15:0	R/O	ANEG Link Partner Next Page Unformatted field [26:11] or [15:0] depending on whether the message page bit is clear or set. (value latched when reg 523 is read)	0x0000

1.7.1.18 aneg_regs:aneg_regs:ANEG_LP_NPAB_REG3

Parent: aneg_regs:aneg_regs

Instances: 1

Table 178. Fields in ANEG_LP_NPAB_REG3

Field Name	Bit	Access	Description	Default
AN_LP_NP_ADV_UNFRMT_HI	15:0	R/O	ANEG Link Partner Next Page Unformatted field [42:27] or [31:0] (value latched when reg 523 is read)	0x0000

1.7.1.19 aneg_regs:aneg_regs:VEND_ARB_STAT

Parent: aneg_regs:aneg_regs

Instances: 1

Table 179. Fields in VEND_ARB_STAT

Field Name	Bit	Access	Description	Default
RESERVED_ARB_15_14_66	15:14	R/O	Spare RO bits	0x0
PMA_LINK_STAT	13:12	R/O	The received pma_link_status. 01 => 100; 10 => 1000	0x0
RESERVED_ARB_11_68	11	R/O	Spare RO bits	0x0
PMA_LINK_CONTROL	10:9	R/O	The resolved pma_link_control (HCD). 01 => 100; 10 => 1000	0x0
AN_LINK_GOOD	8	R/O	Negotiated good link status	0x0
ACK_FINISHED	7	R/O	The TX FSM has sent the final Ack codeword.	0x0
COMPLETE_ACK	6	R/O	The TX FSM is instructed to send the Ack codewords.	0x0
BASE_PAGE	5	R/O	If set, the local device is transmitting a base page. Otherwise, it is transmitting a next page.	0x0

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Field Name	Bit	Access	Description	Default
CONSISTENCY_MATCH	4	R/O	Indicates that the ability_match_word is the same as the link codeword that caused acknowledge_match to be set.	0x0
ACK_NONCE_MATCH	3	R/O	Indicates whether the echoed nonce received from the link partner matches the transmitted nonce field sent by the local device. The echoed nonce value from the DME page that caused acknowledge_match to be set is used for this test.	0x0
ACKNOWLEDGE_MATCH	2	R/O	Indicates that at least one link codeword with the Acknowledge bit set and with good CRC16 was received.	0x0
NONCE_MATCH	1	R/O	Indicates whether the transmitted nonce received from the link partner matches the transmitted nonce field sent by the local device.	0x0
ABILITY_MATCH	0	R/O	Indicates that at least one link codeword with good CRC16 was received.	0x0

1.7.1.20 aneg_regs:aneg_regs:VEND_AB_MATCH_WORD1

Parent: aneg_regs:aneg_regs

Instances: 1

Table 180. Fields in VEND_AB_MATCH_WORD1

Field Name	Bit	Access	Description	Default
ABILITY_MATCH_WORD_15_0	15:0	R/O	Bits [15:0] of the 46-bit ability_match_word. Loaded upon transition to Acknowledge Detect state with the value of the link codeword that caused ability_match = true for that transition	0x0000

1.7.1.21 aneg_regs:aneg_regs:VEND_AB_MATCH_WORD2

Parent: aneg_regs:aneg_regs

Instances: 1

Table 181. Fields in VEND_AB_MATCH_WORD2

Field Name	Bit	Access	Description	Default
ABILITY_MATCH_WORD_31_16	15:0	R/O	Bits [31:16] of the 48-bit ability_match_word.	0x0000

1.7.1.22 aneg_regs:aneg_regs:VEND_AB_MATCH_WORD3

Parent: aneg_regs:aneg_regs

Instances: 1

Table 182. Fields in VEND_AB_MATCH_WORD3

Field Name	Bit	Access	Description	Default
ABILITY_MATCH_WORD_47_32	15:0	R/O	Bits [47:32] of the 48-bit ability_match_word.	0x0000

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1.7.1.23 aneg_regs:aneg_regs:VEND_INTR_MASK

Parent: aneg_regs:aneg_regs

Instances: 1

Table 183. Fields in VEND_INTR_MASK

Field Name	Bit	Access	Description	Default
ANEG_INTERRUPT_MASK	15	R/W	Autoneg block interrupt mask.	0x0
RX_PAGE_TEST_TIMER_MASK	14	R/W	RX page test max timer done mask.	0x0
RX_WAIT_TIMER_MASK	13	R/W	RX wait timer done mask.	0x0
RX_UNKNOWN_TRANS_MASK	12	R/W	Unknown RX transistion (i.e. not clock, data, or start delimiter) mask.	0x0
ANEG_RESERVED11_MASK	11	R/W	Reserved for future enhancement.	0x0
BAD_CRC16_MASK	10	R/W	Bad CRC16 interrupt mask.	0x0
CONSISTENCY_MATCH_FAIL_MASK	9	R/W	Mask for the consistency_match_fail intrerrupt.	0x0
ACK_NONCE_MATCH_FAIL_MASK	8	R/W	Mask for the ack_nonce_match_fail intrerrupt.	0x0
NONCE_MATCH_MASK	7	R/W	Mask for the nonce_match intrerrupt.	0x0
MS_CFG_FAULT_MASK	6	R/W	Master/Slave config fault mask	0x0
HCD_FAIL_MASK	5	R/W	HCD fail mask	0x0
LINK_FAIL_TIMER_MASK	4	R/W	Link fail timer mask	0x0
ANEG_NP_RX_MASK	3	R/W	Next page bit seen in receive code word interrupt mask	0x0
ANEG_COMPLETE_MASK	2	R/W	Autoneg complete mask.	0x0
SPEED_CHANGE_MASK	1	R/W	Speed change relative to previous link-up mask.	0x0
LINK_STATE_CHANGE_MASK	0	R/W	Link state change (up or down) mask. NOTE: The meaning of each interrupt mask bit is: 1: enable the interrupt 0: disable the interrupt NOTE: the mask controls the forwarding of the interrupt (i.e. it won't be forwarded if the mask is clear). However the status will be captured regardless of the mask as additional debug status. This applies to all of the interrupt mask bits in this register.	0x0

1.7.1.24 aneg_regs:aneg_regs:VEND_INTR_STAT

Parent: aneg_regs:aneg_regs

Instances: 1

Table 184. Fields in VEND_INTR_STAT

Field Name	Bit	Access	Description	Default
ANEG_INTERRUPT_STAT	15	Sticky	Autoneg block interrupt status (latch high; write 1 to clear)	0x0
RX_PAGE_TEST_TIMER_STAT	14	Sticky	RX page test max timer done status (latch high; write 1 to clear)	0x0

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Field Name	Bit	Access	Description	Default
RX_WAIT_TIMER_STAT	13	Sticky	RX wait timer done status (latch high; write 1 to clear)	0x0
RX_UNKNOWN_TRANS_STAT	12	Sticky	Unknown RX transistion (i.e. not clock, data, or start delimiter) (latch high; write 1 to clear)	0x0
ANEG_RESERVED11_STAT	11	Sticky	Reserved for future enhancement.	0x0
BAD_CRC16_STAT	10	Sticky	Bad CRC16 interrupt status.	0x0
CONSISTENCY_MATCH_FAIL_STAT	9	Sticky	This will be set if consistency_match is false when acknowledge_match is set. (latch high; write 1 to clear)	0x0
ACK_NONCE_MATCH_FAIL_STAT	8	Sticky	This will be set if ack_nonce_match is false when acknowledge_match is set. (latch high; write 1 to clear)	0x0
NONCE_MATCH_STAT	7	Sticky	This will be set if nonce_match is true when ability_match is set. (latch high; write 1 to clear)	0x0
MS_CFG_FAULT_STAT	6	Sticky	Master/Slave config fault status (latch high; write 1 to clear)	0x0
HCD_FAIL_STAT	5	Sticky	HCD fail status (latch high; write 1 to clear)	0x0
LINK_FAIL_TIMER_STAT	4	Sticky	Link fail timer status (latch high; write 1 to clear)	0x0
ANEG_NP_RX_STAT	3	Sticky	Next page bit seen in receive code word status (useful for SW exchange of next pages) (latch high; write 1 to clear)	0x0
ANEG_COMPLETE_STAT	2	Sticky	Autoneg complete status (latch high; write 1 to clear)	0x0
SPEED_CHANGE_STAT	1	Sticky	Speed change status relative to previous link-up (none/100/1000) (latch high; write 1 to clear)	0x0
LINK_STATE_CHANGE_STAT	0	Sticky	Link state change (up or down) status (latch high; write 1 to clear). NOTE: this will capture the event regardless of the interrup mask as additional debug status. However it won't get forwarded to the interrupt if the mask is clear. This applies to all of the interrupt status bits in this register.	0x0

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1.7.1.25 aneg_regs:aneg_regs:VEND_DBG_CTRL_STAT_REG

Parent: aneg_regs:aneg_regs

Instances: 1

Table 185. Fields in VEND_DBG_CTRL_STAT_REG

Field Name	Bit	Access	Description	Default
AN_LOCAL_DEV_CONFIG_FAULT	10	R/O	Local device has a configuration fault when this bit is set.	0x0
AN_LOCAL_DEV_CONFIG_SLAVE	9	R/O	Local device is configured as Slave when this bit is set.	0x0
AN_LOCAL_DEV_CONFIG_MASTER	8	R/O	Local device is configured as Master when this bit is set.	0x0

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Field Name	Bit	Access	Description	Default
RESERVED	7:4	R/W	Must be set to its default.	0xF
RESERVED	3	R/W	Must be set to its default.	0x1
RESERVED	2	R/W	Must be set to its default.	0x1

1.8 mis_regs

Table 186. Register Groups in MMD mis_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
mis_regs	0x00000000	1		Page 61

1.8.1 mis_regs:mis_regs

Parent: mis_regs

Instances: 1

Table 187. Registers in mis_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
MIS_CFG_REG0	0x00000A00	1		Page 63
MIS_DLL_CFG_REG0	0x00000A01	1		Page 65
MIS_DLL_CFG_REG1	0x00000A02	1		Page 65
MIS_CFG_REG2	0x00000A03	1		Page 66
MIS_INT_EN_REG0	0x00000A04	1		Page 66
MIS_INT_STAT_REG0	0x00000A05	1		Page 67
MIS_PKT_STAT_REG0	0x00000A06	1		Page 67
MIS_PKT_STAT_REG1	0x00000A07	1		Page 68
MIS_PKT_STAT_REG2	0x00000A08	1		Page 68
MIS_PKT_STAT_REG3	0x00000A09	1		Page 68
MIS_PKT_STAT_REG4	0x00000A0A	1		Page 69
MIS_PKT_STAT_REG5	0x00000A0B	1		Page 69
MIS_PKT_STAT_REG6	0x00000A0C	1		Page 69
MIS_EPG_CFG1	0x00000A0D	1		Page 70
MIS_EPG_PYLD	0x00000A11	1		Page 70
MIS_EPG_BRST_CNT	0x00000A12	1		Page 71

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
MIS_EPG_MBR_CNT	0x00000A13	1		Page 71
MIS_EPG_IPG_CFG	0x00000A14	1		Page 71
MIS_EPG_BRST_IPG	0x00000A15	1		Page 71
MIS_EPG_UMB_IPG	0x00000A16	1		Page 71
MIS_EPG_LMB_IPG	0x00000A17	1		Page 72
MIS_EPG_PLEN	0x00000A18	1		Page 72
MIS_EPG_RPKT_LEN	0x00000A19	1		Page 72
MIS_EPG_MIN_RPKT_LEN	0x00000A1B	1		Page 72
MIS_EPG_CFG2	0x00000A1C	1		Page 72
MIS_EPG_JMB_PSIZE	0x00000A1E	1		Page 73
MIS_EPG_ETHR_TYP	0x00000A1F	1		Page 73
MIS_EPG_UW_MSRC	0x00000A20	1		Page 73
MIS_EPG_MW_MSRC	0x00000A21	1		Page 73
MIS_EPG_LW_MSRC	0x00000A22	1		Page 73
MIS_EPG_UW_MDST	0x00000A23	1		Page 74
MIS_EPG_MW_MDST	0x00000A24	1		Page 74
MIS_EPG_LW_MDST	0x00000A25	1		Page 74
MIS_EPG_UW_ISRC	0x00000A26	1		Page 74
MIS_EPG_LW_ISRC	0x00000A27	1		Page 74
MIS_EPG_UW_IDST	0x00000A28	1		Page 75
MIS_EPG_LW_IDST	0x00000A29	1		Page 75
MIS_EPG_UDP_SRC	0x00000A2A	1		Page 75
MIS_EPG_UDP_DST	0x00000A2B	1		Page 75
MIS_EPG_SIG_CNT	0x00000A2C	1		Page 75
MIS_EPG_SIG_CFG	0x00000A2D	1		Page 76
MIS_EPG_IDL_LEN	0x00000A32	1		Page 76
MIS_EPG_UW_LPI	0x00000A33	1		Page 76
MIS_EPG_LW_LPI	0x00000A34	1		Page 76

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
MIS_PKT_CHK_CFG0	0x00000A48	1		Page 76
MIS_PKT_STAT_REG7	0x00000A49	1		Page 77
MIS_WOL_CFG0	0x00000A50	1		Page 77
MIS_WOL_UW_PSWD	0x00000A51	1		Page 77
MIS_WOL_MW_PSWD	0x00000A52	1		Page 77
MIS_WOL_LW_PSWD	0x00000A53	1		Page 78
MIS_WOL_UW_ADDR	0x00000A54	1		Page 78
MIS_WOL_MW_ADDR	0x00000A55	1		Page 78
MIS_WOL_LW_ADDR	0x00000A56	1		Page 78
MIS_SGMII_INTF_CFG0	0x00000A60	1		Page 78
MIS_SGMII_INTF_STS	0x00000A61	1		Page 79
MIS_DBG_CFG_REG0	0x00000A8C	1		Page 79
MIS_DBG_STAT_REG0	0x00000A8D	1		Page 80
MIS_TX_DBG_STAT_REG0	0x00000A8E	1		Page 80
MIS_RX_DBG_STAT_REG0	0x00000A8F	1		Page 81
MIS_FIFO_FLOW_STAT0	0x00000A9C	1		Page 81
MIS_FIFO_FLOW_STAT1	0x00000A9D	1		Page 81
MIS_TX_FIFO_PPM	0x00000AB5	1		Page 82
MIS_RX_FIFO_PPM	0x00000AB6	1		Page 82
MIS_PPM_CLK_CNT_LIMIT	0x00000AB7	1		Page 83
MIS_TX_FIFO_PPM_THR1_2	0x00000AC0	1		Page 83
MIS_TX_FIFO_PPM_THR3	0x00000AC1	1		Page 83
MIS_RX_FIFO_PPM_THR1_2	0x00000AC2	1		Page 84
MIS_RX_FIFO_PPM_THR3	0x00000AC3	1		Page 84

1.8.1.1 mis_regs:mis_regs:MIS_CFG_REG0

Parent: mis_regs:mis_regs

Instances: 1

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Table 188. Fields in MIS_CFG_REG0

Field Name	Bit	Access	Description	Default
MIS_CORE_RESET	15	R/W	MAC Interface Functional core reset - When asserted HIGH - it will cause MIS to be in RESET until the user deasserts by writing 0 to this register field	0x0
FIFO_LATENCY_CFG	13:12	R/W	FIFO Latency Config Select: This register field needs to be configured based on max packet size supported. 2'b00 - pkts with max pktsize 0-1522 bytes; 2'b01 - pkts with max pktsize 1522- 9600 bytes ; 2'b10 - pkts with max pktsize 9600 - 12500 bytes ; 2'b11: pkts with max pktsize 12500-16384 bytes. In case of speed100, this register is used in conjuction with MII FIFO latency config registers to determine the FIFO latency settings. In case of speed1000, this register is used directly to determine FIFO latency settings for both TX and RX FIFOs.	0x0
ISOLATE_MIS_DATA	10	R/W	Isolate data transmitted or received by MIS. When asserted HIGH, causes to block data upstream by shutting down the incoming data and sending 0s in it's place.	0x0
DIS_CRX	9	R/W	Disable carrier extension for GMII and MII modes	0x1
MIS_INBAND_SIG_DIS	7:6	R/W	Disables to send link diagnostic information during inband/idle mode for RGMII. Bit 0 controls RGMII optional inband signalling while Bit 1 controls RGMII required inband signalling.	0x0
RCLKOUT_DIS	5	R/W	Disables Receive Clock Output	0x0
RX_BIT_REV	4	R/W	Control bit to reverse bit order of data received on 1000/100 RX PCS interface. In case of RGMII mode- bit reversal is done within outgoing 4 bits of RGMII RX data	0x0
TX_BIT_REV	3	R/W	Control bit to reverse bit order of data received on TX MIS interface.In case of RGMII mode- bit reversal is done within incoming 4 bits of RGMII TX data	0x0
REV_MII_MODE_EN	2	R/W	Enables Reverse MII mode where the clocks are controlled by MAC side rather than PHY This register field also is used to reflect the EFUSE configuration for rev_mii_mode_en if EFUSE mode is used to configure Reverse MII mode.	0x0
MAC_MODE_SEL	1:0	R/W	MAC Operating Mode Select: 2'b00 - GMII/MII; 2'b01 - RGMII; 2'b10 - RMII; 2'b11 - SGMII This register field also is used to reflect the EFUSE configuration for mac_mode_sel if EFUSE mode is used to configure mac_mode_sel	0x0

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1.8.1.2 mis_regs:mis_regs:MIS_DLL_CFG_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 189. Fields in MIS_DLL_CFG_REG0

Field Name	Bit	Access	Description	Default
TX_DLL_DELAY_EN	15	R/W	Control bit to Enable/disable TX DLL delay line in signal path- When asserted HIGH - 2ns delayed clock is sent as output while when asserted LOW input clock is seamlessly transmitted without any delay element This needs to be asserted HIGH when mac_mode_sel is configured to select RGMII mode. This register field also is used to reflect the EFUSE configuration for rgmii_tx_id if EFUSE mode is used to configure.	0x0
TX_DLL_TAP_ADJ	14:8	R/W	Control register to adjust Transmit DLL tuned tap to compensate output mux delay. The preferred value for default mode of operation is	0x1B
TX_DLL_TAP_SEL	7:1	R/W	Control register to select tap element to prevent glitches that might occur when delay is first locked for Transmit DLL	0x43
TX_DLL_EN	0	R/W	Enable signal for TX DLL tuning FSM for balancing clocks during RGMII mode. This needs to be asserted HIGH when mac_mode_sel is configured to select RGMII mode. This register field also is used to reflect the EFUSE configuration for rgmii_tx_id if EFUSE mode is used to configure.	0x0

1.8.1.3 mis_regs:mis_regs:MIS_DLL_CFG_REG1

Parent: mis_regs:mis_regs

Instances: 1

Table 190. Fields in MIS_DLL_CFG_REG1

Field Name	Bit	Access	Description	Default
RX_DLL_DELAY_EN	15	R/W	Control bit to Enable/disable RX DLL delay line in signal path; When asserted HIGH - 2ns delayed clock is sent as output while when asserted LOW input clock is seamlessly transmitted without any delay element. This needs to be asserted HIGH when mac_mode_sel is configured to select RGMII mode. This register field also is used to reflect the EFUSE configuration for rgmii_rx_id if EFUSE mode is used to configure.	0x0
RX_DLL_TAP_ADJ	14:8	R/W	Control register to adjust Receive DLL tuned tap to compensate output mux delay	0x1B
RX_DLL_TAP_SEL	7:1	R/W	Control register to select tap element to prevent glitches that might occur when delay	0x43

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Field Name	Bit	Access	Description	Default
			is first locked for Receive DLL	
RX_DLL_EN	0	R/W	Enable signal for RX DLL tuning FSM for balancing clocks during RGMII mode.This needs to be asserted HIGH when mac_mode_sel is configured to select RGMII mode. This register field also is used to reflect the EFUSE configuration for rgmii_rx_id if EFUSE mode is used to configure.	0x0

1.8.1.4 mis_regs:mis_regs:MIS_CFG_REG2

Parent: mis_regs:mis_regs

Instances: 1

Table 191. Fields in MIS_CFG_REG2

Field Name	Bit	Access	Description	Default
RESERVED	15	R/W	Must be set to its default.	0x1
MIS_FE_LPBK_CLK_MODE	10	R/W	Far End Loopback Clock Mode Control. When set to 0 RX follows TX (default) and when asserted to 1 TX follows RX	0x1
RESERVED	9	R/W	Must be set to its default.	0x1
RESERVED	7	R/W	Must be set to its default.	0x1
MIS_IO_LPBK_CTRL	5:4	R/W	Pad I/O loopback Mode control for RX to be looped back onto TX MIS interface	0x0
MIS_IO_LPBK_EN	3	R/W	Pad I/O loopback enable for RX to be looped back onto TX MIS interface	0x0
MIS_FE_LPBK_EN	2	R/W	Far End Loopback Enable	0x0
MIS_NE_LPBK_EN	1	R/W	Near End Loopback Enable	0x0
MAC_SUP_CLK_EN	0	R/W	Control to enable MAC supplied clock to drive RX PCS MIS interface	0x0

1.8.1.5 mis_regs:mis_regs:MIS_INT_EN_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 192. Fields in MIS_INT_EN_REG0

Field Name	Bit	Access	Description	Default
MIS_INT_EN	15	R/W	Interrupt enable for MAC Interface Toplevel Interrupt	0x0
RX_PREMPT_PKT_INT_EN	9	R/W	Interrupt enable for tracking preemption packet counter in Receive direction	0x0
WOL_INT_EN	7	R/W	Interrupt enable for wake_on_lan error interrupt	0x0
RX_PKT_ACTIVE_INT_EN	6	R/W	Interrupt enable for receive Packet Counter	0x0
TX_PKT_ACTIVE_INT_EN	5	R/W	Interrupt enable for Transmit Packet Counter	0x0

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Field Name	Bit	Access	Description	Default
MIS_TX_CRCERR_INT_EN	4	R/W	Interrupt enable for indicating TX CRC Error detected by MIS	0x0
MIS_RX_CRCERR_INT_EN	3	R/W	Interrupt enable for indicating RX CRC Error detected by MIS	0x0
MIS_RX_ERR_INT_EN	2	R/W	Interrupt enable for indicating RX_ER transmitted by PCS towards MIS	0x0
MIS_RX_FIFO_STAT_INT_EN	1	R/W	Interrupt enable for overflow/underflow indication triggered by RX FIFOs present in MIS	0x0
MIS_TX_FIFO_STAT_INT_EN	0	R/W	Interrupt enable for overflow/underflow indication triggered by TX FIFOs present in MIS	0x0

1.8.1.6 mis_regs:mis_regs:MIS_INT_STAT_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 193. Fields in MIS_INT_STAT_REG0

Field Name	Bit	Access	Description	Default
MIS_INT	15	Sticky	Interrupt status for MAC Interface Wrapper Toplevel	0x0
TX_PREMPT_PKT_INT	10	Sticky	Interrupt status indicator for TX preemption packet Counter	0x0
RX_PREMPT_PKT_INT	9	Sticky	Interrupt status indicator for RX preemption packet Counter	0x0
WOL_INT	7	Sticky	Interrupt status for wake_on_lan mode	0x0
RX_PKT_ACTIVE_INT	6	Sticky	Interrupt status indicator forReceive Packet Counter	0x0
TX_PKT_ACTIVE_INT	5	Sticky	Interrupt status indicator for Transmit Packet Counter	0x0
MIS_TX_CRCERR_INT	4	Sticky	Interrupt status for indicating CRC Error detected by MIS	0x0
MIS_RX_CRCERR_INT	3	Sticky	Interrupt status for indicating CRC Error detected by MIS	0x0
MIS_RX_ERR_INT	2	Sticky	Interrupt status for indicating RX packet Error detected by MIS	0x0
MIS_RX_FIFO_STAT_INT	1	Sticky	Interrupt status for overflow/underflow indication triggered by RX FIFOs present in MIS	0x0
MIS_TX_FIFO_STAT_INT	0	Sticky	Interrupt status for overflow/underflow indication triggered by TX FIFOs present in MIS	0x0

1.8.1.7 mis_regs:mis_regs:MIS_PKT_STAT_REG0

Parent: mis_regs:mis_regs

Instances: 1

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Table 194. Fields in MIS_PKT_STAT_REG0

Field Name	Bit	Access	Description	Default
TX_PKTCNTR_ACTIVE	15	R/O	Active status indicator for Transmit Packet Counter Note: Reading this field will clear it.	0x0
TX_PKTCNTR	13:0	R/O	Free running Counter (modulo 10K) that tracks good packet status in TX direction. The register status gets cleared on reading the counter value Note: Reading this field will clear it.	0x0000

1.8.1.8 mis_regs:mis_regs:MIS_PKT_STAT_REG1

Parent: mis_regs:mis_regs

Instances: 1

Table 195. Fields in MIS_PKT_STAT_REG1

Field Name	Bit	Access	Description	Default
RX_PKTCNTR_ACTIVE	15	R/O	Active status indicator for Receive Packet Counter Note: Reading this field will clear it.	0x0
RX_ERR_ACTIVE	14	R/O	Active status indicator for Receive Error Counter Note: Reading this field will clear it.	0x0
RX_PKTCNTR	13:0	R/O	Free running Counter (modulo 10K) that tracks good packet status in RX direction . The register status gets cleared on reading the counter value Note: Reading this field will clear it.	0x0000

1.8.1.9 mis_regs:mis_regs:MIS_PKT_STAT_REG2

Parent: mis_regs:mis_regs

Instances: 1

Table 196. Fields in MIS_PKT_STAT_REG2

Field Name	Bit	Access	Description	Default
RX_PREMPT_PKTCNTR_ACTIVE	15	R/O	Active status indicator for counter that tracks Preemption Packets received on RX direction Note: Reading this field will clear it.	0x0
RX_PREMPT_PKTCNTR	13:0	R/O	Free running Counter (modulo 10K) that tracks Preemption Packets received on RX direction. The register status gets cleared on reading the counter value Note: Reading this field will clear it.	0x0000

1.8.1.10 mis_regs:mis_regs:MIS_PKT_STAT_REG3

Parent: mis_regs:mis_regs

Instances: 1

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Table 197. Fields in MIS_PKT_STAT_REG3

Field Name	Bit	Access	Description	Default
RX_ERR_CNTR	15:0	R/O	Counter that tracks errored packets detected by PCS that triggers RX_ERR towards MIS in RX direction. This register saturates at 0xFFFF in the case of more errors.The register status gets cleared on reading the counter value Note: Reading this field will clear it.	0x0000

1.8.1.11 mis_regs:mis_regs:MIS_PKT_STAT_REG4

Parent: mis_regs:mis_regs

Instances: 1

Table 198. Fields in MIS_PKT_STAT_REG4

Field Name	Bit	Access	Description	Default
TX_CRCERR_ACTIVE	15	R/O	Active status indicator for Transmit CRC Error Counter Note: Reading this field will clear it.	0x0
TX_CRCERR_CNTR	7:0	R/O	Counter that tracks CRC errored packets on TX direction. The register status gets cleared on reading the counter value.This register saturates at 0xFF in the case of more errors.	0x00

1.8.1.12 mis_regs:mis_regs:MIS_PKT_STAT_REG5

Parent: mis_regs:mis_regs

Instances: 1

Table 199. Fields in MIS_PKT_STAT_REG5

Field Name	Bit	Access	Description	Default
RX_CRCERR_ACTIVE	15	R/O	Active status indicator for Receive CRC Error Counter Note: Reading this field will clear it.	0x0
RX_CRCERR_CNTR	7:0	R/O	Counter that tracks CRC errored packets on RX direction.The register status gets cleared on reading the counter value.This register saturates at 0xFF in the case of more errors. Note: Reading this field will clear it.	0x00

1.8.1.13 mis_regs:mis_regs:MIS_PKT_STAT_REG6

Parent: mis_regs:mis_regs

Instances: 1

Table 200. Fields in MIS_PKT_STAT_REG6

Field Name	Bit	Access	Description	Default
SGMII_RX_ALIGNERR	15	R/O	Active status indicator for Receive Align Error Counter Note: Reading this field will clear it.	0x0
SGMII_RX_ALIGNERR_CNTR	7:0	R/O	Counter that tracks align errored packets on RX direction for SGMII MII2GMII Translation interface.The register status gets cleared on	0x00

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Field Name	Bit	Access	Description	Default
			reading the counter value.This register saturates at 0xFF in the case of more errors. Note: Reading this field will clear it.	

1.8.1.14 mis_regs:mis_regs:MIS_EPG_CFG1

Parent: mis_regs:mis_regs

Instances: 1

Table 201. Fields in MIS_EPG_CFG1

Field Name	Bit	Access	Description	Default
EPG_RUN_STOP_EN	15	R/W	Run or Stop Control for EPG. 1: Run EPG either in single/continuos mode; 0: Stop EPG operation	0x0
EPG_FCS_GEN_EN	14	R/W	EPG Good/Bad FCS Generation Enable. 0: Good FCS ; 1: Bad FCS	0x0
EPG_SIG_CNT_EN	13	R/W	Signature Counter Enable. When asserted HIGH, EPG signature counter increments the counter on every generated EPG frame.	0x0
EPG_RUN_STATUS	8	R/O	EPG Run Mode Status - Indicates whether EPG packets are currently under transit. Note: Reading this field will clear it.	0x0
EPG_RAND_PKT_LEN_EN	6	R/W	Enables random packet length mode for generating EPG packets	0x0
EPG_TESTMODE_EN	4	R/W	EPG Testmode Enable	0x0
EPG_RAND_PYLD_EN	3	R/W	EPG Random Payload Enable. When set to 1, Random payload is selected and when set to 0, Fixed payload. In case of fixed payload, epg_payload is used to replicate across the generated EPG packet nased on epg_pkt_len configuration.	0x0
EPG_MULTI_BURST_EN	2	R/W	Enables EPG to generate Multiple bursts of EPG packets	0x0
EPG_CONT_RUN_EN	1	R/W	EPG Continuos Run Enable. 1: Run EPG in Continuous mode ; 0: Run EPG for single iteration	0x0
EPG_EN	0	R/W	EPG Enable for all modes	0x0

1.8.1.15 mis_regs:mis_regs:MIS_EPG_PYLD

Parent: mis_regs:mis_regs

Instances: 1

Table 202. Fields in MIS_EPG_PYLD

Field Name	Bit	Access	Description	Default
EPG_PAYLOAD	15:0	R/W	EPG Packet Payload Data. Based on epg_pkt_len this payload data gets replicated across the packet	0x0000

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1.8.1.16 mis_regs:mis_regs:MIS_EPG_BRST_CNT

Parent: mis_regs:mis_regs

Instances: 1

Table 203. Fields in MIS_EPG_BRST_CNT

Field Name	Bit	Access	Description	Default
EPG_BURST_CNT	15:0	R/W	Used to configure the number of packets sent in burst during EPG mode	0x000A

1.8.1.17 mis_regs:mis_regs:MIS_EPG_MBR_CNT

Parent: mis_regs:mis_regs

Instances: 1

Table 204. Fields in MIS_EPG_MBR_CNT

Field Name	Bit	Access	Description	Default
EPG_MBURST_CNT	15:0	R/W	Used to configure the number of bursts sent in a sequence during EPG mode	0x001E

1.8.1.18 mis_regs:mis_regs:MIS_EPG_IPG_CFG

Parent: mis_regs:mis_regs

Instances: 1

Table 205. Fields in MIS_EPG_IPG_CFG

Field Name	Bit	Access	Description	Default
EPG_IPG	15:0	R/W	EPG Inter packet gap - the resolution unit for this is determined by epg_ipg_type	0x000C

1.8.1.19 mis_regs:mis_regs:MIS_EPG_BRST_IPG

Parent: mis_regs:mis_regs

Instances: 1

Table 206. Fields in MIS_EPG_BRST_IPG

Field Name	Bit	Access	Description	Default
EPG_BURST_IPG	15:0	R/W	Inter packet gap between each burst of EPG packets	0x0000

1.8.1.20 mis_regs:mis_regs:MIS_EPG_UMB_IPG

Parent: mis_regs:mis_regs

Instances: 1

Table 207. Fields in MIS_EPG_UMB_IPG

Field Name	Bit	Access	Description	Default
EPG_UP_MBURST_IPG	15:0	R/W	Upper word (bits [31:16]) of Inter-burst gap between multiple bursts	0x0000

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1.8.1.21 mis_regs:mis_regs:MIS_EPG_LMB_IPG

Parent: mis_regs:mis_regs

Instances: 1

Table 208. Fields in MIS_EPG_LMB_IPG

Field Name	Bit	Access	Description	Default
EPG_LOW_MBURST_IPG	15:0	R/W	Lower word (bits [15:0])of Inter-burst gap between multiple bursts	0x0000

1.8.1.22 mis_regs:mis_regs:MIS_EPG_PLEN

Parent: mis_regs:mis_regs

Instances: 1

Table 209. Fields in MIS_EPG_PLEN

Field Name	Bit	Access	Description	Default
EPG_PKT_LEN	15:0	R/W	EPG Packet Length (valid range - 64 bytes to 1536 bytes in case of normal packet size - beyond this size is considered to be jumbo packet size)	0x007D

1.8.1.23 mis_regs:mis_regs:MIS_EPG_RPKT_LEN

Parent: mis_regs:mis_regs

Instances: 1

Table 210. Fields in MIS_EPG_RPKT_LEN

Field Name	Bit	Access	Description	Default
EPG_RAND_PKT_LEN	15:0	R/O	packet length generated by internal LFSR generator and subjected to epg_rand_pkt_len_mask that is used to select random packet size for EPG packets	0x0000

1.8.1.24 mis_regs:mis_regs:MIS_EPG_MIN_RPKT_LEN

Parent: mis_regs:mis_regs

Instances: 1

Table 211. Fields in MIS_EPG_MIN_RPKT_LEN

Field Name	Bit	Access	Description	Default
EPG_MIN_RAND_PKT_LEN	15:0	R/W	Note: epg_pkt_len can be used to determine max pkt length in case of random mode - This register must not be programmed with a value less than 0x12.	0x0040

1.8.1.25 mis_regs:mis_regs:MIS_EPG_CFG2

Parent: mis_regs:mis_regs

Instances: 1

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Table 212. Fields in MIS_EPG_CFG2

Field Name	Bit	Access	Description	Default
RESERVED	15:8	R/W	Must be set to its default.	0x11
EPG_IP_EN	2	R/W	EPG IPV4 Packet Mode Enable	0x0
EPG_UDP_OVR_EN	1	R/W	UDP Packet Override Enable	0x0
EPG_ETHR_TYPE_OVR_EN	0	R/W	EPG Ether type Field Override Enable. TBD for more details	0x0

1.8.1.26 mis_regs:mis_regs:MIS_EPG_JMB_PSIZE

Parent: mis_regs:mis_regs

Instances: 1

Table 213. Fields in MIS_EPG_JMB_PSIZE

Field Name	Bit	Access	Description	Default
EPG_JUMBO_PKT_SIZE	15:0	R/W	EPG Jumbo Packet Size	0x05DC

1.8.1.27 mis_regs:mis_regs:MIS_EPG_ETHR_TYP

Parent: mis_regs:mis_regs

Instances: 1

Table 214. Fields in MIS_EPG_ETHR_TYP

Field Name	Bit	Access	Description	Default
EPG_ETHR_TYPE	15:0	R/W	EPG Ether Type Field (L/T field when epg_ethr_type_ovr_en is enabled)	0x88F7

1.8.1.28 mis_regs:mis_regs:MIS_EPG_UW_MSRC

Parent: mis_regs:mis_regs

Instances: 1

Table 215. Fields in MIS_EPG_UW_MSRC

Field Name	Bit	Access	Description	Default
EPG_MAC_SA_47_32	15:0	R/W	EPG MAC Source Address [47:32]	0xFFFF

1.8.1.29 mis_regs:mis_regs:MIS_EPG_MW_MSRC

Parent: mis_regs:mis_regs

Instances: 1

Table 216. Fields in MIS_EPG_MW_MSRC

Field Name	Bit	Access	Description	Default
EPG_MAC_SA_31_16	15:0	R/W	EPG MAC Source Address [31:16]	0xFFFF

1.8.1.30 mis_regs:mis_regs:MIS_EPG_LW_MSRC

Parent: mis_regs:mis_regs

Instances: 1

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Table 217. Fields in MIS_EPG_LW_MSRC

Field Name	Bit	Access	Description	Default
EPG_MAC_SA_15_0	15:0	R/W	EPG MAC Source Address [15:0]	0xFFF0

1.8.1.31 mis_regs:mis_regs:MIS_EPG_UW_MDST

Parent: mis_regs:mis_regs

Instances: 1

Table 218. Fields in MIS_EPG_UW_MDST

Field Name	Bit	Access	Description	Default
EPG_MAC_DA_47_32	15:0	R/W	EPG MAC Destination Address [47:32]	0xFFFF

1.8.1.32 mis_regs:mis_regs:MIS_EPG_MW_MDST

Parent: mis_regs:mis_regs

Instances: 1

Table 219. Fields in MIS_EPG_MW_MDST

Field Name	Bit	Access	Description	Default
EPG_MAC_DA_31_16	15:0	R/W	EPG MAC Destination Address [31:16]	0xFFFF

1.8.1.33 mis_regs:mis_regs:MIS_EPG_LW_MDST

Parent: mis_regs:mis_regs

Instances: 1

Table 220. Fields in MIS_EPG_LW_MDST

Field Name	Bit	Access	Description	Default
EPG_MAC_DA_15_0	15:0	R/W	EPG MAC Destination Address [15:0]	0xFFF1

1.8.1.34 mis_regs:mis_regs:MIS_EPG_UW_ISRC

Parent: mis_regs:mis_regs

Instances: 1

Table 221. Fields in MIS_EPG_UW_ISRC

Field Name	Bit	Access	Description	Default
EPG_IP_SA_31_16	15:0	R/W	EPG IP Source Address bits [31:16]	0xC0A8

1.8.1.35 mis_regs:mis_regs:MIS_EPG_LW_ISRC

Parent: mis_regs:mis_regs

Instances: 1

Table 222. Fields in MIS_EPG_LW_ISRC

Field Name	Bit	Access	Description	Default
EPG_IP_SA_15_0	15:0	R/W	EPG IP Source Address bits [15:0]	0x7600

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1.8.1.36 mis_regs:mis_regs:MIS_EPG_UW_IDST

Parent: mis_regs:mis_regs

Instances: 1

Table 223. Fields in MIS_EPG_UW_IDST

Field Name	Bit	Access	Description	Default
EPG_IP_DA_31_16	15:0	R/W	EPG IP Destination Address bits [31:16]	0xC0A8

1.8.1.37 mis_regs:mis_regs:MIS_EPG_LW_IDST

Parent: mis_regs:mis_regs

Instances: 1

Table 224. Fields in MIS_EPG_LW_IDST

Field Name	Bit	Access	Description	Default
EPG_IP_DA_15_0	15:0	R/W	EPG IP Destination Address bits [15:0]	0x7601

1.8.1.38 mis_regs:mis_regs:MIS_EPG_UDP_SRC

Parent: mis_regs:mis_regs

Instances: 1

Table 225. Fields in MIS_EPG_UDP_SRC

Field Name	Bit	Access	Description	Default
EPG_UDP_SRC_PORT	15:0	R/W	UDP Source Port (valid only when epg_udp_ovr_en is enabled)	0x0000

1.8.1.39 mis_regs:mis_regs:MIS_EPG_UDP_DST

Parent: mis_regs:mis_regs

Instances: 1

Table 226. Fields in MIS_EPG_UDP_DST

Field Name	Bit	Access	Description	Default
EPG_UDP_DST_PORT	15:0	R/W	UDP Destination Port	0x013F

1.8.1.40 mis_regs:mis_regs:MIS_EPG_SIG_CNT

Parent: mis_regs:mis_regs

Instances: 1

Table 227. Fields in MIS_EPG_SIG_CNT

Field Name	Bit	Access	Description	Default
EPG_SIG_CNT	15:0	R/W	EPG Signature Counter. This register is used to initialise Signature counter when epg_sig_cnt_init_en is asserted HIGH during write operation. This register is also used to track the status of EPG signature counter during read operation.	0x0000

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1.8.1.41 mis_regs:mis_regs:MIS_EPG_SIG_CFG

Parent: mis_regs:mis_regs

Instances: 1

Table 228. Fields in MIS_EPG_SIG_CFG

Field Name	Bit	Access	Description	Default
EPG_SIG_CNT_PYLD_OFFSET	7:0	R/W	Signature Counter Payload Offset	0x1E

1.8.1.42 mis_regs:mis_regs:MIS_EPG_IDL_LEN

Parent: mis_regs:mis_regs

Instances: 1

Table 229. Fields in MIS_EPG_IDL_LEN

Field Name	Bit	Access	Description	Default
EPG_IDLE_LENGTH	15:0	R/W	EPG Idle Length	0x000C

1.8.1.43 mis_regs:mis_regs:MIS_EPG_UW_LPI

Parent: mis_regs:mis_regs

Instances: 1

Table 230. Fields in MIS_EPG_UW_LPI

Field Name	Bit	Access	Description	Default
EPG_LPI_LEN_31_16	15:0	R/W	EPG LPI Length upper word	0x0000

1.8.1.44 mis_regs:mis_regs:MIS_EPG_LW_LPI

Parent: mis_regs:mis_regs

Instances: 1

Table 231. Fields in MIS_EPG_LW_LPI

Field Name	Bit	Access	Description	Default
EPG_LPI_LEN_15_0	15:0	R/W	EPG LPI Length lower word	0x0000

1.8.1.45 mis_regs:mis_regs:MIS_PKT_CHK_CFG0

Parent: mis_regs:mis_regs

Instances: 1

Table 232. Fields in MIS_PKT_CHK_CFG0

Field Name	Bit	Access	Description	Default
TX_PREMPT_PKT_CHK_EN	1	R/W	Enable bit to check for pre-emption packet processing in TX direction	0x1
RX_PREMPT_PKT_CHK_EN	0	R/W	Enable bit to check for pre-emption packet processing in RX direction	0x1

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1.8.1.46 mis_regs:mis_regs:MIS_PKT_STAT_REG7

Parent: mis_regs:mis_regs

Instances: 1

Table 233. Fields in MIS_PKT_STAT_REG7

Field Name	Bit	Access	Description	Default
TX_PREMPT_PKTCNTR_ACTIVE	15	R/O	Active status indicator for counter that tracks Preemption Packets received on TX direction Note: Reading this field will clear it.	0x0
TX_PREMPT_PKTCNTR	13:0	R/O	Free Counter (modulo 10K) that tracks Preemption Packets received on TX direction. The register status gets cleared on reading the counter value Note: Reading this field will clear it.	0x0000

1.8.1.47 mis_regs:mis_regs:MIS_WOL_CFG0

Parent: mis_regs:mis_regs

Instances: 1

Table 234. Fields in MIS_WOL_CFG0

Field Name	Bit	Access	Description	Default
WOL_EN	6	R/W	enable bit for Wake-on-LAN feature	0x0
SECURE_ON_EN	5	R/W	Enable bit for Secure-on Password	0x0
PSWD_LEN	4	R/W	MAC Secure-On Pasword Length Control	0x0
REPT_CNT	3:0	R/W	Wake on LAN Address Repetition counter . This counter is used to trigger WOL Interrupt if Wake-on-LAN MAC address repeats for the value programmed in this counter within the payload of magic packet.	0xF

1.8.1.48 mis_regs:mis_regs:MIS_WOL_UW_PSWD

Parent: mis_regs:mis_regs

Instances: 1

Table 235. Fields in MIS_WOL_UW_PSWD

Field Name	Bit	Access	Description	Default
MAC_SEC_UP_PSWD	15:0	R/W	Wake on LAN Secure-On Password[47:32]	0x0000

1.8.1.49 mis_regs:mis_regs:MIS_WOL_MW_PSWD

Parent: mis_regs:mis_regs

Instances: 1

Table 236. Fields in MIS_WOL_MW_PSWD

Field Name	Bit	Access	Description	Default
MAC_SEC_MID_PSWD	15:0	R/W	Wake on LAN Secure-On Password[31:16]	0x0000

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1.8.1.50 mis_regs:mis_regs:MIS_WOL_LW_PSWD

Parent: mis_regs:mis_regs

Instances: 1

Table 237. Fields in MIS_WOL_LW_PSWD

Field Name	Bit	Access	Description	Default
MAC_SEC_LOW_PSWD	15:0	R/W	Wake on LAN Secure-On Password[15:0]	0x0000

1.8.1.51 mis_regs:mis_regs:MIS_WOL_UW_ADDR

Parent: mis_regs:mis_regs

Instances: 1

Table 238. Fields in MIS_WOL_UW_ADDR

Field Name	Bit	Access	Description	Default
MAC_UP_ADDR	15:0	R/W	Upper Word of Wake on LAN MAC Address[47:32] used for determining WOL interrupt	0x0000

1.8.1.52 mis_regs:mis_regs:MIS_WOL_MW_ADDR

Parent: mis_regs:mis_regs

Instances: 1

Table 239. Fields in MIS_WOL_MW_ADDR

Field Name	Bit	Access	Description	Default
MAC_MID_ADDR	15:0	R/W	Middle word of Wake on LAN MAC Address[31:16] used for determining WOL interrupt	0x0000

1.8.1.53 mis_regs:mis_regs:MIS_WOL_LW_ADDR

Parent: mis_regs:mis_regs

Instances: 1

Table 240. Fields in MIS_WOL_LW_ADDR

Field Name	Bit	Access	Description	Default
MAC_LOW_ADDR	15:0	R/W	Lower word of Wake on LAN MAC Address[15:0] used for determining WOL interrupt	0x0000

1.8.1.54 mis_regs:mis_regs:MIS_SGMII_INTF_CFG0

Parent: mis_regs:mis_regs

Instances: 1

Table 241. Fields in MIS_SGMII_INTF_CFG0

Field Name	Bit	Access	Description	Default
SGMII_RX_PREAMBLE_MODE	3:2	R/W	Control bits for mode	0x0

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Field Name	Bit	Access	Description	Default
SGMII_RX_ALIGNERR_CTRL	1:0	R/W	Control bits for alignerr mode	0x0

1.8.1.55 mis_regs:mis_regs:MIS_SGMII_INTF_STS

Parent: mis_regs:mis_regs

Instances: 1

Table 242. Fields in MIS_SGMII_INTF_STS

Field Name	Bit	Access	Description	Default
RSVD_SGMII_INTF_STS0_0	15:1	R/O	Reserved status field for SGMII interface	0x0000
SGMII_NIBL_INSERTED	0	R/O	Status bit to indicate nibble insertion during MII2GMII	0x0

1.8.1.56 mis_regs:mis_regs:MIS_DBG_CFG_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 243. Fields in MIS_DBG_CFG_REG0

Field Name	Bit	Access	Description	Default
MIS_DBG_SMI_MODE_EN	15	R/W	MIS critical debug mode enable. . When asserted HIGH, the lower 8 bits of both TX and RX debug bus are used to be sent as part of 16-bit crtical debug MIS data. When set to 0, mis_dbg_smi_mode_sel is used to select the 16-bit TX or RX debug information. mis_dbg_smi_mode_sel is used to determine the respective debug points of intrest.	0x0
TX_MIS_DBG_MODE_FRZ	7	R/W	control bit to freeze debug data output on TX direction. Read operation of tx_dbg_intf_data will cause to clear this register.	0x0
RX_MIS_DBG_MODE_FRZ	5	R/W	Control bit to freeze debug data output on RX direction. Read operation of rx_dbg_intf_data will cause to clear this register.	0x0
MIS_DBG_SMI_MODE_SEL	4:0	R/W	Debug mode select bus . MSB bit of this register filed is used to select between 16-bit TX and RX debug data. MSB = 1 indicates to select debug data from TX direction and MSB=0 for RX direction. In case of 64-bit debug mode, bit 4 is used to select TX or RX direction while bits [3:2] of this register are used to transmit 64-bits of TX and RX debug data as part of 64bit debug bus. Following is the encoding for each of the TX or RX direction that uses mis_dbg_smi_mode_sel[3:0]. 5b00000 - PMAC_RX_INTF Debug status 1(MISC) - 5'b10000 - PMAC_TX_INTF Debug Status 1(MISC) 5b00001 - PMAC_RX_INTF Debug status 2(MISC) - 5'b10001 - PMAC_TX_INTF Debug Status 2(RMII_RGMII)	0x00

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Field Name	Bit	Access	Description	Default
			5b00010 - PMAC_RX_INTF Debug status 3(RGMII) - 5'b10010 - PMAC_TX_INTF Debug Status 3(RGMII) 5b00011 - PMAC_RX_INTF Debug status 4(SGMII) - 5'b10011 - PMAC_TX_INTF Debug Status 4(SGMII) 5b00100 - RX Pkt Checker Debug status 1(Good) - 5'b10100 - TX Pkt Checker Debug Status 1 (Good) 5b00101 - RX Pkt Checker Debug status 2(Bad) - 5'b10101 - TX Pkt Checker Debug Status 2 (Bad) 5b00110 - RX Pkt Checker Debug status 3(Misc) - 5'b10110 - TX Pkt Checker Debug Status 3 (Misc) 5b00111 - RX Pkt Checker Debug status 4 (Misc) - 5'b10111 - TX Pkt Checker Debug Status 4 (Misc) 5b00100 - 16'h2222 - 5'b10100 - EPG Debug Status 1 5b00101 - WOL Debug status 1 - 5'b10101 - EPG Debug Status 2 5b00110 - WOL Debug status 2 - 5'b10110 - EPG Debug Status 3 5b00111 - 16'h5555 - 5'b10111 - 16'hCCCC 5b00100 - RX FIFO Debug status 1 (Wr) - 5'b10100 - TX FIFO Debug Status 1 (Wr) 5b00101 - RX FIFO Debug status 2 (Rd) - 5'b10101 - TX FIFO Debug Status 2 (Rd) 5b00110 - RX FIFO Debug status 3 (Misc) - 5'b10110 - TX FIFO Debug Status 3 (Misc) 5b00111 - RX FIFO Debug status 4 (Misc) - 5'b10111 - TX FIFO Debug Status 4 (Misc)	

1.8.1.57 mis_regs:mis_regs:MIS_DBG_STAT_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 244. Fields in MIS_DBG_STAT_REG0

Field Name	Bit	Access	Description	Default
RX_DBG_INTF_RD_RDY	1	R/O	Control signal to indicate ready status of Receive debug data bus for read operation. Note: Reading this field will clear it.	0x0
TX_DBG_INTF_RD_RDY	0	R/O	Control signal to indicate ready status of Transmit debug data bus for read operation. Note: Reading this field will clear it.	0x0

1.8.1.58 mis_regs:mis_regs:MIS_TX_DBG_STAT_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 245. Fields in MIS_TX_DBG_STAT_REG0

Field Name	Bit	Access	Description	Default
TX_DBG_INTF_DATA	15:0	R/O	16-bit data snapshot of Transmit debug data bus	0x0000

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Field Name	Bit	Access	Description	Default
			Note: Reading this field will clear it.	

1.8.1.59 mis_regs:mis_regs:MIS_RX_DBG_STAT_REG0

Parent: mis_regs:mis_regs

Instances: 1

Table 246. Fields in MIS_RX_DBG_STAT_REG0

Field Name	Bit	Access	Description	Default
RX_DBG_INTF_DATA	15:0	R/O	16-bit data snapshot of Receive debug data bus Note: Reading this field will clear it.	0x0000

1.8.1.60 mis_regs:mis_regs:MIS_FIFO_FLOW_STAT0

Parent: mis_regs:mis_regs

Instances: 1

Table 247. Fields in MIS_FIFO_FLOW_STAT0

Field Name	Bit	Access	Description	Default
GMII_RX_FIFO_UFLOW	7	R/O	FIFO underflow status detected in RX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
GMII_RX_FIFO_OFLOW	6	R/O	FIFO overflow status detected in RX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
MII_RX_FIFO_UFLOW	5	R/O	FIFO underflow status detected in RX direction for 100-PCS mode Note: Reading this field will clear it.	0x0
MII_RX_FIFO_OFLOW	4	R/O	FIFO overflow status detected in RX direction for 100-PCS mode Note: Reading this field will clear it.	0x0
GMII_TX_FIFO_UFLOW	3	R/O	FIFO underflow status detected in TX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
GMII_TX_FIFO_OFLOW	2	R/O	FIFO overflow status detected in TX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
MII_TX_FIFO_UFLOW	1	R/O	FIFO underflow status detected in TX direction for 100-PCS mode Note: Reading this field will clear it.	0x0
MII_TX_FIFO_OFLOW	0	R/O	FIFO overflow status detected in TX direction for 100-PCS mode Note: Reading this field will clear it.	0x0

1.8.1.61 mis_regs:mis_regs:MIS_FIFO_FLOW_STAT1

Parent: mis_regs:mis_regs

Instances: 1

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Table 248. Fields in MIS_FIFO_FLOW_STAT1

Field Name	Bit	Access	Description	Default
GMII_RX_FIFO_EMPTY	7	R/O	FIFO Empty status detected in RX direction for 1000-PCS mode Note: Reading this field will clear it.	0x1
GMII_RX_FIFO_FULL	6	R/O	FIFO Full status detected in RX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
MII_RX_FIFO_EMPTY	5	R/O	FIFO Empty status detected in RX direction for 100-PCS mode Note: Reading this field will clear it.	0x1
MII_RX_FIFO_FULL	4	R/O	FIFO Full status detected in RX direction for 100-PCS mode Note: Reading this field will clear it.	0x0
GMII_TX_FIFO_EMPTY	3	R/O	FIFO Empty status detected in TX direction for 1000-PCS mode Note: Reading this field will clear it.	0x1
GMII_TX_FIFO_FULL	2	R/O	FIFO Full status detected in TX direction for 1000-PCS mode Note: Reading this field will clear it.	0x0
MII_TX_FIFO_EMPTY	1	R/O	FIFO Empty status detected in TX direction for 100-PCS mode Note: Reading this field will clear it.	0x1
MII_TX_FIFO_FULL	0	R/O	FIFO Full status detected in TX direction for 100-PCS mode Note: Reading this field will clear it.	0x0

1.8.1.62 mis_regs:mis_regs:MIS_TX_FIFO_PPM

Parent: mis_regs:mis_regs

Instances: 1

Table 249. Fields in MIS_TX_FIFO_PPM

Field Name	Bit	Access	Description	Default
MIS_TX_FIFO_PPM	9:0	R/O	This register field reflects PPM offset reported by FIFO adjustment block for TX FIFO (speed100/1000) based on MAC operating mode.	0x000

1.8.1.63 mis_regs:mis_regs:MIS_RX_FIFO_PPM

Parent: mis_regs:mis_regs

Instances: 1

Table 250. Fields in MIS_RX_FIFO_PPM

Field Name	Bit	Access	Description	Default
MIS_RX_FIFO_PPM	9:0	R/O	This register field reflects PPM offset reported by FIFO adjustment block for RX FIFO (speed100/1000) based on MAC operating mode.	0x000

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1.8.1.64 mis_regs:mis_regs:MIS_PPM_CLK_CNT_LIMIT

Parent: mis_regs:mis_regs

Instances: 1

Table 251. Fields in MIS_PPM_CLK_CNT_LIMIT

Field Name	Bit	Access	Description	Default
MIS_PPM_CLK_CNT_LIMIT	15:0	R/W	This register field reflects max limit of 1000 clock intervals of the clock used in determining measured ppm for TX and RX FIFOs. Default of this register is 100 and the threshold limits are scaled with ref to this register field.	0x0064

1.8.1.65 mis_regs:mis_regs:MIS_TX_FIFO_PPM_THR1_2

Parent: mis_regs:mis_regs

Instances: 1

Table 252. Fields in MIS_TX_FIFO_PPM_THR1_2

Field Name	Bit	Access	Description	Default
MIS_TX_FIFO_PPM_THR2	15:8	R/W	This register field reflects PPM threshold limit 2 for TX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range between fifo_ppm_thresh2 <= measured_ppm <= fifo_ppm_thresh3, then optimal FIFO latency setting is set to 2. Default setting is for jumbo packet size of 16K	0x0C
MIS_TX_FIFO_PPM_THR1	7:0	R/W	This register field reflects PPM threshold limit 1 for TX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range between fifo_ppm_thresh1 <= measured_ppm <= fifo_ppm_thresh2, then optimal FIFO latency setting is set to 1.Default setting is for jumbo packet size of 16K	0x06

1.8.1.66 mis_regs:mis_regs:MIS_TX_FIFO_PPM_THR3

Parent: mis_regs:mis_regs

Instances: 1

Table 253. Fields in MIS_TX_FIFO_PPM_THR3

Field Name	Bit	Access	Description	Default
MIS_TX_FIFO_PPM_THR3	8:0	R/W	This register field reflects PPM threshold limit 3 for TX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range where measured_ppm >= fifo_ppm_thresh3, then optimal FIFO latency setting is set to 3.Default setting is for jumbo packet size of 16K	0x012

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1.8.1.67 mis_regs:mis_regs:MIS_RX_FIFO_PPM_THR1_2

Parent: mis_regs:mis_regs

Instances: 1

Table 254. Fields in MIS_RX_FIFO_PPM_THR1_2

Field Name	Bit	Access	Description	Default
MIS_RX_FIFO_PPM_THR2	15:8	R/W	This register field reflects PPM threshold limit 2 for RX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range between fifo_ppm_thresh2 <= measured_ppm <= fifo_ppm_thresh3, then optimal FIFO latency setting is set to 2. Default setting is for jumbo packet size of 16K for GMII mode	0x0C
MIS_RX_FIFO_PPM_THR1	7:0	R/W	This register field reflects PPM threshold limit 1 for RX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range between fifo_ppm_thresh1 <= measured_ppm <= fifo_ppm_thresh2, then optimal FIFO latency setting is set to 1.Default setting is for jumbo packet size of 16K	0x06

1.8.1.68 mis_regs:mis_regs:MIS_RX_FIFO_PPM_THR3

Parent: mis_regs:mis_regs

Instances: 1

Table 255. Fields in MIS_RX_FIFO_PPM_THR3

Field Name	Bit	Access	Description	Default
MIS_RX_FIFO_PPM_THR3	8:0	R/W	This register field reflects PPM threshold limit 3 for RX FIFO to compare the measured ppm for determining the optimal FIFO latency setting. If measured ppm is in the range where measured_ppm >= fifo_ppm_thresh3, then optimal FIFO latency setting is set to 3.Default setting is for jumbo packet size of 16K	0x012

1.9 dev30_common

Table 256. Register Groups in MMD dev30_common

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
dev30_common	0x00000000	1		Page 84

1.9.1 dev30_common:dev30_common

Parent: dev30_common

Instances: 1

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Table 257. Registers in dev30_common

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
COMMON_REG_1	0x00000C00	1		Page 85
COMMON_DLL_CTRL	0x00000C01	1		Page 85
COMMON_DLL_STATUS	0x00000C02	1		Page 86
COMMON_LED	0x00000C03	1		Page 86
COMMON_LED1_LED0	0x00000C04	1		Page 86
COMMON_LED3_LED2	0x00000C05	1		Page 87
COMMON_TESTBUS	0x00000C06	1		Page 87
COMMON_PGEN_PURPOSE	0x00000C07	1		Page 88
COMMON_PORT_INT_EN_CFG	0x00000C10	1		Page 88
COMMON_PORT_INT_STAT	0x00000C11	1		Page 88
DEBUG_MODE_VIA_SMI	0x0000FFFF	1		Page 89

1.9.1.1 dev30_common:dev30_common:COMMON_REG_1

Parent: dev30_common:dev30_common

Instances: 1

Table 258. Fields in COMMON_REG_1

Field Name	Bit	Access	Description	Default
RSVD_CMN_REG1_0	15:10	R/W	Reserved common Config Field	0x00
RSVD_CMN_REG2_1	9:6	R/W	Reserved common Config Field	0x0
DBG_CLK_SEL	5:2	R/W	Debug clock selection for dbg_clk0,dbg_clk1,dbg_clk2 and dbg_clk3.Each bit is exclusively for one debug clk. When debug_clk_sel[X] = 1 , debug_clk_1 is selected When debug_clk_sel[X]= 0, denug_clk_0 is selected as dbg_clkX where X= [3..0]	0x0
AFE_LPBK_EN	1	R/W	AFE Loopback mode to manipulate some control logic in PCS/PMA/DSP.	0x0
RSVD_CMN_REG3_4	0	R/W	Reserved Common Config field for ADC Test Mode Enable	0x0

1.9.1.2 dev30_common:dev30_common:COMMON_DLL_CTRL

Parent: dev30_common:dev30_common

Instances: 1

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Table 259. Fields in COMMON_DLL_CTRL

Field Name	Bit	Access	Description	Default
RSVD_DLL_CONTROL_5	15:2	R/O	Reserved DLL Control configuration	0x0000
RX_DLL_BIST_START	1	R/W	RX DLL Bist Start	0x0
TX_DLL_BIST_START	0	R/W	TX DLL Bist Start	0x0

1.9.1.3 dev30_common:dev30_common:COMMON_DLL_STATUS

Parent: dev30_common:dev30_common

Instances: 1

Table 260. Fields in COMMON_DLL_STATUS

Field Name	Bit	Access	Description	Default
RSVD_DLL_STATUS_8	15:4	R/O	Reserved DLL status field	0x000
TX_DLL_BIST_END	3	R/O	TX DLL Bist End Status	0x0
TX_DLL_BIST_PASS	2	R/O	TX DLL Bist Pass Status	0x0
RX_DLL_BIST_END	1	R/O	RX DLL Bist End Status	0x0
RX_DLL_BIST_PASS	0	R/O	RX DLL Bist Pass Status	0x0

1.9.1.4 dev30_common:dev30_common:COMMON_LED

Parent: dev30_common:dev30_common

Instances: 1

Table 261. Fields in COMMON_LED

Field Name	Bit	Access	Description	Default
RSVD_CMN_LED_13	15:9	R/O	Reserved LED Config Field	0x00
LED_FAST_BLINK	8	R/W	When this bit is asserted, Blink gets generated once every 100 us instead of 100 ms. It is intended only for debug purposes to speed up blink rate for simulations. This setting is global for all LEDs.	0x0
LED_INVERT	7:4	R/W	Invert the polarity of the LED whose bit is set (LED default polarity is active-low). bit 3 => LED3 bit 2 => LED2 bit 1 => LED1 bit 0 => LED0	0x0
LED_TRI_HIGH	3:0	R/W	When set, tri-state each LED when the LED level is high (otherwise actively drive it high for high levels). Each LED is always driven low when the LED level is low. bit 3 => LED3 bit 2 => LED2 bit 1 => LED1 bit 0 => LED0	0x0

1.9.1.5 dev30_common:dev30_common:COMMON_LED1_LED0

Parent: dev30_common:dev30_common

Instances: 1

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Table 262. Fields in COMMON_LED1_LED0

Field Name	Bit	Access	Description	Default
RSVD_LED1_17	15:13	R/O	Reserved Config Field for LED1	0x0
LED1_MODE_SEL	12:8	R/W	Mode select for LED1. Mode select definitions are the same as for LED0.	0x01
RSVD_LED0_19	7:5	R/O	Reserved Config Field for LED0	0x0
LED0_MODE_SEL	4:0	R/W	Mode select for LED0. For each of the modes, the respective status will propagate to LED0. 5h0 => Link Status with Activity for any speed. 5'h1 => Link Status with Activity for speed 1000. 5'h2 => Link Status with Activity for speed 100 5'h3 => Link Status (without Activity blinking) for any speed. 5'h4 => RESERVED 5'h5 => Local Receiver Status 5'h6 => Remote Receiver Status 5'h7 => Negotiated Speed (ON: 1000; OFF: 100). 5'h8 => Master/Slave Mode (ON: Master; OFF: Slave) 5'h9 => PCS TX Error Status 5'hA => PCS RX Error Status 5'hB => PCS TX Activity 5'hC => PCS RX Activity 5'hD => Wake-on-LAN 5'hE => Force LED OFF 5'hF => Force LED ON 5'h10-5'h1F => RESERVED	0x00

1.9.1.6 dev30_common:dev30_common:COMMON_LED3_LED2

Parent: dev30_common:dev30_common

Instances: 1

Table 263. Fields in COMMON_LED3_LED2

Field Name	Bit	Access	Description	Default
RSVD_LED3_21	15:13	R/O	Reserved Config Field for LED3	0x0
LED3_MODE_SEL	12:8	R/W	Mode select for LED3. Mode select definitions are the same as for LED0.	0x0A
RSVD_LED2_23	7:5	R/O	Reserved Config Field for LED2	0x0
LED2_MODE_SEL	4:0	R/W	Mode select for LED2. Mode select definitions are the same as for LED0.	0x08

1.9.1.7 dev30_common:dev30_common:COMMON_TESTBUS

Parent: dev30_common:dev30_common

Instances: 1

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Table 264. Fields in COMMON_TESTBUS

Field Name	Bit	Access	Description	Default
TESTBUS_SEL_CLK2	15:12	R/W	debug clock selection 2	0x0
TESTBUS_SEL_CLK1	11:8	R/W	debug clock selection 1	0x0
TESTBUS_SEL_DATA	7:3	R/W	test bus source selection	0x00
RSVD_TESTBUS_1_28	2:1	R/O	Reserved	0x0
TESTBUS_EN	0	R/W	testbus enable	0x0

1.9.1.8 dev30_common:dev30_common:COMMON_PGEN_PURPOSE

Parent: dev30_common:dev30_common

Instances: 1

Table 265. Fields in COMMON_PGEN_PURPOSE

Field Name	Bit	Access	Description	Default
RSVD_PGEN_PURPOSE_30	15:1	R/W	Reserved Port General Purpose Register	0x0000
PORT_HW_RESETB_TO_CHIP	0	R/W	HW Reset towards Port	0x1

1.9.1.9 dev30_common:dev30_common:COMMON_PORT_INT_EN_CFG

Parent: dev30_common:dev30_common

Instances: 1

Table 266. Fields in COMMON_PORT_INT_EN_CFG

Field Name	Bit	Access	Description	Default
PHY_INT_EN	15	R/W	Interrupt enable for Phy Port Interrupt	0x0
TC10_INT_EN	14	R/W	Interrupt enable for TC10 Port interrupt	0x0
RSVD_PORT_INT_EN_34	13:10	R/W	Reserved Interrupt Enable	0x0
SQI_ERR_INT_EN	9	R/W	Interrupt enable for SQI Error Interrupt	0x0
ED_OFF_INT_EN	8	R/W	Interrupt enable for Energy Detection ON Status	0x0
ED_ON_INT_EN	7	R/W	Interrupt enable for Energy Detection OFF Status	0x0
RCV_WAKE_INT_EN	6	R/W	Interrupt enable for Receive Wake Interrupt	0x0
RECV_LPS_INT_EN	5	R/W	Interrupt enable for Recovered LPS Interrupt	0x0
WAKE_IN_INT_EN	4	R/W	Interrupt enable for Wake In Interrupt	0x0
ANEG_INT_EN	3	R/W	Interrupt enable for ANEG toplevel Interrupt	0x0
MIS_INT_EN	2	R/W	Interrupt enable for MIS Toplevel Interrupt	0x0
SPD100_PHY_DIG_INT_EN	1	R/W	Interrupt enable for Speed 100 PCS PMA Toplevel Interrupt	0x0
SPD1000_PHY_DIG_INT_EN	0	R/W	Interrupt enable for Speed 1000 PCS PMA Toplevel Interrupt	0x0

1.9.1.10 dev30_common:dev30_common:COMMON_PORT_INT_STAT

Parent: dev30_common:dev30_common

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Instances: 1

Table 267. Fields in COMMON_PORT_INT_STAT

Field Name	Bit	Access	Description	Default
PHY_INT	15	Sticky	Phy Port Interrupt	0x0
TC10_INT	14	Sticky	TC10 Port Interrupt	0x0
RSVD_PORT_INT_47	13:10	Sticky	Reserved Port Interrupt	0x0
SQI_ERR_INT	9	Sticky	SQI Error Interrupt	0x0
ED_OFF_INT	8	Sticky	Energy Detection ON Status	0x0
ED_ON_INT	7	Sticky	Energy Detection OFF Status	0x0
RCV_WAKE_INT	6	Sticky	Receive Wake Interrupt	0x0
RECV_LPS_INT	5	Sticky	Recovered LPS Interrupt	0x0
WAKE_IN_INT	4	Sticky	Wake In Interrupt	0x0
ANEG_INT	3	Sticky	ANEG toplevel Interrupt	0x0
MIS_INT	2	Sticky	MIS Toplevel Interrupt	0x0
SPD100_PHY_DIG_INT	1	Sticky	Speed 100 PCS PMA Toplevel Interrupt	0x0
SPD1000_PHY_DIG_INT	0	Sticky	Speed 1000 PCS PMA Toplevel Interrupt	0x0

1.9.1.11 dev30_common:dev30_common:DEBUG_MODE_VIA_SMI

Parent: dev30_common:dev30_common

Instances: 1

Table 268. Fields in DEBUG_MODE_VIA_SMI

Field Name	Bit	Access	Description	Default
RSVD2_DBG_MODE_58	15:6	R/W	Reserved Config Register	0x000
SMI_TIMEOUT_ENABLE	5	R/W	Once this bit has been enabled timeout counter starts working on ref25 clock. Anytime SMI state machine stays out of PREAM_ST for over 10ms, reset asserts.	0x0
RST_AMBA_COMP	4	R/W	Resets AHB Master Slave Bus Decoder when set to 1. This is not self clearing.Writing 0 is required to clear the reset. Also make sure that debug mode is enabled when accessing this field.	0x0
REG_UNAUTHR_ACCESS	3	R/W	Status register that indicates unauthorized register access. When asserted HIGH, it indicates that one or more registers excluding Common AFE registers or dbg_mode_via_smi register were accessed using MDC clock prior to pll_clk_stable=1. This register field is not self-clearing and needs to be written to 0 to be cleared.	0x0
RSVD1_DBG_MODE_62	2	R/W	Reserved Config Register for Preamble Suppression Enable	0x0
DBG_MODE_FRC_EN	1	R/W	Debug Mode Force Enable . When set to 1, reg_clk represents the clock defined by dbg_mode_ena configuration irrespective of whether pll_clk is stable or not.	0x0

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Field Name	Bit	Access	Description	Default
DBG_MODE_ENA	0	R/W	Debug Mode Enable. This is not a self clearing bit. When configured to 1, MDC clock is used as reg_clk (debug_mode) until pll_clk_stable is asserted. After pll_clk_stable gets asserted, txpll_clk125 is used as reg_clk. When configured to 0, txpll_clk125 Is used as reg_clk (functional mode) via SMI Core.	0x1

1.10 mx_chip_top_regs

Table 269. Register Groups in MMD mx_chip_top_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
mx_chip_top_regs	0x00000000	1		Page 90

1.10.1 mx_chip_top_regs:mx_chip_top_regs

Parent: mx_chip_top_regs

Instances: 1

Table 270. Registers in mx_chip_top_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
INT_STS	0x0000F000	1	Interrupt Status Register	Page 95
INT_MSK	0x0000F001	1	Interrupt Mask Register	Page 96
CONTROL1	0x0000F002	1	Control1 Register	Page 97
CONTROL2	0x0000F003	1	Control2 Register	Page 98
COMMON_EFUSE_CFG1	0x0000F004	1	eFUSE Configuration 1 Register	Page 98
COMMON_EFUSE_CFG2	0x0000F005	1	eFUSE Configuration 2 Register	Page 98
COMMON_EFUSE_STAT1	0x0000F006	1	eFUSE Status Register	Page 99
COMMON_EFUSE_DATA	0x0000F007	1	eFUSE Data Register	Page 99
DBG_BUS_CTL_HI	0x0000F008	1	Debug Bus Control High Regsister.	Page 100
DBG_BUS_CTL_LO	0x0000F009	1	Debug Bus Control Low Regsister.	Page 100
GPIO_DIR	0x0000F00A	1	GPIO Direction Register	Page 100
GPIO_BUF	0x0000F00B	1	GPIO Buffer Register	Page 101
GPIO_DATA	0x0000F00C	1	GPIO Data Register	Page 101
GPIO_INT_STS	0x0000F00D	1	GPIO Interrupt Status Register	Page 101
GPIO_INT_MSK	0x0000F00E	1	GPIO Interrupt Mask Register	Page 102
GPIO_INT_POL	0x0000F00F	1	GPIO Interrupt Polarity Register	Page 102
GPIO_PU_PD_OVR	0x0000F010	1	GPIO Interrupt PU and PD Override Register	Page 102

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
GPIO_SOF_SEL_EN	0x0000F011	1	GPIO SOF Selection and Enable Register	Page 103
IO_CONTROL	0x0000F012	1	IO Control Register	Page 103
RGMII1_PAD_TUNE	0x0000F013	1	RGMII1 Pad Tune Register	Page 104
RGMII2_PAD_TUNE	0x0000F014	1	RGMII2 Pad Tune Register	Page 104
WDOG_TIMER0	0x0000F015	1	Watchdog Timer 0 Register	Page 105
WDOG_TIMER1	0x0000F016	1	Watchdog Timer 1 Register	Page 105
WDOG_TIMER2	0x0000F017	1	Watchdog Timer 2 Register	Page 106
WDOG_TIMER3	0x0000F018	1	Watchdog Timer 3 Register	Page 106
WDOG_TIMER4	0x0000F019	1	Watchdog Timer 4 Register	Page 106
SGMII_CTL	0x0000F01A	1	SGMII Control Register	Page 107
QSGMII_EXT_CFG	0x0000F01B	1	QSGMII Extender Configuration and Status Register This register configures the QSGMII Extender.	Page 107
QSGMII_STATUS	0x0000F01C	1	QSGMII Status Register	Page 108
SGMII_HRST	0x0000F01D	1	SGMII Hard Reset Register	Page 108
SGMII_SRST	0x0000F01E	1	SGMII Soft Reset Register	Page 108
QSGMII_MUX_CTL	0x0000F01F	1	QSGMII Mux Control Register	Page 108
QSGMII_ANEG_EN	0x0000F020	1	QSGMII Auto ANEG Enable Register	Page 109
SGMII_RX_LANE_STATUS	0x0000F021	1	Serdes Receive Lane Status Register This register provides XAUI PHY receive lane status. It is analogous to the Synopsys VR_MII_RX_LSTS PCS register.	Page 109
SGMII_TX_LVL_CTL	0x0000F022	1	Serdes Transmit Level Control Register This register provides QSGMII serdes transmit level adjustment. It is analogous to the Synopsys VR_MII_RX_Gen2_Gen4_TX_BST CTRL0 and VR_MII_Gen2_Gen4_TX_LVLCTR L0 PCS registers.	Page 110
SGMII_TX_CTL	0x0000F023	1	Serdes Transmit General Control Register This register provides general configuration for the QSGMII serdes transmitter. It is analogous to the Synopsys VR_MII_Gen2_Gen4_TXGENCTR L0, VR_MII_Gen2_Gen4_TXGENCTR L1, and VR_MII_Gen2_Gen4_TXGENCTR	Page 110

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
			L2 PCS register2.	
SGMII_TX_STS	0x0000F024	1	Serdes Transmit Status Register This register provides status of the QSGMII serdes transmitter. It is analogous to the Synopsys VR_MII_Gen2_Gen4_TX_STS PCS register.	Page 111
SGMII_RX_CTL	0x0000F025	1	Serdes Receive General Control Register This register provides general configuration for the QSGMII serdes receiver. It is analogous to the Synopsys VR_MII_Gen2_Gen4_RXGENCTRL0 and VR_MII_Gen2_Gen4_RXGENCTRL1 PCS registers.	Page 111
SGMII_RX_EQU_CTL	0x0000F026	1	Serdes Receive Equalization Control Register This register configures QSGMII serdes receiver equalization. It is analogous to the Synopsys VR_MII_Gen4_RXEQ_CTRL PCS register.	Page 112
SGMII_RX_LOSS_CTL	0x0000F027	1	Serdes Receive Loss of Signal Control Register This register configures the QSGMII serdes receiver Loss of Signal detection. It is analogous to the Synopsys VR_MII_Gen4_RXLOS_CTRL0 PCS register.	Page 112
SGMII_MPLL_CTL	0x0000F028	1	Serdes MPLL Control and Status Register This register provides control and status of the QSGMII serdes MPLL. It is analogous to the Synopsys VR_MII_Gen2_Gen4_MPLL_CTRL0, VR_MII_Gen2_Gen4_MPLL_CTRL1, and VR_MII_Gen2_Gen4_MPLL_STS PCS registers.	Page 113
SGMII_RX_LVL_CTL	0x0000F029	1	Serdes Receive Level Control Register This register configures receiver levels for the QSGMII serdes. It is analogous to the Synopsys VR_MII_Gen2_Gen4_LVL_CTRL PCS register.	Page 113
SGMII__MISC_CTL	0x0000F02A	1	Serdes Miscellaneous Control Register This register provides	Page 113

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
			miscellaneous control for the QSGMII serdes. It is analogous to the Synopsys VR_MII_Gen2_Gen4_MISC_CTRL 0 and VR_MII_Gen2_Gen4_MISC_CTRL 1 PCS registers.	
SGMII_CTL_STS	0x0000F02B	1	Serdes Control and Status Register This register provides powerup / powerdown control and status for the serdes	Page 114
SGMII_CR_CTL	0x0000F02C	1	Serdes Synopsys CR Control Register	Page 115
SGMII_CR_ADDR	0x0000F02D	1	Serdes Synopsys CR Address Register The Synopsis CR Control, Address, and Data Registers provide access to additional advanced serdes registers within the Synopsys 6G CEI serdes IP (see Synopsis data book for details). This register is analogous to the Synopsys VR_MII_SNPS_CR_ADDR PCS register.	Page 115
SGMII_CR_DATA	0x0000F02E	1	Serdes Synopsis CR Data Register The Synopsis CR Control, Address, and Data Registers provide access to additional advanced serdes registers within the Synopsys 6G CEI serdes IP (see Synopsis data book for details). This register is analogous to the Synopsys VR_MII_SNPS_CR_DATA PCS register.	Page 115
SGMII_VREG_BYP	0x0000F02F	1	Serdes VREG_BYP Control Register	Page 115
REC_CLK_OUT_SEL	0x0000F030	1	Recovered Clock Output Selector Register This register configures the source for Recovered Clock Output 1. To use a Recovered Clock Input or Output it must be enabled as a GPIO Alternate Function.	Page 116
REC_CLK_OUT_DIV	0x0000F031	1	Recovered Clock Output 1 Divider Register This register configures the divider for Recovered Clock Output 1.	Page 117
SGMII_PCS_HRST	0x0000F032	1	SGMII PCS Hard Reset Register	Page 117
SGMII_PCS_SRST	0x0000F033	1	SGMII PCS Soft Reset Register	Page 117

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
SGMII_PCS_CFG	0x0000F034	1	SGMII PCS Configuration and Status Register	Page 117
PCS_ANEG_CFG	0x0000F035	1	PCS1G ANEG Configuration Register This register provides Autonegotiation configuration and status for the QSGMII PCS1G. Note: RC, LL and LH bits are updated at the end of the read command. Depending on the timing, a serial read can finish simultaneously with an event (start read, capture data, event & end read) or it can overlap an event (start read, capture data, event then end read). To ensure that an event is never missed, the clear on read logic is gated with the read data, bit for bit, such that a bit is only cleared if it was read as active. This has a natural consequence that in the event of a simultaneousness event and end read, the event will have priority (if the bit wasnt previously active).	Page 119
PCS_ANEG_TX_ADV_ABLT	0x0000F036	1	PCS1G ANEG Tx Advertised Abilities Register This register provides the ANEG Advertised Abilities to be sent to the QSGMII PCS1G Link Partner.	Page 119
PCS_ANEG_TX_NP_DATA	0x0000F037	1	PCS1G ANEG Tx NP Data Register This register provides the ANEG NP data to be sent to the QSGMII PCS1G Link Partner	Page 120
PCS_ANEG_LP_ADV_ABLT	0x0000F038	1	PCS1G ANEG LP Advertised Abilities Register This register provides the ANEG Advertised Abilities data received from the QSGMII PCS1G Link Partner.	Page 120
PCS_ANEG_LP_NP_DATA	0x0000F039	1	PCS1G ANEG LP NP Data Register This register provides the ANEG Next Page data from the QSGMII PCS1G Link Partner.	Page 120
PCS_DEBUG	0x0000F03A	1	PCS1G Debug Register This register provides debug and loopback control for the QSGMII PCS1G.	Page 121
PCS_LPI_CFG_STS	0x0000F03B	1	PCS1G LPI Configuration and	Page 121

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
			Status Register This register provides Low Power Idle control and status for the QSGMII PCS1G.	
PCS_TEST_CFG_STS	0x0000F03C	1	PCS1G Test Pattern Configuration and Status Register This register provides Jitter test pattern control and status for the QSGMII PCS1G.	Page 122
CHIP_HARD_RST	0x0000F03E	1	Chip Hard Reset Register	Page 123
CHIP_SOFT_RST	0x0000F03F	1	Chip Soft Reset Register	Page 123
REGULATOR_CTL	0x0000F040	1	Regulator and SGMII/RGMII Pad Control Register	Page 123
SKU_DBG_STS	0x0000F041	1	SKU Select and DBG_EN Status register. This returns the status of the SKU_SEL0/1, DBG_EN, and STRAP pins for debug information.	Page 124
EFUSE_READ_DAT0	0x0000F200	1	16 bits of EFUSE Read Data.	Page 124
EFUSE_READ_DAT1	0x0000F201	1	16 bits of EFUSE Read Data.	Page 124
EFUSE_READ_DAT2	0x0000F202	1	16 bits of EFUSE Read Data.	Page 124
EFUSE_READ_DAT3	0x0000F203	1	16 bits of EFUSE Read Data.	Page 125
EFUSE_READ_DAT4	0x0000F204	1	16 bits of EFUSE Read Data.	Page 125
EFUSE_READ_DAT5	0x0000F205	1	16 bits of EFUSE Read Data.	Page 125
EFUSE_READ_DAT6	0x0000F206	1	16 bits of EFUSE Read Data.	Page 125
EFUSE_READ_DAT7	0x0000F207	1	16 bits of EFUSE Read Data.	Page 126
EFUSE_READ_DAT8	0x0000F208	1	16 bits of EFUSE Read Data.	Page 126
EFUSE_READ_DAT9	0x0000F209	1	16 bits of EFUSE Read Data.	Page 126
EFUSE_READ_DAT10	0x0000F20A	1	16 bits of EFUSE Read Data.	Page 127
EFUSE_READ_DAT11	0x0000F20B	1	16 bits of EFUSE Read Data.	Page 127
EFUSE_READ_DAT12	0x0000F20C	1	16 bits of EFUSE Read Data.	Page 128
EFUSE_READ_DAT13	0x0000F20D	1	16 bits of EFUSE Read Data.	Page 128
EFUSE_READ_DAT14	0x0000F20E	1	16 bits of EFUSE Read Data.	Page 128
EFUSE_READ_DAT15	0x0000F20F	1	16 bits of EFUSE Read Data.	Page 128

1.10.1.1 mx_chip_top_regs:mx_chip_top_regs:INT_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 271. Fields in INT_STS

Field Name	Bit	Access	Description	Default
RESERVED_1_0	15:12	R/O		0x0
WDT_INT_STS	11	R/O	watchdog time interrupt status Note: Reading this field will clear it.	0x0
GPIOS_INT_STS	10	R/O	gpio interrupt status	0x0
PVT_INT_STS	9	R/O	pvt interrupt status	0x0
P1588_COM_INT_STS	8	R/O	1588 common interrupt status	0x0
RESERVED_2_5	7:6	R/O		0x0
UVOV_CHIP_INT_STS	5	R/O	UVOV module interrupt status	0x0
WDT_CHIP_INT_STS	4	R/O	WDT module interrupt status	0x0
P1588_MOD_INT_STS	3	R/O	1588 module interrupt status	0x0
T1_PHY_INT_STS	2	R/O	T1 PHY interrupt status	0x0
LINK_UP_STS	1	R/O	Link up interrupt status Note: Reading this field will clear it.	0x0
LINK_DOWN_STS	0	R/O	Link down interrupt status Note: Reading this field will clear it.	0x0

1.10.1.2 mx_chip_top_regs:mx_chip_top_regs:INT_MSK

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 272. Fields in INT_MSK

Field Name	Bit	Access	Description	Default
RESERVED_1_12	15:12	R/O		0x0
WDT_INT_MSK	11	R/W	watchdog time interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
GPIOS_INT_MSK	10	R/W	gpio interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
PVT_INT_MSK	9	R/W	pvt interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
P1588_COM_INT_MSK	8	R/W	1588 common interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
RESERVED_2_17	7:6	R/O		0x0
UVOV_CHIP_INT_MSK	5	R/W	UVOV module interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
WDT_CHIP_INT_MSK	4	R/W	WDT module interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
P1588_MOD_INT_MSK	3	R/W	1588 module interrupt mask	0x1

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Field Name	Bit	Access	Description	Default
			When 0, this interrupt is enabled. When 1, this interrupt is disabled.	
T1_PHY_INT_MSK	2	R/W	T1 PHY interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
LINK_UP_MSK	1	R/W	Link up interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1
LINK_DOWN_MSK	0	R/W	Link down interrupt mask When 0, this interrupt is enabled. When 1, this interrupt is disabled.	0x1

1.10.1.3 mx_chip_top_regs:mx_chip_top_regs:CONTROL1

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 273. Fields in CONTROL1

Field Name	Bit	Access	Description	Default
IRQN_PIN_DISABLE	15	R/W	Setting this bit to 1 Disables IRQN/LED4/DEBUG29 pad OEN. Only one pad of IRQN/LED4/DEBUG29, TX_TCLK (controlled by port_testbus_in[94]), and ATP (controlled by com_testbus_in[0]) should be enabled at a time.	0x0
CLKOUT_SEL	14:13	R/W	Determine the frequency of the CLKOUT pin. 00: CLK_XTAL (25 MHz) 01: 25 MHz 10: 50 MHz 11: 125 MHz The default value for this field is determined by the respective eFUSE field. Note: Options 01b, 10b and 11b are derived from the intenral PLL.	0x0
CLKOUT_EN	12	R/W	When set the CLKOUT pin is enabled. The default value for this field is determined by the respective eFUSE field.	0x0
RESERVED_27	11	R/O		0x0
P1588_PIN_SEL	10:8	R/W	Each bit enables the corresponding 1588 function in the pin mux. 1588_pin_sel[2] = 1588_CLK_LD_ADJ 1588_pin_sel[1] = 1588_REF_CLK 1588_pin_sel[0] = 1588_EVT	0x0
RESERVED_29	7	R/O		0x0
GPIO_PIN_SEL	6:4	R/W	Each bit enables the corresponding gpio in the pin mux. gpio_pin_sel[2] = GPIO3 gpio_pin_sel[1] = GPIO2 gpio_pin_sel[0] = GPIO1	0x0
RESERVED_31	3:2	R/O		0x0
LED_PIN_SEL	1:0	R/W	Each bit enables the corresponding LED in the	0x0

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Field Name	Bit	Access	Description	Default
			pin mux. led_pin_sel[0] = LED1 led_pin_sel[1] = LED4	

1.10.1.4 mx_chip_top_regs:mx_chip_top_regs:CONTROL2

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 274. Fields in CONTROL2

Field Name	Bit	Access	Description	Default
RESERVED_33	15:7	R/O		0x000
MAGIC	6	R/W	Magic Bit When set fixes any digital HW bug. Note: Bit is only usable during waxing crescent moon phase.	0x0
RESERVED_5	5	R/W	Spare R/W bit for future use.	0x0
RESERVED_4	4	R/W	Spare R/W bit for future use.	0x0
EF_EM_EN	3	R/W	eFUSE emulation enable. When set eFUSE emulation mode is enabled. This bit is non-functional if the EF_EM_DIS bit is set in the eFUSE.	0x0
EF_DP_EN	2	R/W	eFUSE Dataport Enable When set the eFUSE dataport is enabled.	0x0
PWR_FET_GD	1	R/O	Power FET Good Reports the status of the FETPW_GD_CR signal from the integrated eFUSE power FET.	0x0
PWR_FET_EN	0	R/W	Power FET Enable When set the eFUSE power FET is forced on.	0x0

1.10.1.5 mx_chip_top_regs:mx_chip_top_regs:COMMON_EFUSE_CFG1

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 275. Fields in COMMON_EFUSE_CFG1

Field Name	Bit	Access	Description	Default
RSVD_EFUSE_CFG1_41	15:4	R/W	Reserved EFUSE Override Configuration 1	0x000
EFUSE_BKDR_REG_ADDR	3:0	R/W	Address to access EFUSE using Backdoor Indirect access	0x0

1.10.1.6 mx_chip_top_regs:mx_chip_top_regs:COMMON_EFUSE_CFG2

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 276. Fields in COMMON_EFUSE_CFG2

Field Name	Bit	Access	Description	Default
RSVD_EFUSE_CFG2_43	15:5	R/W	Reserved EFUSE Override Configuration 2	0x000
EF_EM_DONE	4	R/W	This bit is used in conjunction with EFUSE emulation mode to manually assert the efuse_done signal.	0x0
EFUSE_PRGM_EN	3	R/W	Writing a 1 to this register will trigger an EFUSE program operation. This bit will self clear when the programming operation has completed.	0x0
EFUSE_READ_EN	2	R/W	Writing a 1 to this register will trigger a manual EFUSE read operation. This bit will self clear when the read operation has completed.	0x0
EFUSE_BKDR_REG_RD	1	R/W	Read operation to access EFUSE using Backdoor Indirect access. When asserted HIGH, this will cause Read operation by EFUSE Backdoor Register Controller. This bit will self clear when the Backdoor Read operation has completed.	0x0
EFUSE_BKDR_REG_WR	0	R/W	Write operation to access EFUSE using Backdoor Indirect access. When asserted HIGH, this will cause Write operation by EFUSE Backdoor Register Controller. This bit will self clear when the Backdoor Write operation has completed.	0x0

1.10.1.7 mx_chip_top_regs:mx_chip_top_regs:COMMON_EFUSE_STAT1

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 277. Fields in COMMON_EFUSE_STAT1

Field Name	Bit	Access	Description	Default
RSVD_EFUSE_STAT1_49	15:4	R/W	Reserved EFUSE Override Status register	0x000
EFUSE_READ_RUNNING	3	R/O	When this bit is 1, the EFUSE read operation is in progress.	0x0
EFUSE_PRGM_RUNNING	2	R/O	When this bit is 1, the EFUSE programming operation is in progress.	0x0
EFUSE_DONE	1	R/O	The EFUSE operation is done.	0x0
EFUSE_BKDR_REG_RDY	0	R/O	Indicates completion status of EFUSE Backdoor register operation. In case of read opertion, this bit indicates when read data has been stored in the efuse_bkdr_reg_data register. 1 = EFUSE Register Backdoor Controller is ready for usage 0= EFUSE Register Backdoor controller is busy	0x0

1.10.1.8 mx_chip_top_regs:mx_chip_top_regs:COMMON_EFUSE_DATA

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 278. Fields in COMMON_EFUSE_DATA

Field Name	Bit	Access	Description	Default
EFUSE_BKDR_REG_DATA	15:0	R/W	In case of write operation, this register field indicates data to be written into the respective address location specified using efuse_bkdr_reg_addr. In case of read operation, this register field reflects the read data. Before reading this registter for read operation, user needs to read efuse_bkdr_reg_rdy and ensure it is HIGH. Otherwise the read operation is still in progress.	0x0000

1.10.1.9 mx_chip_top_regs:mx_chip_top_regs:DBG_BUS_CTL_HI

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 279. Fields in DBG_BUS_CTL_HI

Field Name	Bit	Access	Description	Default
DB_PIN_SEL_HI	15:0	R/W	Enables the debug bus on the respecitve pin. This field controls the upper 16-bits of the DEBUG bus select control. db_pin_sel[31:16] = db_pin_sel_hi db_pin_sel[31] = DEBUG31 .. db_pin_sel[0] = DEBUG16 This field should only be used in the engineering SKU. Note: Do not show this register to customers.	0x0000

1.10.1.10 mx_chip_top_regs:mx_chip_top_regs:DBG_BUS_CTL_LO

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 280. Fields in DBG_BUS_CTL_LO

Field Name	Bit	Access	Description	Default
DB_PIN_SEL_LO	15:0	R/W	Enables the debug bus on the respecitve pin. This field control the lower 16-bits of the DEBUG bus select control. db_pin_sel[15:0] = db_pin_sel_lo db_pin_sel[15] = DEBUG15 .. db_pin_sel[0] = DEBUG0 This field should only be used in the engineering SKU. Note: Do not show this register to customers.	0x0000

1.10.1.11 mx_chip_top_regs:mx_chip_top_regs:GPIO_DIR

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 281. Fields in GPIO_DIR

Field Name	Bit	Access	Description	Default
RESERVED_57	15:3	R/O		0x0000
GPIO_DIR	2:0	R/W	GPIO Direction When set, enables the corresponding GPIO as an output. When cleared the GPIO is enabled as an input.	0x0

1.10.1.12 mx_chip_top_regs:mx_chip_top_regs:GPIO_BUF

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 282. Fields in GPIO_BUF

Field Name	Bit	Access	Description	Default
RESERVED_59	15:3	R/O		0x0000
GPIO_BUFFER	2:0	R/W	GPIO Buffer Type When set, the output buffer for the corresponding GPIO signal is configured as a push/pull driver. When cleared, the corresponding GPIO signal is configured as an open-drain driver.	0x0

1.10.1.13 mx_chip_top_regs:mx_chip_top_regs:GPIO_DATA

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 283. Fields in GPIO_DATA

Field Name	Bit	Access	Description	Default
RESERVED_61	15:3	R/O		0x0000
GPIO_DATA	2:0	R/W	GPIO Data When enabled as an output, the value written is reflected on the GPIO. When read, the value always reflects the current state of the corresponding GPIO pin, regardless of the value written or the GPIO direction.	0x0

1.10.1.14 mx_chip_top_regs:mx_chip_top_regs:GPIO_INT_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 284. Fields in GPIO_INT_STS

Field Name	Bit	Access	Description	Default
RESERVED_63	15:3	R/O		0x0000
GPIO_INT_STS	2:0	R/O	GPIO Interrupt Status Interrupts generated from the GPIOs. Note: The sources for these interrupts are	0x0

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Field Name	Bit	Access	Description	Default
			level sensitive. Note: The GPIO inputs must be stable for ~16ns (2 consecutive 125MHz edges) to be recognized. Note: Reading this field will clear it.	

1.10.1.15 mx_chip_top_regs:mx_chip_top_regs:GPIO_INT_MSK

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 285. Fields in GPIO_INT_MSK

Field Name	Bit	Access	Description	Default
RESERVED_65	15:3	R/O		0x0000
GPIO_INT_MSK	2:0	R/W	GPIO Interrupt Mask When set, the respective GPIO interrupt is disabled. When clear, the respective GPIO interrupt is enabled.	0x7

1.10.1.16 mx_chip_top_regs:mx_chip_top_regs:GPIO_INT_POL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 286. Fields in GPIO_INT_POL

Field Name	Bit	Access	Description	Default
RESERVED_67	15:3	R/O		0x0000
GPIO_INT_POL	2:0	R/W	GPIO Interrupt Polarity When clear, an interrupt is triggered when the GPIO input is low. When set, an interrupt is triggered when the GPIO input is high.	0x0

1.10.1.17 mx_chip_top_regs:mx_chip_top_regs:GPIO_PU_PD_OVR

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 287. Fields in GPIO_PU_PD_OVR

Field Name	Bit	Access	Description	Default
RESERVED_69	15:6	R/O		0x000
GPIO_INT_PU_PD_OVR3	5:4	R/W	GPIO3 PU PD Override 00 = No Override (PU or PD determined by hardware) 01 = GPIO has Pullup, no Pulldown 10 = GPIO has Pulldown, no Pullup 11 = GPIO has no Pullup, no Pulldown	0x0
GPIO_INT_PU_PD_OVR2	3:2	R/W	GPIO2 PU PD Override 00 = No Override (PU or PD determined by	0x0

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Field Name	Bit	Access	Description	Default
			hardware) 01 = GPIO has Pullup, no Pulldown 10 = GPIO has Pulldown, no Pullup 11 = GPIO has no Pullup, no Pulldown	
GPIO_INT_PU_PD_OVR1	1:0	R/W	GPIO1 PU PD Override 00 = No Override (PU or PD determined by hardware) 01 = GPIO has Pullup, no Pulldown 10 = GPIO has Pulldown, no Pullup 11 = GPIO has no Pullup, no Pulldown	0x0

1.10.1.18 mx_chip_top_regs:mx_chip_top_regs:GPIO_SOF_SEL_EN

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 288. Fields in GPIO_SOF_SEL_EN

Field Name	Bit	Access	Description	Default
RESERVED_73	15:7	R/O		0x000
SOF_ALT_FUNC_EN	6:4	R/W	SOF Alternate Function Enable When set the SOF alternate function is enabled on the respective GPIO.	0x0
RESERVED_75	3	R/O		0x0
SOF3_SEL	2	R/W	SOF3 Select 0: Output is TX SOF 1: Output is RX SOF	0x0
SOF2_SEL	1	R/W	SOF2 Select 0: Output is TX SOF 1: Output is RX SOF	0x0
SOF1_SEL	0	R/W	SOF1 Select 0: Output is TX SOF 1: Output is RX SOF	0x0

1.10.1.19 mx_chip_top_regs:mx_chip_top_regs:IO_CONTROL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 289. Fields in IO_CONTROL

Field Name	Bit	Access	Description	Default
RESERVED_79	15:2	R/O		0x0000
RGMII2_SLEW	1	R/W	RGMII2 Slew Rate Enable When set a faster slew rate is enabled on the RGMII pads.	0x1
RGMII1_SLEW	0	R/W	RGMII1 Slew Rate Enable When set a faster slew rate is enabled on the RGMII pads.	0x1

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1.10.1.20 mx_chip_top_regs:mx_chip_top_regs:RGMII1_PAD_TUNE

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 290. Fields in RGMII1_PAD_TUNE

Field Name	Bit	Access	Description	Default
RGMII1_TUNE	15	R/W	TUNE When set IO pads are tuned by asserting TUNE_B to TUNE_3VO cell for 1 us. This bit self clears after the tuning is complete.	0x0
RGMII_1TUNE_SEL	14	R/W	Tune Select 0 = Tune bits set inside of the TUNE_3VO cell are used. 1 = Tune bits defined by CAp and CAn are used.	0x0
RESERVED_84	13:12	R/O		0x0
RGMII1_TAP	11:9	R/O	Reports the values of A0p, A1p and A2p from TUNE_3VO.	0x0
RGMII1_TAN	8:6	R/O	Reports the values of A0n, A1n and A2n from TUNE_3VO.	0x0
RGMII1_CAP	5:3	R/W	Used to set the tune bits CA0p, CA1p and CA2p in the TUNE_3VO cell when TUNE_SEL is set.	0x0
RGMII1_CAN	2:0	R/W	Used to set the tune bits CA0n, CA1n and CA2n in the TUNE_3VO cell when TUNE_SEL is set.	0x0

1.10.1.21 mx_chip_top_regs:mx_chip_top_regs:RGMII2_PAD_TUNE

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 291. Fields in RGMII2_PAD_TUNE

Field Name	Bit	Access	Description	Default
RGMII2_TUNE	15	R/W	TUNE When set IO pads are tuned by asserting TUNE_B to TUNE_3VO cell for 1 us. This bit self clears after the tuning is complete.	0x0
RGMII_2TUNE_SEL	14	R/W	Tune Select 0 = Tune bits set inside of the TUNE_3VO cell are used. 1 = Tune bits defined by CAp and CAn are used.	0x0
RESERVED_91	13:12	R/O		0x0
RGMII2_TAP	11:9	R/O	Reports the values of A0p, A1p and A2p from TUNE_3VO.	0x0
RGMII2_TAN	8:6	R/O	Reports the values of A0n, A1n and A2n from TUNE_3VO.	0x0
RGMII2_CAP	5:3	R/W	Used to set the tune bits CA0p, CA1p and CA2p in the TUNE_3VO cell when TUNE_SEL is	0x0

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Field Name	Bit	Access	Description	Default
			set.	
RGMII2_CAN	2:0	R/W	Used to set the tune bits CA0n, CA1n and CA2n in the TUNE_3VO cell when TUNE_SEL is set.	0x0

1.10.1.22 mx_chip_top_regs:mx_chip_top_regs:WDOG_TIMER0

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 292. Fields in WDOG_TIMER0

Field Name	Bit	Access	Description	Default
RESERVED_96	15:6	R/O		0x000
WDGBLSLEEPEN	5	R/W	Watchdog timer global sleep enable 0: A watchdog fault (underflow or error) has no direct effect on the device's TC10 state. 1: A watchdog fault (underflow or error) forces the device into the TC10 SLEEP state. In a typical TC10 compliant configuration this device will undergo a power-on-reset after a wakeup event is received.	0x0
WDSYSRSTEN	4	R/W	Watchdog timer system reset enable When set a watchdog fault (underflow or error) forces the device into reset.	0x0
WDMWOE	3	One-shot	Watchdog Mode Write Once Enable Once set this bit is only cleared by a reset (POR/UVOV/Pin/Soft/WDT). 0: Contents of WDOG_TIMER1 and WDOG_TIMER2 are RW. 1: Contents of WDOG_TIMER1 and WDOG_TIMER2 are RWrite-Once	0x0
WDDIS	2	R/W	Watchdog Disable 0: Enables the Watchdog Timer. 1: Disables the Watchdog Timer. Note: When setting the WDDIS bit, and while it is set, the fields WDV and WDD must not be modified	0x1
WDRSTEN	1	R/W	Watchdog Reset Enable 0: A watchdog fault (underflow or error) has no effect on the resets. 1: A watchdog fault (underflow or error) triggers a watchdog reset.	0x0
WDFIEN	0	R/W	Watchdog Fault Interrupt Enable 0: A watchdog fault (underflow or error) has no effect on interrupt. 1: A watchdog fault (underflow or error) asserts interrupt.	0x0

1.10.1.23 mx_chip_top_regs:mx_chip_top_regs:WDOG_TIMER1

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 293. Fields in WDOG_TIMER1

Field Name	Bit	Access	Description	Default
WDV	15:0	R/W	Watchdog Counter Value Defines the value loaded into the watchdog timer.	0x0000

1.10.1.24 mx_chip_top_regs:mx_chip_top_regs:WDOG_TIMER2

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 294. Fields in WDOG_TIMER2

Field Name	Bit	Access	Description	Default
WDD	15:0	R/W	Watchdog Delta Value Defines the permitted range for reloading the Watchdog Timer. If the Watchdog Timer value is less than or equal to WDD, setting bit WDRSTT restarts the timer. If the Watchdog Timer value is greater than WDD, setting bit WDRSTT causes a watchdog error. Defines the value loaded into the watchdog timer.	0x0000

1.10.1.25 mx_chip_top_regs:mx_chip_top_regs:WDOG_TIMER3

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 295. Fields in WDOG_TIMER3

Field Name	Bit	Access	Description	Default
KEY	15:8	R/W	Password Writing any other value than 0xA5 in this field aborts the write operation.	0x00
RESERVED_106	7:1	R/O		0x00
WDRSTT	0	R/W	Watchdog Restart 0: No effect. 1: Restarts the watchdog if KEY is written to 0xA5	0x0

1.10.1.26 mx_chip_top_regs:mx_chip_top_regs:WDOG_TIMER4

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 296. Fields in WDOG_TIMER4

Field Name	Bit	Access	Description	Default
RESERVED_108	15:2	R/O		0x0000
WDUNF	1	R/O	Watchdog Underflow (cleared on read) 0: No watchdog underflow occurred since the last read of WDOG_TIMER4.	0x0

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Field Name	Bit	Access	Description	Default
			1: At least one watchdog underflow occurred since the last read of WDOG_TIMER4. Note: Reading this field will clear it.	
WDERR	0	R/O	Watchdog Error (cleared on read) 0: No watchdog error occurred since the last read of WDOG_TIMER4. 1: At least one watchdog error occurred since the last read of WDOG_TIMER4. Note: Reading this field will clear it.	0x0

1.10.1.27 mx_chip_top_regs:mx_chip_top_regs:SGMII_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 297. Fields in SGMII_CTL

Field Name	Bit	Access	Description	Default
RESERVED_111	15:3	R/O		0x0000
XAUI_RST	2	R/W	When set the XAUI PHY is reset. This bit is self-cleaing.	0x0
VREG_BYP_CTL	1	R/W	This bit controls the bypassing of the votlage regulator inside of the XAUI PHY. Note: This bit functions as a strap into the PHY. It will not affect the state of the PHY unless the SerDes is also reset. Note: The default value is impactd by eFUSE settings.	0x0
SGMII_MUX_EN	0	R/W	This bit controls the RGMII_SGMII mux. When set the mux enables the SGMII SerDes. Note: The default value depends on the eFUSE settings.	0x0

1.10.1.28 mx_chip_top_regs:mx_chip_top_regs:QSGMII_EXT_CFG

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 298. Fields in QSGMII_EXT_CFG

Field Name	Bit	Access	Description	Default
RESERVED_115	15:8	R/O		0x00
FLIP_LANES_116	7	R/W	Flip or swap lanes (lane 0 with lane 3, lane 1 with lane 2). Note: Not used in this device.	0x0
SHYST_DIS_117	6	R/W	Disable hysteresis of synchronization state machine	0x0
E_DET_ENA_118	5	R/W	Enable 8b10b error propagation (8b10b errors code-groups are replaced by K30.7 error symbols)	0x1
USE_I1_ENA_119	4	R/W	Use /I1/ during Idle sequencing only	0x0
RESERVED_120	3:0	R/O		0x0

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1.10.1.29 mx_chip_top_regs:mx_chip_top_regs:QSGMII_STATUS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 299. Fields in QSGMII_STATUS

Field Name	Bit	Access	Description	Default
RESERVED_121	15:5	R/O		0x000
SYNC_STAT_122	4	R/O	Set when QSGMII channel has successfully synchronized on K28.1. This field is only valid This field is only valid when SHYST_DIS is 1.	0x0
SYNC_STAT_PORT_123	3	R/O	Synchronization status of Port 0. Note this status is also available in PCS1G.	0x0
RESERVED_124	2:0	R/O		0x0

1.10.1.30 mx_chip_top_regs:mx_chip_top_regs:SGMII_HRST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 300. Fields in SGMII_HRST

Field Name	Bit	Access	Description	Default
RESERVED_125	15:1	R/O		0x0000
SGMII_HRST	0	R/W	When set this resets the QSGMII Extender, XAUI PHY, and the SGMII PCS1G blocks and sets all related configuration registers to their default state.	0x0

1.10.1.31 mx_chip_top_regs:mx_chip_top_regs:SGMII_SRST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 301. Fields in SGMII_SRST

Field Name	Bit	Access	Description	Default
RESERVED_127	15:1	R/O		0x0000
SGMII_SRST	0	R/W	When set this resets the QSGMII Extender, XAUI PHY, and the SGMII PCS1G blocks but does not affect the state of all related configuration registers.	0x0

1.10.1.32 mx_chip_top_regs:mx_chip_top_regs:QSGMII_MUX_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 302. Fields in QSGMII_MUX_CTL

Field Name	Bit	Access	Description	Default
RESERVED_129	15:0	R/O		0x0000

1.10.1.33 mx_chip_top_regs:mx_chip_top_regs:QSGMII_ANEG_EN

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 303. Fields in QSGMII_ANEG_EN

Field Name	Bit	Access	Description	Default
RESERVED_130	15:3	R/O		0x0000
NP_DISABLE	2	R/W	1 NP feature is disabled. This bit controls the PCS1G ANEG SM to support as per S/QS/U-QS/GMII spec, next page is not supported. 0 NP feature is enabled. This bit controls the PCS1G ANEG SM as per 802.3 AN spec, next page is supported.	0x1
SGMII_USGMII_TX_CONFIG_EN	1	R/W	When bit is 1 -> SGMII/QSGMII (2-bit speed indicator mode) is selected. 0 -> U-QSGMII (3- bitspeed indictor mode) is selected. This bit is valid if QSGMII_AUTO_ANEG_ENA is set to 1.	0x1
QSGMII_AUTO_ANEG_ENA	0	R/W	When this bit is 1, the device will automatically configure the QSGMII operation based on each PHY IP speed status, then perform auto-negotiation over the QSGMII link. 0 = QSGMII Auto ANEG not enabled 1 = QSGMII Auto ANEG enabled This bit enables the QSGMII interface to automatically come up running out of reset. For non-SGMII SKUs this bit shall default to 0b.	0x0

1.10.1.34 mx_chip_top_regs:mx_chip_top_regs:SGMII_RX_LANE_STATUS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 304. Fields in SGMII_RX_LANE_STATUS

Field Name	Bit	Access	Description	Default
RESERVED_134	15:3	R/O		0x0000
RX_VALID	2	R/O	DPLL lock status 0 = not locked 1 = locked	0x0
RX_PLL_STATE	1	R/O	Receive DPLL state 0 = not in Power_Good state 1 = in Power_Good state	0x0
SIG_DET	0	R/O	Receive signal detect 0 = receive signal not detected	0x0

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Field Name	Bit	Access	Description	Default
			1 = receive signal detected	

1.10.1.35 mx_chip_top_regs:mx_chip_top_regs:SGMII_TX_LVL_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 305. Fields in SGMII_TX_LVL_CTL

Field Name	Bit	Access	Description	Default
RESERVED_138	15:14	R/O		0x0
TX_LEVEL	13:7	R/W	This field sets the peak-to-peak differential output level of the transmitter driver output. The amplitude is specified in volts according to following formula: 1.33*TX_LEVEL[6:0]/64	0x55
TX_BOOST	6:0	R/W	This field sets the pre-emphasis level of the transmitter driver output. The amplitude is specified in decibels according to following formula: (TX_BOOST[6:0]+1)/4	0x02

1.10.1.36 mx_chip_top_regs:mx_chip_top_regs:SGMII_TX_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 306. Fields in SGMII_TX_CTL

Field Name	Bit	Access	Description	Default
TX_VBOOST_EN	15	R/W	Enables the current mode Tx Swing boost 0 = disabled 1 = enabled	0x0
TX_VBOOST_LVL	14:12	R/W	This field sets the launch amplitude of the transmit driver output.	0x4
TX_DISABLE_L0	11	R/W	0 = Tx serdes lane enabled 1 = Tx serdes lane disabled	0x0
TX_DT_EN	10	R/W	Output driver enable 0 = disabled 1 = enabled	0x1
TX_RST	9	R/W	Transmitter reset 0 = not reset 1 = reset	0x0
TX_INV	8	R/W	Inverts the logic levels on the differential transmitter output 0 = not inverted 1 = inverted Note: this bit is inverted by default to align with the package pinout.	0x0
TX_DET_RX_REQ	7	R/W	When asserted, this bit initiates the Receiver_Detection operation. This bit should be set to 1 only when TX_DET_RX_ACK in the QSGMII Serdes Transmit Status Register is set	0x0

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Field Name	Bit	Access	Description	Default
			to 0. When TX_DET_RX_ACK is set to 1, it should not be set to 0 until TX_DET_RX_ACK is set to 1	
TX_CM_EN	6	R/W	Transmit Common Mode enable 1 = Enable Transmit Common Mode to be held at half of vptx 0 = Disable Transmit Common Mode to be held at half of vptx. Transmit Common Mode will be weakly pulled to ground.	0x1
RESERVED_149	5	R/O		0x0
TX_TERM_OFFSET	4:0	R/W	Transmit termination offset This field applies an offset to the resistor calibration value.	0x00

1.10.1.37 mx_chip_top_regs:mx_chip_top_regs:SGMII_TX_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 307. Fields in SGMII_TX_STS

Field Name	Bit	Access	Description	Default
RESERVED_151	15:4	R/O		0x000
TX_DET_RX_RESLT	3	R/O	Transmit Receiver_Detection result. 0 = no receiver detected 1 = receiver detected	0x0
TX_DET_RX_ACK	2	R/O	Transmit Receiver_Detection acknowledgement When set to 1, this field indicates that the Receiver_Detection operation, requested by asserting TX_DET_RX_REQ, is complete. RX_DET_RX_ACK is de-asserted only when TX_DET_RX_REQ is deasserted.	0x0
TX_CMNST	1	R/O	Transmit Common Mode State 0 = Transmit Common Mode level has not reached its intended value 1 = Transmit Common Mode level has reached its intended value	0x0
TX_ST	0	R/O	Transmit Active State 0 = Transmitter is not active 1 = Transmitter is active. Transmitter has reached its intended level and is able to transmit data.	0x0

1.10.1.38 mx_chip_top_regs:mx_chip_top_regs:SGMII_RX_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 308. Fields in SGMII_RX_CTL

Field Name	Bit	Access	Description	Default
RESERVED_156	15:7	R/O		0x000

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Field Name	Bit	Access	Description	Default
RX_DET_EN	6	R/W	Receiver clock and data output enable 0 = receiver clock and data outputs not enabled 1 = receiver clock and data outputs enabled	0x1
RX_ALIGN_EN	5	R/W	Receiver alignment enable When set to 1, this bit enables 10-bit word alignment based on the COMMA character. This bit must only be asserted when RX_DT_EN is deasserted.	0x0
RX_TRMEN	4	R/W	Receiver termination enable When set to 1, this bit enables the 100 Ohm termination on the differential receiver input. This bit must only be asserted when RX_DT_EN is deasserted.	0x1
RX_LOS_EN	3	R/W	Receiver Loss of Signal enable 0 = Rx LOS detection disabled 1 = Rx LOS detection enabled	0x1
RX_PLL_EN	2	R/W	Receiver DPLL enable 0 = Rx DPLL disabled 1 = Rx DPLLenabled	0x1
RX_RST	1	R/W	Receiver reset 0 = not reset 1 = reset	0x0
RX_INV	0	R/W	Inverts the logic levels on the differential receiver input 0 = not inverted 1 = inverted	0x0

1.10.1.39 mx_chip_top_regs:mx_chip_top_regs:SGMII_RX_EQU_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 309. Fields in SGMII_RX_EQU_CTL

Field Name	Bit	Access	Description	Default
RESERVED_164	15:3	R/O		0x0000
RX_EQ_VAL0	2:0	R/W	This field controls the equalization boost level of the receiver	0x5

1.10.1.40 mx_chip_top_regs:mx_chip_top_regs:SGMII_RX_LOSS_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 310. Fields in SGMII_RX_LOSS_CTL

Field Name	Bit	Access	Description	Default
RESERVED_166	15:7	R/O		0x000
RX_LOS_FLTR	6:0	R/W	This field provides the LOS filter value.	0x0E

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1.10.1.41 mx_chip_top_regs:mx_chip_top_regs:SGMII_MPLL_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 311. Fields in SGMII_MPLL_CTL

Field Name	Bit	Access	Description	Default
RESERVED_168	15:12	R/O		0x0
MPLL_STATE	11	R/O	This bit indicates state of the MPLL 0 = MPLL is phase-locked 1 = MPLL is not phase-locked	0x1
REF_CLK_DIV2	10	R/W	Divides the reference clock by two before passing it to the MPLL 0 = reference clock is 25 MHz or 62.5 MHz 1 = reference clock is 50 MHz, 125 MHz, or 156.25 MHz	0x0
LANE_10BIT_SEL	9	R/W	Lane width control 0 = 20-bit lane 1 = 10-bit lane	0x1
REF_SSP_EN	8	R/W	Enables the reference clock 0 = reference clock not enabled 1 = reference clock enabled	0x1
SERDES_MPLL_OVERRIDE	7	R/W	It is used to select the reference clock source for serdes (0 = Internal, 1 = External) and also overrides mpll_multiplier, lane_10bit_sel, and ref_clk_div2 values in MPLL_OFF state 0 = MPLL override not enabled 1 = MPLL override enabled	0x0
MPLL_MULTIPLIER	6:0	R/W	MPLL frequency multiplier control	0x64

1.10.1.42 mx_chip_top_regs:mx_chip_top_regs:SGMII_RX_LVL_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 312. Fields in SGMII_RX_LVL_CTL

Field Name	Bit	Access	Description	Default
RESERVED_175	15:13	R/O		0x0
ACJT_LEVEL	12:8	R/W	Receiver sensitivity level control	0x00
LOS_BIAS	7:5	R/W	Loss of Signal detector threshold control A positive binary bit setting change results in a +15 mVp incremental change in the LOS threshold. A negative binary setting change results in a -15 mVp incremental change in the LOS threshold. The setting 3'b000 is reserved and must not be used	0x1
LOS_LEVEL	4:0	R/W	Loss of Signal detector sensitivity control	0x09

1.10.1.43 mx_chip_top_regs:mx_chip_top_regs:SGMII__MISC_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

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Table 313. Fields in SGMII__MISC_CTL

Field Name	Bit	Access	Description	Default
RESERVED_179	15:14	R/O		0x0
SSC_REF_CLK_SEL	13:5	R/W	Spread Spectrum Reference Clock Shifting Control This field provides control to help non-standard oscillator frequencies generate targeted MPLL output rates.	0x000
LB_MODE	4	R/W	Analog loopback, from Tx serial output to Rx serial input. 0 = loopback disabled 1 = loopback enabled. LOS not detected when LB_MODE=1.	0x0
RX_RATE	3:2	R/W	Selects receiver baud rate, as follows: 00 = mpll_baud_clk 01 = mpll_baud_clk/2 10 = mpll_baud_clk/4 11 = RESERVED	0x2
TX_RATE	1:0	R/W	Selects transmitter baud rate, as follows: 00 = mpll_baud_clk 01 = mpll_baud_clk/2 10 = mpll_baud_clk/4 11 = RESERVED	0x2

1.10.1.44 mx_chip_top_regs:mx_chip_top_regs:SGMII_CTL_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 314. Fields in SGMII_CTL_STS

Field Name	Bit	Access	Description	Default
RESERVED_184	15:6	R/O		0x000
POWER_DOWN	5	R/W	0 = normal operation 1 = powerdown mode, turns off transmitter, receiver, and all clocks	0x0
PSAVE	4	R/W	0 = normal operation 1 = powersave mode, turns off transmitter and receiver	0x0
BYP_SEQ	3	R/W	0 = the normal powerup flow is followed, where the PCS1G waits for serdes PLLs to be stable before becoming operational. 1 = the normal powerup flow is bypassed, where the PCS1G does not wait for serdes PLLs to be stable before becoming operational. This mode is useful mainly for emulation.	0x0
PSEQ_STATE	2:0	R/O	This field indicates the state variable value of the power-up sequence module. 000 = Wait for MPLL On 001 = Wait for Tx Up 010 = Wait for Rx Up 011/100 = Tx/Rx Stable	0x0

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Field Name	Bit	Access	Description	Default
			101 = Wait for Rx Down (MPLL still ON) 110 = MPLL OFF 111 = Reserved	

1.10.1.45 mx_chip_top_regs:mx_chip_top_regs:SGMII_CR_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 315. Fields in SGMII_CR_CTL

Field Name	Bit	Access	Description	Default
RESERVED_189	15:2	R/O		0x0000
WR_RDN	1	R/W	This bit controls whether a Read or Write operation is to be performed 0 = read 1 = write	0x0
START_BUSY	0	R/W	Assert this bit to initiate a CR read or write transfer. The bit self-clears when the transfer is completed. This bit must be deasserted before writing to the Synopsys CR Control Register, Synopsys CR Address Register, or the Synopsys CR Data Register. 0 = CR interface not busy 1 = CR interface busy	0x0

1.10.1.46 mx_chip_top_regs:mx_chip_top_regs:SGMII_CR_ADDR

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 316. Fields in SGMII_CR_ADDR

Field Name	Bit	Access	Description	Default
ADDRESS	15:0	R/W	This field provides the register address for the CR read or write transfer. This field must not be changed when START_BUSY is asserted.	0x0000

1.10.1.47 mx_chip_top_regs:mx_chip_top_regs:SGMII_CR_DATA

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 317. Fields in SGMII_CR_DATA

Field Name	Bit	Access	Description	Default
DATA	15:0	R/W	This field provides the data for the CR read or write transfer. This field must not be changed when START_BUSY is asserted.	0x0000

1.10.1.48 mx_chip_top_regs:mx_chip_top_regs:SGMII_VREG_BYP

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 318. Fields in SGMII_VREG_BYP

Field Name	Bit	Access	Description	Default
RESERVED_194	15:5	R/O		0x000
REFCLK_SEL	4:3	R/O	Read-only status of the REFCLK_SEL[1:0] input pins 00 = SYSPLL ref is 25 MHz from XI/XO QSGMII ref is 25 MHz from XI/XO 01 = SYSPLL ref is 25 MHz from XI/XO QSGMII ref is from QSGMII_EXT_REF input pins 10 = SYSPLL ref is 25 MHz from CK125_REF_IN input pins QSGMII ref is 125 MHz from CK125_REF_IN input pins 11 = RESERVED	0x0
TX_MSB_TO_LSB	2	R/W	Controls Tx data bit order from QSGMII Extender to serdes 1:Swaps the Tx data bit order 0: Does not swap the Tx data bit order	0x1
RX_MSB_TO_LSB	1	R/W	Controls Rx data bit order from serdes to QSGMII Extender 1:Swaps the Rx data bit order 0: Does not swap the Rx data bit order	0x1
VREG_BYP	0	R/W	Controls the VREG_BYPASS input pin into the XAUI PHY. 1 = serdes will be powered directly from 2.5V (regulator bypassed) 0 = serdes will be powered from internal 3.3V->2.5V regulator (regulator in use) Note: Default value depends on eFUSE setting. Note: This signal is implemented as a strap into the XAUI PHY. It will not take affect until after a reset is issue to the PHY.	0x0

1.10.1.49 mx_chip_top_regs:mx_chip_top_regs:REC_CLK_OUT_SEL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 319. Fields in REC_CLK_OUT_SEL

Field Name	Bit	Access	Description	Default
RESERVED_199	15:3	R/O		0x0000
REC_CLK_OUT_SELECT_200	2:0	R/W	Recovered Clock Output 1 Selector [2:0] This field selects the source for Recovered Clock Output 1. 000 : PHY 001 : Reserved 010 : Reserved 011 : Reserved3 100 : Reserved1 101 : Reserved 110 : Reserved 111 : Zero. Forces Recovered Clock Output 1 to 0.	0x0

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1.10.1.50 mx_chip_top_regs:mx_chip_top_regs:REC_CLK_OUT_DIV

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 320. Fields in REC_CLK_OUT_DIV

Field Name	Bit	Access	Description	Default
RESERVED_201	15:6	R/O		0x000
REC_CLK_OUT_DIV_202	5:0	R/W	Recovered Clock Output 1 Divider [5:0] This field controls the divider for Recovered Clock Output 1. 000000 : Reserved 000001 : Output Clock frequency is the Input Clock Frequency divided by 1 | | 111111 : Output Clock frequency is the Input clock Frequency divided by 63	0x00

1.10.1.51 mx_chip_top_regs:mx_chip_top_regs:SGMII_PCS_HRST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 321. Fields in SGMII_PCS_HRST

Field Name	Bit	Access	Description	Default
RESERVED_203	15:1	R/O		0x0000
PCS_HRST	0	R/W	When set the XAUI PHY's PCS block and all registers revert to their default state. Writing a zero has no effect	0x0

1.10.1.52 mx_chip_top_regs:mx_chip_top_regs:SGMII_PCS_SRST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 322. Fields in SGMII_PCS_SRST

Field Name	Bit	Access	Description	Default
RESERVED_205	15:1	R/O		0x0000
PCS_SRST	0	R/W	When set the XAUI PHY's PCS block is reset except for the configuration registers. Writing a zero has no effect	0x0

1.10.1.53 mx_chip_top_regs:mx_chip_top_regs:SGMII_PCS_CFG

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 323. Fields in SGMII_PCS_CFG

Field Name	Bit	Access	Description	Default
RESERVED_207	15:12	R/O		0x0

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Field Name	Bit	Access	Description	Default
SAVE_PREAMBLE_ENA	11	R/W	Preserves preamble when using PCH/MCH or preemption. When SAVE_PREAMBLE_ENA = 0, the PCS1G may occasionally drop a preamble byte due to even/odd byte alignment requirements 0 = Preamble Preservation disabled 1 = Preamble Preservation enabled	0x1
LINK_STATUS_TYPE	10	R/W	Define type of QSGMII link status indication 1 = QSGMII Link Partner up/down indication from lp_adv_ability[15] 0 = Sync Status from QSGMII PCS1G state machine	0x0
PCS_ENA	9	R/W	1 = enable QSGMII PCS1G 0 = disable QSGMII PCS1G Default value depends on eFUSE settings.	0x0
RESERVED_211	8	R/O		0x0
SD_POL	7	R/W	Selects the polarity of the signal_detect line 0 = a 0 on the signal_detect line indicates active signal 1 = a 1 on the signal_detect line indicates active signal	0x1
SD_ENA	6	R/W	Signal detect enable 0 = the signal_detect line is ignored. The QSGMII PCS1G assumes an active signal_detect at all times. 1 = the signal_detect line is used to determine if a signal is detected	0x1
SIGNAL_DETECT	5	R/O	Provides the current status of the QSGMII serdes signal detect. 0 = no signal is detected 1= signal is detected	0x0
LINK_STATUS	4	R/O	Indicates whether the QSGMII PCS1G link is up or down. The QSGMII PCS1G link is up when the ANEG state machine is in either the LINK_OK state, or the ANEG_DISABLE_LINK_OK state. 0 = QSGMII PCS1G link is down 1 = QSGMII PCS1G is up	0x0
SYNC_STATUS	3	R/O	Indicates if the QSGMII PCS1G has successfully synchronized 0 = QSGMII PCS1G is not synchronized 1 = QSGMII PCS1G is synchronized	0x0
LT_TESTMODE	2	R/W	Enables test mode with link timer divided by 1024 1 = enable link timer test mode 0 = disable link timer test mode (normal operation)	0x0
LINK_STATUS_CHANGE	1	R/O	This bit indicates a change of QSGMII PCS1G Link status. 0 = no change of QSGMII PCS1G Link status 1 = QSGMII PCS1G Link status has changed Note: Reading this field will clear it.	0x0
SYNC_STATUS_CHANGE	0	R/O	This bit indicates a change of QSGMII PCS1G Sync status. 0 = no change of QSGMII PCS1G Sync status	0x0

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Field Name	Bit	Access	Description	Default
			1 = QSGMII PCS1G Sync status has changed Note: Reading this field will clear it.	

1.10.1.54 mx_chip_top_regs:mx_chip_top_regs:PCS_ANEG_CFG

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 324. Fields in PCS_ANEG_CFG

Field Name	Bit	Access	Description	Default
RESERVED_220	15:7	R/O		0x000
AN_COMPLETE	6	R/O	0 = ANEG not complete 1 = ANEG complete	0x0
SW_RESOLVE_PRIORITY	5	R/W	Resolve priority via software 0 = software does not resolve priority 1 = software resolves priority	0x1
ANEG_RESTART_ONE_SHOT	4	R/W	Initiates restart of Autonegotiation with the QSGMII Link Partner 0 = no action 1 = restart autonegotiation	0x0
ANEG_ENA	3	R/W	0 = disable Autonegotiation with the QSGMII PCS1G Link Partner 1 = enable Autonegotiation with the QSGMII PCS1G Link Partner	0x1
NP_LOADED_ONE_SHOT	2	R/W	TX_NP[15:0] must first be written with the desired NP data, then assert NP_LOADED_ONE_SHOT for this data to be loaded (take effect). 0 = no action 1 = load the contents of TX_NP[15:0]	0x0
PRIORITY_RESOLVE	1	R/O	Indicates ANEG priority can now be resolved by software 0 = ANEG is in process 1 = ANEG is nearly completed and priority can be resolved by software	0x0
NEW_PAGE_RX	0	R/O	0 = no new page received from the QSGMII PCS1G Link Partner 1 = new page received from the QSGMII PCS1G Link Partner Note: Reading this field will clear it.	0x0

1.10.1.55 mx_chip_top_regs:mx_chip_top_regs:PCS_ANEG_TX_ADV_ABLT

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 325. Fields in PCS_ANEG_TX_ADV_ABLT

Field Name	Bit	Access	Description	Default
TX_ADV_ABILITY	15:0	R/W	When QSGMII_AUTO_ANEG_ENA=0, this register must be fully software-configured with the ANEG Abilities to be sent to the QSGMII PCS1G Link Partner.	0x1401

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Field Name	Bit	Access	Description	Default
			When QSGMII_AUTO_ANEG_ENA=1 LINKUP/DOWN (bit 15) , SPEED (bits 11:9) when SGMII_USGMII_TX_CONFIG_EN = 0, SPEED (bits 11:10) when SGMII_USGMII_TX_CONFIG_EN = 1, and DUPLEX (bit 12) are automatically configured by hardware based on PHY settings. Writing to this bits has no effect, however the automatically-determined settings can be read here. All other bits must be software-configured as normal. EEE Capability (bit 8 when SGMII_USGMII_TX_CONFIG_EN = 0, bit 9 when SGMII_USGMII_TX_CONFIG_EN = 1) and EEE Clock Stop Capability (bit 7 when SGMII_USGMII_TX_CONFIG_EN = 0, bit 8 when SGMII_USGMII_TX_CONFIG_EN = 1) are both set to 0 (not supported) by default. Note: In Molineux, EEE is not supported. Bit 0 (defined only as 1 in the SGMII, QSGMII, and USGMII standards, defined as 0 in the IEEE 802.3 Clause 37 standard) is set to 1 by default.	

1.10.1.56 mx_chip_top_regs:mx_chip_top_regs:PCS_ANEG_TX_NP_DATA

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 326. Fields in PCS_ANEG_TX_NP_DATA

Field Name	Bit	Access	Description	Default
NP_TX	15:0	R/W	ANEG Next Page data to be sent to the QSGMII PCS1G Link Partner	0x0000

1.10.1.57 mx_chip_top_regs:mx_chip_top_regs:PCS_ANEG_LP_ADV_ABLT

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 327. Fields in PCS_ANEG_LP_ADV_ABLT

Field Name	Bit	Access	Description	Default
LP_RX_ADV_ABILITY	15:0	R/O	ANEG Advertised Abilities data received from the QSGMII PCS1G Link Partner	0x0000

1.10.1.58 mx_chip_top_regs:mx_chip_top_regs:PCS_ANEG_LP_NP_DATA

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 328. Fields in PCS_ANEG_LP_NP_DATA

Field Name	Bit	Access	Description	Default
LP_NP_RX	15:0	R/O	ANEG Next Page data received from the QSGMII PCS1G Link Partner	0x0000

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1.10.1.59 mx_chip_top_regs:mx_chip_top_regs:PCS_DEBUG

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 329. Fields in PCS_DEBUG

Field Name	Bit	Access	Description	Default
RESERVED_232	15:9	R/O		0x00
QSGMII_RX_CAR_EXT_FILTER_ENA	8	R/W	0 = Do not filter Carrier Extend symbols after end of packet (normal mode) 1 = Filter Carrier Extend symbols after end of packet If enabled, filtering is from QSGMII Rx toward the Tx media interface	0x0
QSGMII_TX_CAR_EXT_FILTER_ENA	7	R/W	0 = Do not filter Carrier Extend symbols after end of packet (normal mode) 1 = Filter Carrier Extend symbols after end of packet If enabled, filtering is from Rx media interface toward QSGMII Tx	0x0
UNIDIR_MODE	6	R/W	Enables 802.3 Clause 66 Unidirectional Mode 0 = Unidirectional Mode disabled 1 = Unidirectional Mode enabled	0x0
EN_CDET	5	R/W	Enables Comma Detection in PCS1G. Comma Detection is normally performed by the QSGMII Extender. 0 = Comma Detection disabled 1 = Comma Detection enabled	0x1
RA_ENA	4	R/W	Enables Rate Adaptation in Rx direction, required when GMII_LB_ENA = 1 0 = Rate Adaptation disabled 1 = Rate Adaptation enabled	0x0
XMIT_MODE	3:2	R/O	Indicates the current QSGMII PCS1G TBI mode 00 = Idle mode 01 = Configuration mode 10 = RESERVED 11 = Data mode	0x0
TBI_HOST_LB_ENA	1	R/W	Enables QSGMII PCS1G TBI loopback (TBI Tx to TBI Rx direction) 0 = TBI loopback disabled 1 = TBI loopback enabled	0x0
GMII_LB_ENA	0	R/W	Enables QSGMII PCS1G GMII loopback (GMIII Rx to GMII Tx direction) 0 = GMII loopback disabled 1 = GMII loopback enabled	0x0

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1.10.1.60 mx_chip_top_regs:mx_chip_top_regs:PCS_LPI_CFG_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

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Table 330. Fields in PCS_LPI_CFG_STS

Field Name	Bit	Access	Description	Default
RESERVED_241	15:5	R/O		0x000
TX_ASSERT_LPIDL	4	R/W	Asserts Low Power Idle control toward the QSGMII Link Partner 0 = disable LPI in the QSGMII transmit direction 1 = enable LPI in the QSGMII transmit direction	0x0
RX_LPI_OUT_DIS	3	R/W	Disable LPI output from the QSGMII PCS1G toward the GPHY 0 = LPI output enabled 1 = LPI output disabled	0x0
LPI_TESTMODE	2	R/W	Enables LPI timer test mode (reduced test time) 0 = LPI timer test mode disabled (normal operation) 1 = LPI timer test mode enabled	0x0
RX_LPI_MODE	1	R/O	Indicates receiver is in low-power idle mode 0 = receiver is not in low-power idle mode 1 = receiver is in low-power idle mode	0x0
TX_LPI_MODE	0	R/O	Indicates transmitter is in low-power idle mode 0 = transmitter is not in low-power idle mode 1 = transmitter is in low-power idle mode	0x0

1.10.1.61 mx_chip_top_regs:mx_chip_top_regs:PCS_TEST_CFG_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 331. Fields in PCS_TEST_CFG_STS

Field Name	Bit	Access	Description	Default
JTP_ERR_CNT	15:8	R/O	JTP Error Count Count of Jitter Test Pattern errors. Counter saturates at FFh and is cleared to zero upon read. Note: Reading this field will clear it.	0x00
RESERVED_248	7:5	R/O		0x0
JTP_SEL	4:2	R/W	JTP_SEL[2:0] Selects the jitter test pattern. Patterns are according to IEEE 802.3 Annex 36A. 000 = Disable transmission of test patterns 001 = High Frequency Test Pattern repeated transmission of D21.5 code group 010 = Low Frequency Test Pattern repeated transmission of K28.7 code group 011 = Mixed Frequency Test Pattern repeated transmission of K28.5 code group 100 = Long Continuous Random Test Pattern 1524 byte frames 101 = Short Continuous Random Test Pattern 360 byte frames 110 = RESERVED 111 = RESERVED	0x0

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Field Name	Bit	Access	Description	Default
JTP_ERR	1	R/O	JTP_ERR Indicates jitter pattern checker has found an error 0 = jitter pattern checker has not found an error 1 = jitter pattern checker has found an error	0x0
JTP_LOCK	0	R/O	JTP_LOCK Indicates jitter pattern checker is locked 0 = jitter pattern checker is not locked 1 = jitter pattern checker is locked	0x0

1.10.1.62 mx_chip_top_regs:mx_chip_top_regs:CHIP_HARD_RST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 332. Fields in CHIP_HARD_RST

Field Name	Bit	Access	Description	Default
RESERVED_252	15:1	R/O		0x0000
CHIP_HARD_RESET	0	R/W	Write a one to this bit to perform a chipwide hard reset.	0x0

1.10.1.63 mx_chip_top_regs:mx_chip_top_regs:CHIP_SOFT_RST

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 333. Fields in CHIP_SOFT_RST

Field Name	Bit	Access	Description	Default
RESERVED_254	15:1	R/O		0x0000
CHIP_SOFT_RESET	0	R/W	Write a one to this bit to perform a chipwide soft reset.	0x0

1.10.1.64 mx_chip_top_regs:mx_chip_top_regs:REGULATOR_CTL

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 334. Fields in REGULATOR_CTL

Field Name	Bit	Access	Description	Default
RESERVED_256	15:5	R/O		0x000
SGMII_RGMII_CP_ENB	4	R/W	Controls the SGMII/RGMII pad cp_enb signal.	0x0
SGMII_RGMII_RBULK_SHORT	3	R/W	Controls the SGMII/RGMII pad rbulk_short signal.	0x0
REG_CLIMIT_ENB_CR	2	R/W	Controls the regulator climit_enb_cr signal. Active low.	0x0
REG_CLIMIT_TEST	1:0	R/W	Controls the regulator climit_test[1:0] signals. Startup Current Limiter Threshold settings:	0x0

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Field Name	Bit	Access	Description	Default
			00: 702mA-1000mA 01: 635mA-894mA 10: 767mA-1118mA 11: 833mA-1200mA	

1.10.1.65 mx_chip_top_regs:mx_chip_top_regs:SKU_DBG_STS

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 335. Fields in SKU_DBG_STS

Field Name	Bit	Access	Description	Default
RESERVED_261	15:11	R/O		0x00
STRAP_STS	10:3	R/O	Returns the status of the STRAP pins. STRAP_STS[7]: SKUSEL1_STRP STRAP_STS[6]: REFCLK_SEL0_STRP STRAP_STS[5]: SPEED_SEL_STRP/PHYAD2_STRP STRAP_STS[4]: PHYAD0_STRP STRAP_STS[3]: AUTO_MODE_EN_STRP STRAP_STS[2]: AUTO_NEG_EN_STRP/PHYAD3_STRP STRAP_STS[1]: PHYAD1_STRP STRAP_STS[0]: MODE_STRP/PHYAD4_STRP NOTE: the reset value depends on the strap values.	0x00
DEBUG_EN	2	R/O	Returns the status of the DBG_EN pin.	0x0
SKU_SEL	1:0	R/O	Returns the status of the SKU_SEL0/1 pins.	0x0

1.10.1.66 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT0

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 336. Fields in EFUSE_READ_DAT0

Field Name	Bit	Access	Description	Default
DIE_DATA_0	15:0	R/O	DIE_DATA[15:0]. Device die data programmed at wafer probe.	0x0000

1.10.1.67 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT1

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 337. Fields in EFUSE_READ_DAT1

Field Name	Bit	Access	Description	Default
DIE_DATA_1	15:0	R/O	DIE_DATA[31:16]. Device die data programmed at wafer probe.	0x0000

1.10.1.68 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT2

Parent: mx_chip_top_regs:mx_chip_top_regs

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Instances: 1

Table 338. Fields in EFUSE_READ_DAT2

Field Name	Bit	Access	Description	Default
DIE_DATA_2	15:0	R/O	DIE_DATA[47:32]. Device die data programmed at wafer probe.	0x0000

1.10.1.69 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT3

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 339. Fields in EFUSE_READ_DAT3

Field Name	Bit	Access	Description	Default
DIE_DATA_3	15:0	R/O	DIE_DATA[63:48]. Device die data programmed at wafer probe.	0x0000

1.10.1.70 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT4

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 340. Fields in EFUSE_READ_DAT4

Field Name	Bit	Access	Description	Default
UUID_0	15:0	R/O	UUID[15:0]. Device UUID programmed at final test.	0x0000

1.10.1.71 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT5

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 341. Fields in EFUSE_READ_DAT5

Field Name	Bit	Access	Description	Default
UUID_1	15:0	R/O	UUID[31:16]. Device UUID programmed at final test.	0x0000

1.10.1.72 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT6

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 342. Fields in EFUSE_READ_DAT6

Field Name	Bit	Access	Description	Default
RESERVED_DAT6_271	15:13	R/O	Reserved for future use.	0x0
DEVAD	12:8	R/O	Default PHY address. Defines default for PHY DEVAD. Configuration straps further modify this value.	0x00
UUID_2	7:0	R/O	UUID[39:32]. Device UUID programmed at final test.	0x00

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1.10.1.73 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT7

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 343. Fields in EFUSE_READ_DAT7

Field Name	Bit	Access	Description	Default
DEV_ID_HI	15:12	R/O	This field represents Device ID bits [11:8].	0x0
DEV_ID_LO	11:8	R/O	This field represents Device ID bits [3:0]. NOTE: Device ID bits [15:12] are always equal to 8, and Device ID bits [7:4] are always equal to 7 for Molineux.	0x0
RESERVED_DAT7_276	7:6	R/O	Reserved for future use.	0x0
MODEL_NUM	5:0	R/O	Vendor Model Number	0x00

1.10.1.74 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT8

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 344. Fields in EFUSE_READ_DAT8

Field Name	Bit	Access	Description	Default
RESERVED_DAT8_278	15:0	R/O	Reserved for future use.	0x0000

1.10.1.75 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT9

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 345. Fields in EFUSE_READ_DAT9

Field Name	Bit	Access	Description	Default
EFUSE_LOCK	15	R/O	If this bit is set the eFUSE controller shall block any future attempts to program the eFUSE.	0x0
EF_EM_DIS	14	R/O	When set eFUSE emulation mode is permanently disabled. Note: This bit must be set in all production parts.	0x0
VREG_BYP_CFG_EF	13	R/O	If this bit is set and SGMII is enabled, HW shall set VREG_BYP based upon the voltage detected in the UVOV instance assigned to the VDD_RGMII pad ring segment.	0x0
SEND_WUP	12	R/O	When set a WUP will always be sent immediately after a system reset event completes. Sets default for respective register.	0x0
DIS_100BT1	11	R/O	When set, 100Base-T1 operation is permanently disabled. This can not be overridden by any CSR bit. This and 1000BT_DIS are mutually exclusive.	0x0
DIS_1000BT1	10	R/O	When set, 1000Base-T1 operation is	0x0

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Field Name	Bit	Access	Description	Default
			permanently disabled. This can not be overridden by any CSR bit. This and 100BT_DIS are mutually exclusive.	
SGMII_DIS	9	R/O	Permanently disables the XAUI PHY.	0x0
EF_CLKOUT_EN	8	R/O	Enables clock on CLKOUT pin.	0x0
CLKOUT_FREQ	7:6	R/O	Default frequency for CLKOUT pin.	0x0
RGMII_RX_ID	5	R/O	Enable RGMII Internal Delay mode for the RX clock.	0x0
RGMII_TX_ID	4	R/O	Enable RGMII Internal Delay mode for the TX clock.	0x0
NC_SI_EN	3	R/O	Enable NC_SI mdoe for RMII.	0x0
REV_MII_EN	2	R/O	Enable Reverse MII mode.	0x0
MAC_MODE	1:0	R/O	Default MAC Mode. 00: GMII/MII 01: RGMII 10: RMII 11: SGMII	0x0

1.10.1.76 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT10

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 346. Fields in EFUSE_READ_DAT10

Field Name	Bit	Access	Description	Default
RESERVED_DAT10_15_293	15	R/O	Reserved for future use.	0x0
TUNE_M1	14:9	R/O	TUNE_M for UVOV in VDDIO segment of pad ring.	0x00
RESERVED_DAT10_3_295	8:3	R/O	Reserved for future use.	0x00
TUNE_T	2:0	R/O	UVOV temerature trim. Same for all UVOV instances.	0x0

1.10.1.77 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT11

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 347. Fields in EFUSE_READ_DAT11

Field Name	Bit	Access	Description	Default
RESERVED_DAT11_297	15:12	R/O	Reserved for future use.	0x0
TRIM	11:7	R/O	Trim for PVT sensor.	0x00
PSEL_25	6	R/O	Selects external voltage use by PVT sensor. This defines the default for the register bit that controls the voltage selection for the PVT sensor IP. 0: 3.3V 1: 2.5V	0x0

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Field Name	Bit	Access	Description	Default
TUNE_M2	5:0	R/O	TUNE_M for UVOV in VDD_RGMII segment of pad ring.	0x00

1.10.1.78 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT12

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 348. Fields in EFUSE_READ_DAT12

Field Name	Bit	Access	Description	Default
RESERVED_DAT12_301	15:11	R/O	Reserved for future use.	0x00
BG_TC_TRIM	10:6	R/O	Band gap temperature trim in AFE.	0x00
BG_MAG_TRIM	5:0	R/O	Band gap magnitude trim in AFE.	0x00

1.10.1.79 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT13

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 349. Fields in EFUSE_READ_DAT13

Field Name	Bit	Access	Description	Default
RESERVED_DAT13_304	15:0	R/O	Reserved for future use.	0x0000

1.10.1.80 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT14

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 350. Fields in EFUSE_READ_DAT14

Field Name	Bit	Access	Description	Default
RESERVED_DAT14_305	15:0	R/O	Reserved for future use.	0x0000

1.10.1.81 mx_chip_top_regs:mx_chip_top_regs:EFUSE_READ_DAT15

Parent: mx_chip_top_regs:mx_chip_top_regs

Instances: 1

Table 351. Fields in EFUSE_READ_DAT15

Field Name	Bit	Access	Description	Default
RESERVED_DAT15_306	15:0	R/O	Reserved for future use.	0x0000

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1.11 mx_uvov_regs

Table 352. Register Groups in MMD mx_uvov_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
mx_uvov_regs	0x00000000	1		Page 129

1.11.1 mx_uvov_regs:mx_uvov_regs

Parent: mx_uvov_regs

Instances: 1

Table 353. Registers in mx_uvov_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
UVOV_CTL	0x0000F220	1	Under and over voltage control register	Page 130
UVOV_INT_STS	0x0000F221	1	Under and Over Voltage Interrupt Status Register	Page 131
UVOV_INT_EN	0x0000F222	1	Under and Over Voltage Interrupt Enable Register	Page 131
UVOV_INT_DIS	0x0000F223	1	Under and Over Voltage Interrupt Disable Register	Page 131
UVOV_CFG0_0	0x0000F224	1	Under and Over Voltage Configuration 0 Register Instance 0 NOTE: UVOV Instance 0 deals with the VDD_RGMII domain.	Page 132
UVOV_CFG0_1	0x0000F225	1	Under and Over Voltage Configuration 0 Register Instance 1 NOTE: UVOV Instance 1 deals with the VDDIO domain.	Page 133
UVOV_CFG0_2	0x0000F226	1	Under and Over Voltage Configuration 0 Register Instance 2 NOTE: UVOV Instance 2 deals with the VDDA domain in the AFE.	Page 134
UVOV_CFG0_3	0x0000F227	1	Under and Over Voltage Configuration 0 Register Instance 3 NOTE: UVOV Instance 3 deals with the VDDCORE domain in the AFE.	Page 135
UVOV_CFG1_0	0x0000F228	1	Under and Over Voltage Configuration 1 Register Instance 0	Page 135
UVOV_CFG1_1	0x0000F229	1	Under and Over Voltage Configuration 1 Register Instance 1	Page 136

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
UVOV_CFG1_2	0x0000F22A	1	Under and Over Voltage Configuration 1 Register Instance 2	Page 137
UVOV_CFG1_3	0x0000F22B	1	Under and Over Voltage Configuration 1 Register Instance 3	Page 137
UVOV_CFG2_0	0x0000F22C	1	Under and Over Voltage Configuration 2 Register Instance 0	Page 138
UVOV_CFG2_1	0x0000F22D	1	Under and Over Voltage Configuration 2 Register Instance 1	Page 139
UVOV_CFG2_2	0x0000F22E	1	Under and Over Voltage Configuration 2 Register Instance 2	Page 139
UVOV_CFG2_3	0x0000F22F	1	Under and Over Voltage Configuration 2 Register Instance 3	Page 140
UVOV_CFG3_0	0x0000F230	1	Under and Over Voltage Configuration 3 Register Instance 0	Page 141
UVOV_CFG3_1	0x0000F231	1	Under and Over Voltage Configuration 3 Register Instance 1	Page 141
UVOV_CFG3_2	0x0000F232	1	Under and Over Voltage Configuration 3 Register Instance 2	Page 141
UVOV_CFG3_3	0x0000F233	1	Under and Over Voltage Configuration 3 Register Instance 3	Page 142

1.11.1.1 mx_uvov_regs:mx_uvov_regs:UVOV_CTL

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 354. Fields in UVOV_CTL

Field Name	Bit	Access	Description	Default
RESERVED_0	15:2	R/O		0x0000
MST_OVR_DET_EN	1	R/W	Master over voltage enable. This bit enables over voltage detection. This bit should be set only after all over voltage registers have been configured.	0x0
MST_UVR_DET_EN	0	R/W	Master under voltage enable. This bit enables under voltage detection. This bit should be set only after all under voltage registers have been configured.	0x0

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1.11.1.2 mx_uvov_regs:mx_uvov_regs:UVOV_INT_STS

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 355. Fields in UVOV_INT_STS

Field Name	Bit	Access	Description	Default
RESERVED_3	15:12	R/O		0x0
OV_INT_STS	11:8	Sticky	over voltage interrupt status	0x0
UV_INT_STS	7:4	Sticky	under voltage interrupt status	0x0
TRIM_INT_STS	3:0	Sticky	trim interrupt status	0x0

1.11.1.3 mx_uvov_regs:mx_uvov_regs:UVOV_INT_EN

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 356. Fields in UVOV_INT_EN

Field Name	Bit	Access	Description	Default
RESERVED_7	15:12	R/O		0x0
OV_INT_EN	11:8	R/W	over voltage interrupt enable. When set the respective interrupt is enabled. This bit is cleared by the corresponding interrupt disable bit.	0x0
UV_INT_EN	7:4	R/W	under voltage interrupt enable. When set the respective interrupt is enabled. This bit is cleared by the corresponding interrupt disable bit.	0x0
TRIM_INT_EN	3:0	R/W	trim interrupt enable. When set the respective interrupt is enabled. This bit is cleared by the corresponding interrupt disable bit.	0x0

1.11.1.4 mx_uvov_regs:mx_uvov_regs:UVOV_INT_DIS

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 357. Fields in UVOV_INT_DIS

Field Name	Bit	Access	Description	Default
RESERVED_11	15:12	R/O		0x0
OV_INT_DIS	11:8	R/W	over voltage interrupt disable. When set the respective interrupt is disabled. Note: This is Write-Only from Software perspective.	0x0
UV_INT_DIS	7:4	R/W	under voltage interrupt disable. When set the respective interrupt is disabled. Note: This is Write-Only from Software perspective.	0x0

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Field Name	Bit	Access	Description	Default
TRIM_INT_DIS	3:0	R/W	trim interrupt disable. When set the respective interrupt is disabled. Note: This is Write-Only from Software perspective.	0x0

1.11.1.5 mx_uvov_regs:mx_uvov_regs:UVOV_CFG0_0

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 358. Fields in UVOV_CFG0_0

Field Name	Bit	Access	Description	Default
RESERVED_15	15:12	R/O		0x0
OV_RST_EN_0	11	R/W	When set the respective over voltage detection channel is capable of generating a system reset.	0x0
UV_RST_EN_0	10	R/W	When set the respective under voltage detection channel is capable of generating a system reset.	0x0
VDD_REF_SEL_0	9:8	R/W	When suplvl_det is cleared this field specifies the reference voltage to be used by UV/OV module. 00b: 1.8V 01b: 2.5V 1Xb: 3.3V This field has no meaning for the core voltage UV/OV detector. This should be written to 00b for that instance.	0x0
SUPLVL_DET_0	7	R/W	When set the supply voltage is internally detected for the UV/OV module. Otherwise the UV/OV detector will use the value specified in VDD_REF_SEL. 0b: Use value specified in VDD_REF_SEL. 1b: Use internally detected VDD. Note: When a value of one is written to this bit the SUPPLVL_DET input to the UV/OV module shall drive 1b for 10us.	0x0
TRIM_EN_0	6	R/W	When set this bit enables the trimming function. This bit self clears after the trim function completes.	0x0
OVR_RNG_SEL_0	5:4	R/W	Over-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
UVR_RNG_SEL_0	3:2	R/W	Under-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
OVR_DET_EN_0	1	R/W	This bit enables over voltage detection for this detector.	0x0

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Field Name	Bit	Access	Description	Default
UVR_DET_EN_0	0	R/W	This bit enables under voltage detection for this detector.	0x0

1.11.1.6 mx_uvov_regs:mx_uvov_regs:UVOV_CFG0_1

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 359. Fields in UVOV_CFG0_1

Field Name	Bit	Access	Description	Default
RESERVED_25	15:12	R/O		0x0
OV_RST_EN_1	11	R/W	When set the respective over voltage detection channel is capable of generating a system reset.	0x0
UV_RST_EN_1	10	R/W	When set the respective under voltage detection channel is capable of generating a system reset.	0x0
VDD_REF_SEL_1	9:8	R/W	When suplvl_det is cleared this field specifies the reference voltage to be used by UV/OV module. 00b: 1.8V 01b: 2.5V 1Xb: 3.3V This field has no meaning for the core voltage UV/OV detector. This should be written to 00b for that instance.	0x0
SUPLVL_DET_1	7	R/W	When set the supply voltage is internally detected for the UV/OV module. Otherwise the UV/OV detector will use the value specified in VDD_REF_SEL. 0b: Use value specified in VDD_REF_SEL. 1b: Use internally detected VDD. Note: When a value of one is written to this bit the SUPPLVL_DET input to the UV/OV module shall drive 1b for 10us.	0x0
TRIM_EN_1	6	R/W	When set this bit enables the trimming function. This bit self clears after the trim function completes.	0x0
OVR_RNG_SEL_1	5:4	R/W	Over-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
UVR_RNG_SEL_1	3:2	R/W	Under-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
OVR_DET_EN_1	1	R/W	This bit enables over voltage detection for this detector.	0x0
UVR_DET_EN_1	0	R/W	This bit enables under voltage detection for	0x0

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Field Name	Bit	Access	Description	Default
			this detector.	

1.11.1.7 mx_uvov_regs:mx_uvov_regs:UVOV_CFG0_2

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 360. Fields in UVOV_CFG0_2

Field Name	Bit	Access	Description	Default
RESERVED_35	15:12	R/O		0x0
OV_RST_EN_2	11	R/W	When set the respective over voltage detection channel is capable of generating a system reset.	0x0
UV_RST_EN_2	10	R/W	When set the respective under voltage detection channel is capable of generating a system reset.	0x0
VDD_REF_SEL_2	9:8	R/W	When suplvl_det is cleared this field specifies the reference voltage to be used by UV/OV module. 00b: 1.8V 01b: 2.5V 1Xb: 3.3V This field has no meaning for the core voltage UV/OV detector. This should be written to 00b for that instance.	0x0
SUPLVL_DET_2	7	R/W	When set the supply voltage is internally detected for the UV/OV module. Otherwise the UV/OV detector will use the value specified in VDD_REF_SEL. 0b: Use value specified in VDD_REF_SEL. 1b: Use internally detected VDD. Note: When a value of one is written to this bit the SUPPLVL_DET input to the UV/OV module shall drive 1b for 10us.	0x0
TRIM_EN_2	6	R/W	When set this bit enables the trimming function. This bit self clears after the trim function completes.	0x0
OVR_RNG_SEL_2	5:4	R/W	Over-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
UVR_RNG_SEL_2	3:2	R/W	Under-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
OVR_DET_EN_2	1	R/W	This bit enables over voltage detection for this detector.	0x0
UVR_DET_EN_2	0	R/W	This bit enables under voltage detection for this detector.	0x0

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1.11.1.8 mx_uvov_regs:mx_uvov_regs:UVOV_CFG0_3

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 361. Fields in UVOV_CFG0_3

Field Name	Bit	Access	Description	Default
RESERVED_45	15:12	R/O		0x0
OV_RST_EN_3	11	R/W	When set the respective over voltage detection channel is capable of generating a system reset.	0x0
UV_RST_EN_3	10	R/W	When set the respective under voltage detection channel is capable of generating a system reset.	0x0
VDD_REF_SEL_3	9:8	R/W	When suplvl_det is cleared this field specifies the reference voltage to be used by UV/OV module. 00b: 1.8V 01b: 2.5V 1Xb: 3.3V This field has no meaning for the core voltage UV/OV detector. This should be written to 00b for that instance.	0x0
SUPLVL_DET_3	7	R/W	When set the supply voltage is internally detected for the UV/OV module. Otherwise the UV/OV detector will use the value specified in VDD_REF_SEL. 0b: Use value specified in VDD_REF_SEL. 1b: Use internally detected VDD. Note: When a value of one is written to this bit the SUPPLVL_DET input to the UV/OV module shall drive 1b for 10us.	0x0
TRIM_EN_3	6	R/W	When set this bit enables the trimming function. This bit self clears after the trim function completes.	0x0
OVR_RNG_SEL_3	5:4	R/W	Over-voltage range select. Where X is VDD. 00b: X+X*(5/100) 01b: X+X*(7.5/100) 10b: X+X*(10/100) 11b: X+X*(2.5/100)	0x1
UVR_RNG_SEL_3	3:2	R/W	Under-voltage range select. Where X is VDD. 00b: X-X*(5/100) 01b: X-X*(7.5/100) 10b: X-X*(10/100) 11b: X-X*(2.5/100)	0x1
OVR_DET_EN_3	1	R/W	This bit enables over voltage detection for this detector.	0x0
UVR_DET_EN_3	0	R/W	This bit enables under voltage detection for this detector.	0x0

1.11.1.9 mx_uvov_regs:mx_uvov_regs:UVOV_CFG1_0

Parent: mx_uvov_regs:mx_uvov_regs

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Instances: 1

Table 362. Fields in UVOV_CFG1_0

Field Name	Bit	Access	Description	Default
RESERVED_55	15	R/O		0x0
TUNE_CDR_0	14:12	R/W	Temperature trim tune input. This register will be loaded with the respective trim value in OTP/eFUSE (TUNE_T).	0x0
TUNE_M_0	11:6	R/W	Magnitude trim tune input. This register will be loaded with the respective trim value in OTP/eFUSE (TUNE_M2).	0x00
TRIM_MAG_0	5:0	R/W	This field is used to report the result of the trim sequence. When TRIM_EN is asserted the value in this register is used by the internal HW FSM as the starting point for the trim sequence. This value is decremented automatically until the UV output of this UVOV module toggles. After the trim process completes the new value is loaded into this register. The TRIM_EN bit then self clears and the respective interrupt asserts. Note: The trim process is only run once at ATE. Note: The value in this field is stored in OTP/eFUSE and must be written into the respective TUNE_M register before using the UVOV function.	0x3F

1.11.1.10 mx_uvov_regs:mx_uvov_regs:UVOV_CFG1_1

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 363. Fields in UVOV_CFG1_1

Field Name	Bit	Access	Description	Default
RESERVED_59	15	R/O		0x0
TUNE_CDR_1	14:12	R/W	Temperature trim tune input. This register will be loaded with the respective trim value in OTP/eFUSE (TUNE_T).	0x0
TUNE_M_1	11:6	R/W	Magnitude trim tune input. This register will be loaded with the respective trim value in OTP/eFUSE (TUNE_M1).	0x00
TRIM_MAG_1	5:0	R/W	This field is used to report the result of the trim sequence. When TRIM_EN is asserted the value in this register is used by the internal HW FSM as the starting point for the trim sequence. This value is decremented automatically until the UV output of this UVOV module toggles. After the trim process completes the new value is loaded into this register. The TRIM_EN bit then self clears and the respective interrupt asserts. Note: The trim process is only run once at ATE.	0x3F

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Field Name	Bit	Access	Description	Default
			Note: The value in this field is stored in OTP/eFUSE and must be written into the respective TUNE_M register before using the UVOV function.	

1.11.1.11 mx_uvov_regs:mx_uvov_regs:UVOV_CFG1_2

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 364. Fields in UVOV_CFG1_2

Field Name	Bit	Access	Description	Default
RESERVED_63	15	R/O		0x0
TUNE_CDR_2	14:12	R/W	Temperature trim tune input. Not used. Replaced by bg_tc_trim in afe_common_block_control_2.	0x0
TUNE_M_2	11:6	R/W	Magnitude trim tune input. Not used. Replaced by bg_mag_trim in afe_common_block_control_2.	0x00
TRIM_MAG_2	5:0	R/W	This field is used to report the result of the trim sequence. When TRIM_EN is asserted the value in this register is used by the internal HW FSM as the starting point for the trim sequence. This value is decremented automatically until the UV output of this UVOV module toggles. After the trim process completes the new value is loaded into this register. The TRIM_EN bit then self clears and the respective interrupt asserts. Note: The trim process is only run once at ATE. Note: The value in this field is stored in OTP/eFUSE and must be written into the respective TUNE_M register before using the UVOV function.	0x3F

1.11.1.12 mx_uvov_regs:mx_uvov_regs:UVOV_CFG1_3

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 365. Fields in UVOV_CFG1_3

Field Name	Bit	Access	Description	Default
RESERVED_67	15	R/O		0x0
TUNE_CDR_3	14:12	R/W	Temperature trim tune input. Not used. Replaced by bg_tc_trim in afe_common_block_control_2.	0x0
TUNE_M_3	11:6	R/W	Magnitude trim tune input. Not used. Replaced by bg_mag_trim in afe_common_block_control_2.	0x00
TRIM_MAG_3	5:0	R/W	This field is used to report the result of the	0x3F

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Field Name	Bit	Access	Description	Default
			trim sequence. When TRIM_EN is asserted the value in this register is used by the internal HW FSM as the starting point for the trim sequence. This value is decremented automatically until the UV output of this UVOV module toggles. After the trim process completes the new value is loaded into this register. The TRIM_EN bit then self clears and the respective interrupt asserts. Note: The trim process is only run once at ATE. Note: The value in this field is stored in OTP/eFUSE and must be written into the respective TUNE_M register before using the UVOV function.	

1.11.1.13 mx_uvov_regs:mx_uvov_regs:UVOV_CFG2_0

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 366. Fields in UVOV_CFG2_0

Field Name	Bit	Access	Description	Default
UVOV_DEB_LO_0	15:8	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. A value of 0h disables the debouncer.	0xFF
OV_SLEEP_EN_0	7	R/W	When set the respective over voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
UV_SLEEP_EN_0	6	R/W	When set the respective under voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
OV_DEB_EN_0	5	R/W	When set the respective over voltage detection channel is de-bounced.	0x0
UV_DEB_EN_0	4	R/W	When set the respective under voltage detection channel is de-bounced.	0x0
V33_STS_0	3	R/O	Provides the real time status of the UVOV detectors V33_CR output signal. This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
V25_STS_0	2	R/O	Provides the real time status of the UVOV detectors V25_CR output signal. Note: This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
OV_STS_0	1	R/O	Provides the real time status of the UVOV detectors OV_CR output signal. Note: The default is Xb for this bit.	0x0
UV_STS_0	0	R/O	Provides the real time status of the UVOV detectors UV_CR output signal. Note: The default is Xb for this bit.	0x0

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1.11.1.14 mx_uvov_regs:mx_uvov_regs:UVOV_CFG2_1

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 367. Fields in UVOV_CFG2_1

Field Name	Bit	Access	Description	Default
UVOV_DEB_LO_1	15:8	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. A value of 0h disables the debouncer.	0xFF
OV_SLEEP_EN_1	7	R/W	When set the respective over voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
UV_SLEEP_EN_1	6	R/W	When set the respective under voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
OV_DEB_EN_1	5	R/W	When set the respective over voltage detection channel is de-bounced.	0x0
UV_DEB_EN_1	4	R/W	When set the respective under voltage detection channel is de-bounced.	0x0
V33_STS_1	3	R/O	Provides the real time status of the UVOV detectors V33_CR output signal. This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
V25_STS_1	2	R/O	Provides the real time status of the UVOV detectors V25_CR output signal. Note: This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
OV_STS_1	1	R/O	Provides the real time status of the UVOV detectors OV_CR output signal. Note: The default is Xb for this bit.	0x0
UV_STS_1	0	R/O	Provides the real time status of the UVOV detectors UV_CR output signal. Note: The default is Xb for this bit.	0x0

1.11.1.15 mx_uvov_regs:mx_uvov_regs:UVOV_CFG2_2

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 368. Fields in UVOV_CFG2_2

Field Name	Bit	Access	Description	Default
UVOV_DEB_LO_2	15:8	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. A value of 0h disables the debouncer.	0xFF
OV_SLEEP_EN_2	7	R/W	When set the respective over voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
UV_SLEEP_EN_2	6	R/W	When set the respective under voltage	0x0

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Field Name	Bit	Access	Description	Default
			detection channel is capable of forcing the device into the TC10 SLEEP state.	
OV_DEB_EN_2	5	R/W	When set the respective over voltage detection channel is de-bounced.	0x0
UV_DEB_EN_2	4	R/W	When set the respective under voltage detection channel is de-bounced.	0x0
V33_STS_2	3	R/O	Provides the real time status of the UVOV detectors V33_CR output signal. This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
V25_STS_2	2	R/O	Provides the real time status of the UVOV detectors V25_CR output signal. Note: This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
OV_STS_2	1	R/O	Provides the real time status of the UVOV detectors OV_CR output signal. Note: The default is Xb for this bit.	0x0
UV_STS_2	0	R/O	Provides the real time status of the UVOV detectors UV_CR output signal. Note: The default is Xb for this bit.	0x0

1.11.1.16 mx_uvov_regs:mx_uvov_regs:UVOV_CFG2_3

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 369. Fields in UVOV_CFG2_3

Field Name	Bit	Access	Description	Default
UVOV_DEB_LO_3	15:8	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. A value of 0h disables the debouncer.	0xFF
OV_SLEEP_EN_3	7	R/W	When set the respective over voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
UV_SLEEP_EN_3	6	R/W	When set the respective under voltage detection channel is capable of forcing the device into the TC10 SLEEP state.	0x0
OV_DEB_EN_3	5	R/W	When set the respective over voltage detection channel is de-bounced.	0x0
UV_DEB_EN_3	4	R/W	When set the respective under voltage detection channel is de-bounced.	0x0
V33_STS_3	3	R/O	Provides the real time status of the UVOV detectors V33_CR output signal. This bit does not apply for the core voltage detector. Note: The default is Xb for this bit.	0x0
V25_STS_3	2	R/O	Provides the real time status of the UVOV detectors V25_CR output signal. Note: This bit does not apply for the core	0x0

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Field Name	Bit	Access	Description	Default
			voltage detector. Note: The default is Xb for this bit.	
OV_STS_3	1	R/O	Provides the real time status of the UVOV detectors OV_CR output signal. Note: The default is Xb for this bit.	0x0
UV_STS_3	0	R/O	Provides the real time status of the UVOV detectors UV_CR output signal. Note: The default is Xb for this bit.	0x0

1.11.1.17 mx_uvov_regs:mx_uvov_regs:UVOV_CFG3_0

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 370. Fields in UVOV_CFG3_0

Field Name	Bit	Access	Description	Default
UVOV_DEB_UP_0	15:0	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. The register specifies bit [23:8] of the repsective debouncer. A value of 0h disables the debouncer.	0x0007

1.11.1.18 mx_uvov_regs:mx_uvov_regs:UVOV_CFG3_1

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 371. Fields in UVOV_CFG3_1

Field Name	Bit	Access	Description	Default
UVOV_DEB_UP_1	15:0	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. The register specifies bit [23:8] of the repsective debouncer. A value of 0h disables the debouncer.	0x0007

1.11.1.19 mx_uvov_regs:mx_uvov_regs:UVOV_CFG3_2

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 372. Fields in UVOV_CFG3_2

Field Name	Bit	Access	Description	Default
UVOV_DEB_UP_2	15:0	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. The register specifies bit [23:8] of the repsective debouncer. A value of 0h disables the debouncer.	0x0000

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1.11.1.20 mx_uvov_regs:mx_uvov_regs:UVOV_CFG3_3

Parent: mx_uvov_regs:mx_uvov_regs

Instances: 1

Table 373. Fields in UVOV_CFG3_3

Field Name	Bit	Access	Description	Default
UVOV_DEB_UP_3	15:0	R/W	Specifies the debounce value used or UV and OV channels that have debounce enabled. This register has units of 8 ns. The register specifies bit [23:8] of the repsective debouncer. A value of 0h disables the debouncer.	0x0000

1.12 mx_pvt_regs

Table 374. Register Groups in MMD mx_pvt_regs

Register Group Name	Start Address in MMD	Instances and Address Spacing	Description	Details
mx_pvt_regs	0x00000000	1		Page 142

1.12.1 mx_pvt_regs:mx_pvt_regs

Parent: mx_pvt_regs

Instances: 1

Table 375. Registers in mx_pvt_regs

Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PVT_CTL1	0x0000F240	1	PVT Control1 Regsiter	Page 143
PVT_CTL2	0x0000F241	1	PVT Control2 Register	Page 144
PVT_STS	0x0000F242	1	PVT Status Register	Page 144
PVT_RESERVED1	0x0000F243	1	Reserved (do not use)	Page 144
PVT_INT_STS	0x0000F244	1	PVT Interrupt Status Register	Page 144
PVT_RESERVED2	0x0000F245	1	Reserved (do not use)	Page 145
PVT_INT_MSK	0x0000F246	1	PVT Interrupt Mask Register	Page 145
PVT_RESERVED3	0x0000F247	1	Reserved (do not use)	Page 146
PVT_DATA	0x0000F248	1	PVT Data Register	Page 146
PVT_RESERVED4	0x0000F249	1	Reserved (do not use)	Page 146
PVT_SAMP_TIME	0x0000F24A	1	PVT Sample Time Register	Page 146
PVT_RESERVED5	0x0000F24B	1	Reserved (do not use)	Page 146
PVT_THERM_COMP_CTL	0x0000F24C	1	PVT Thermal Comparator Control Register	Page 147
PVT_RESERVED6	0x0000F24D	1	Reserved (do not use)	Page 147

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Register Name	Offset within Register Group	Instances and Address Spacing	Description	Details
PVT_THERM_COMP_THR0	0x0000F24E	1	PVT Thermal Comparator Threshold0 Register	Page 147
PVT_RESERVED7	0x0000F24F	1	Reserved (do not use)	Page 147
PVT_THERM_COMP_THR1	0x0000F250	1	PVT Thermal Comparator Threshold1 Register	Page 148
PVT_RESERVED8	0x0000F251	1	Reserved (do not use)	Page 148
PVT_THERM_COMP_THR2	0x0000F252	1	PVT Thermal Comparator Threshold2 Register	Page 148
PVT_RESERVED9	0x0000F253	1	Reserved (do not use)	Page 148
PVT_SAMP_CLK_DIV	0x0000F254	1	PVT Sample Clock Divider Register	Page 148
PVT_RESERVED10	0x0000F255	1	Reserved (do not use)	Page 149
PVT_PSEL25	0x0000F256	1	PVT PSEL25 Register	Page 149
PVT_HARD_RST	0x0000F257	1	PVT Hard Reset Register	Page 149
PVT_SOFT_RST	0x0000F258	1	PVT Soft Reset Register	Page 149
PVT_CLK_DIVIDER	0x0000F259	1	PVT Clock Divider Register	Page 150

1.12.1.1 mx_pvt_regs:mx_pvt_regs:PVT_CTL1

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 376. Fields in PVT_CTL1

Field Name	Bit	Access	Description	Default
RESERVED_0	15	R/O		0x0
PVT_RST_EN	14	R/W	When set a Threshold2 Match will force a chip level reset on the device.	0x0
PVT_SLEEP_EN	13	R/W	When set a Threshold2 Match will force the device into the TC10 Sleep state.	0x0
SEL_TRIM	12:8	R/W	TRIM[4:0] input pins of the AB PVT IP Default value for this register field is loaded from the respective bits in the eFUSE.	0x0F
PROBE_VREG_EN	7	R/W	PROBE_VREG_ENA input pin of the AB PVT IP	0x0
SEL_VSEL	6:5	R/W	Selects the VIN[3:0] input, this is an input to the VSEL[1:0] input pins of the AB PVT IP. In Molinux these inputs are tied to VDDCORE.	0x0
SEL_VSAMPLE	4	R/W	Select for the Voltage Sample Mode. VSAMPLE input pin of the AB PVT IP	0x0
SEL_PSAMPLE	3:2	R/W	Select for the Process Sample Mode. PSAMPLE[1:0] input pins of the AB PVT IP	0x0
PVT_EN	1	R/W	If SEL_ENA=1, this is connected to the ENA input pin of the AB PVT IP	0x0
SEL_ENA	0	R/W	Selects if the ENA input pin to the AB PVT IP is	0x1

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Field Name	Bit	Access	Description	Default
			controlled by HW or SW 0: ENA is controlled by HW 1: ENA is controlled by SW	

1.12.1.2 mx_pvt_regs:mx_pvt_regs:PVT_CTL2

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 377. Fields in PVT_CTL2

Field Name	Bit	Access	Description	Default
RESERVED_10	15:6	R/O		0x000
SEL_VTRIM	5:0	R/W	VTRIM[5:0] input pins of the AB PVT IP	0x00

1.12.1.3 mx_pvt_regs:mx_pvt_regs:PVT_STS

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 378. Fields in PVT_STS

Field Name	Bit	Access	Description	Default
RESERVED_12	15:5	R/O		0x000
THERM2_THR_MATCH	4	R/O	Thermal Threshold 2 Match. Thermal overload mode signal	0x0
THERM1_THR_MATCH	3	R/O	Thermal Threshold 1 Match. Thermal warning mode signal.	0x0
THERM0_THR_MATCH	2	R/O	THERM0_THR_MATCH Thermal Threshold 0 Match. Thermal warning mode signal with hysteresis.	0x0
DATA_VALID_STAT	1	R/O	DATA_VALID output pin of the AB PVT IP.	0x0
ENA_STAT	0	R/O	Value of the ENA input pin of the AB PVT IP.	0x0

1.12.1.4 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED1

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 379. Fields in PVT_RESERVED1

Field Name	Bit	Access	Description	Default
RESERVED_18	15:0	R/O		0x0000

1.12.1.5 mx_pvt_regs:mx_pvt_regs:PVT_INT_STS

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

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Table 380. Fields in PVT_INT_STS

Field Name	Bit	Access	Description	Default
RESERVED_19	15:5	R/O		0x000
THERM2_CHANGE_INT	4	Sticky	1 = change in value of THERM2_THR_MATCH 0 = no change in value of THERM2_THR_MATCH	0x0
THERM1_CHANGE_INT	3	Sticky	1 = change in value of THERM1_THR_MATCH 0 = no change in value of THERM1_THR_MATCH	0x0
THERM0_CHANGE_INT	2	Sticky	1 = change in value of THERM0_THR_MATCH 0 = no change in value of THERM0_THR_MATCH	0x0
DATA_VALID_STAT_INT	1	Sticky	1 = change in value of DATA_VALID_STAT 0 = no change in value of DATA_VALID_STAT	0x0
ENA_STAT_INT	0	Sticky	1 = change in value of ENA_STAT 0 = no change in value of ENA_STAT	0x0

1.12.1.6 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED2

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 381. Fields in PVT_RESERVED2

Field Name	Bit	Access	Description	Default
RESERVED_25	15:0	R/O		0x0000

1.12.1.7 mx_pvt_regs:mx_pvt_regs:PVT_INT_MSK

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 382. Fields in PVT_INT_MSK

Field Name	Bit	Access	Description	Default
RESERVED_26	15:5	R/O		0x000
THERM2_CHANGE_MSK	4	R/W	When set to 0 generates an interrupt when THERM2_CHANGE_INT is set. When 1, this interrupt is disabled.	0x1
THERM1_CHANGE_MSK	3	R/W	When set to 0 generates an interrupt when THERM1_CHANGE_INT is set. When 1, this interrupt is disabled.	0x1
THERM0_CHANGE_MSK	2	R/W	When set to 0 generates an interrupt when THERM0_CHANGE_INT is set. When 1, this interrupt is disabled.	0x1
DATA_VALID_MASK	1	R/W	When set to 0 generates an interrupt when DATA_VALID_INT is set. When 1, this interrupt is disabled.	0x1
ENA_STAT_MASK	0	R/W	When set to 0 generates an interrupt when ENA_INT is set. When 1, this interrupt is disabled.	0x1

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1.12.1.8 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED3

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 383. Fields in PVT_RESERVED3

Field Name	Bit	Access	Description	Default
RESERVED_32	15:0	R/O		0x0000

1.12.1.9 mx_pvt_regs:mx_pvt_regs:PVT_DATA

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 384. Fields in PVT_DATA

Field Name	Bit	Access	Description	Default
RESERVED_33	15:10	R/O		0x00
PVT_DATA	9:0	R/O	The last sensed value from the DATA_OUT[9:0] pins of the AB PVT IP	0x000

1.12.1.10 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED4

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 385. Fields in PVT_RESERVED4

Field Name	Bit	Access	Description	Default
RESERVED_35	15:0	R/O		0x0000

1.12.1.11 mx_pvt_regs:mx_pvt_regs:PVT_SAMP_TIME

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 386. Fields in PVT_SAMP_TIME

Field Name	Bit	Access	Description	Default
SAMPLE_TIME	15:0	R/W	Time between samples taken from the AB PVT IP. 0000h = HW takes a sample whenever DATA_VALID is asserted 0001h-FFFFh: sample time in ms	0x0000

1.12.1.12 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED5

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

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Table 387. Fields in PVT_RESERVED5

Field Name	Bit	Access	Description	Default
RESERVED_37	15:0	R/O		0x0000

1.12.1.13 mx_pvt_regs:mx_pvt_regs:PVT_THERM_COMP_CTL

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 388. Fields in PVT_THERM_COMP_CTL

Field Name	Bit	Access	Description	Default
RESERVED_38	15:2	R/O		0x0000
THERM_COMP_CTL	1:0	R/W	This field controls the Thermal Comparator. It is sampled every 100 us. 00 = Comparator is disabled 01 = Comparator is enabled 10 - RESERVED 11 = RESERVED	0x0

1.12.1.14 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED6

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 389. Fields in PVT_RESERVED6

Field Name	Bit	Access	Description	Default
RESERVED_40	15:0	R/O		0x0000

1.12.1.15 mx_pvt_regs:mx_pvt_regs:PVT_THERM_COMP_THR0

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 390. Fields in PVT_THERM_COMP_THR0

Field Name	Bit	Access	Description	Default
RESERVED_41	15:10	R/O		0x00
THERM_COMP_THRESHOLD0	9:0	R/W	Comparator threshold 0 value, units are VDDA/1024.	0x000

1.12.1.16 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED7

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 391. Fields in PVT_RESERVED7

Field Name	Bit	Access	Description	Default
RESERVED_43	15:0	R/O		0x0000

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1.12.1.17 mx_pvt_regs:mx_pvt_regs:PVT_THERM_COMP_THR1

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 392. Fields in PVT_THERM_COMP_THR1

Field Name	Bit	Access	Description	Default
RESERVED_44	15:10	R/O		0x00
THERM_COMP_THRESHOLD1	9:0	R/W	Comparator threshold 1 value, units are VDDA/1024.	0x000

1.12.1.18 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED8

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 393. Fields in PVT_RESERVED8

Field Name	Bit	Access	Description	Default
RESERVED_46	15:0	R/O		0x0000

1.12.1.19 mx_pvt_regs:mx_pvt_regs:PVT_THERM_COMP_THR2

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 394. Fields in PVT_THERM_COMP_THR2

Field Name	Bit	Access	Description	Default
RESERVED_47	15:10	R/O		0x00
THERM_COMP_THRESHOLD2	9:0	R/W	Comparator threshold 2 value, units are VDDA/1024.	0x000

1.12.1.20 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED9

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 395. Fields in PVT_RESERVED9

Field Name	Bit	Access	Description	Default
RESERVED_49	15:0	R/O		0x0000

1.12.1.21 mx_pvt_regs:mx_pvt_regs:PVT_SAMP_CLK_DIV

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 396. Fields in PVT_SAMP_CLK_DIV

Field Name	Bit	Access	Description	Default
SAMPLE_CLK_DIV	15:0	R/W	This register is used with the AB PVT Clock	0x04B0

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Field Name	Bit	Access	Description	Default
			Divider Register to generate the 1 kHz AB PVT sampling clock. SAMPLE_CLK_DIV provides the divisor value to divide the 1.15-1.25 MHz AB PVT clock down to 1 kHz. Value is set based on frequency of AB PVT sampling clock, as follows. AB PVT sampling clock = 1.15 MHz, SAMPLE_CLK_DIV = d1150 (047Eh). | AB PVT sampling clock = 1.25 MHz, SAMPLE_CLK_DIV = d1250 (04E2h).	

1.12.1.22 mx_pvt_regs:mx_pvt_regs:PVT_RESERVED10

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 397. Fields in PVT_RESERVED10

Field Name	Bit	Access	Description	Default
RESERVED_51	15:0	R/O		0x0000

1.12.1.23 mx_pvt_regs:mx_pvt_regs:PVT_PSEL25

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 398. Fields in PVT_PSEL25

Field Name	Bit	Access	Description	Default
RESERVED_52	15:1	R/O		0x0000
PSEL_25	0	R/W	Controls whether the AB PVT IP runs from 2.5V or 3.3V. The default is 3.3V, set this bit to 1 if VDDAH is connected to 2.5V. The default value is loaded from the respective bit in the eFUSE.	0x0

1.12.1.24 mx_pvt_regs:mx_pvt_regs:PVT_HARD_RST

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 399. Fields in PVT_HARD_RST

Field Name	Bit	Access	Description	Default
RESERVED_54	15:1	R/O		0x0000
PVT_HARD_RST	0	One-shot	1 = Reset the AB PVT and set all configuration registers to their default state Writing a zero has no effect	0x0

1.12.1.25 mx_pvt_regs:mx_pvt_regs:PVT_SOFT_RST

Parent: mx_pvt_regs:mx_pvt_regs

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Table 400. Fields in PVT_SOFT_RST

Field Name	Bit	Access	Description	Default
RESERVED_56	15:1	R/O		0x0000
PVT_SOFT_RST	0	One-shot	1 = Reset the AB PVT except all configuration registers Writing a zero has no effect	0x0

1.12.1.26 mx_pvt_regs:mx_pvt_regs:PVT_CLK_DIVIDER

Parent: mx_pvt_regs:mx_pvt_regs

Instances: 1

Table 401. Fields in PVT_CLK_DIVIDER

Field Name	Bit	Access	Description	Default
RESERVED_58	15:8	R/O		0x00
PVT_CLK_DIV	7:0	R/W	Value is set based on desired value of AB PVT sampling clock, as follows. For 200 MHz system clock: PVT_CLK_DIV = 'd160 (A0h) results in AB PVT clock frequency of 1.25 MHz PVT_CLK_DIV = 'd167 (A7h) results in AB PVT clock frequency of 1.2 MHz approx (1.197M in exact) PVT_CLK_DIV = 'd173 (ADh) results in AB PVT clock frequency of 1.156 MHz	0xA7

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